

A Possible Answer For C. Wu, Methods of Mathematical Physics, 101 Edition

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Started in the morning of 2026/2/8

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Abstract—This is a possible answer by a poor sophomore who was tortured by Methods of Mathematical Physics last semester. I am a student from Hangzhou No. 2 High School who entered through the Gaokao and have no prior experience with academic competitions. A lot of the time, I can't even identify the mathematical or physical context behind the problems. Therefore, my solution may have some flaws, and I welcome your corrections. Below is the table of contents for my solution.

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Index Terms—Morning, Methods of Mathematical Physics, 101 Edition.

I. Complex Numbers and Complex Functions

1. Evaluate the following expressions.

(1)

$$\left(\frac{1+i}{2-i}\right)^2 = \left(\frac{1+3i}{5}\right)^2 = \frac{-8+6i}{25} \quad (\text{I.1})$$

(2) Note that $1+i = \sqrt{2}e^{i\frac{\pi}{4}}, 1-i = \sqrt{2}e^{-i\frac{\pi}{4}}$, thus

$$\begin{aligned} (1+i)^n + (1-i)^n &= 2^{n/2} [e^{i\frac{n\pi}{4}} + e^{-i\frac{n\pi}{4}}] \\ &= 2^{\frac{n}{2}+1} \cos\left(\frac{n\pi}{4}\right) \end{aligned} \quad (\text{I.2})$$

2. Find the real part, imaginary part, modulus, and argument of the following complex numbers.

- (1) The real part is 1, the imaginary part is $-\sqrt{3}$, the modulus is 2, the argument is $-\pi/3 + 2k\pi, k \in \mathbb{Z}$.
- (2) The real part is $\cos(\cos x)$, the imaginary part is $\sin(\cos x)$, the modulus is 1, the argument is $\cos x + 2k\pi, k \in \mathbb{Z}$.
- (3) Note $e^{ix} + e^{-ix} = 2\cos x$, thus the real part is $2\cos x$, the imaginary part is 0, the modulus is $2|\cos x|$, the argument is $2k\pi, k \in \mathbb{Z}$, if $\cos x > 0$, $2k\pi + \pi, k \in \mathbb{Z}$, if $\cos x < 0$, *Not Defined*, if $\cos x = 0$.
- (4) The real part is $e^x - e^{-x}$, the imaginary part is 0, the modulus is $|e^x - e^{-x}|$, the argument is $2k\pi, k \in \mathbb{Z}$, if $x > 0$, $\pi + 2k\pi, k \in \mathbb{Z}$, if $x < 0$, *Not Defined*, if $x = 0$.
- (5) The real part is $\cos(\phi(x))$, the imaginary part is $\sin(\phi(x))$, the modulus is 1, the argument is $\phi(x) + 2k\pi, k \in \mathbb{Z}$.
- (6) (Supposing $\alpha \in [0, \pi]$) The real part is $1 + \cos \alpha$, the imaginary part is $-\sin \alpha$, the modulus is $\sqrt{2+2\cos \alpha}$, the argument is $\pi - \frac{\alpha}{2} + 2k\pi, k \in \mathbb{Z}$.

3. Determine what geometric shape each of the following relations represents.

- (1) This relation equals to $z - z_1 = k(z - z_2), k \in \mathbb{R}$, which means z lies on the line determined by z_1 and z_2 .
- (2) This relation equals to $z - z_1 = ik(z - z_2), k \in \mathbb{R}$, which means the vector $\overrightarrow{zz_1}$ is perpendicular to $\overrightarrow{zz_2}$. Thus, z lies on the circle with diameter z_1z_2 .
- (3) Note that $|z| = \sqrt{x^2 + y^2}$, where x, y is the real part and imaginary part of z , respectively. Thus, the relation equals to

$$y^2 < 2x + 1. \quad (\text{I.3})$$

- (4) Following the approach from the previous problem, we have $x + y < 1$.
- (5) From the relation we have

$$x^4 + y^4 - 2x^2 - 6y^2 + 2x^2y^2 + 1 < 0. \quad (\text{I.4})$$

Note that

$$\begin{aligned} x^4 + y^4 - 2x^2 - 6y^2 + 2x^2y^2 + 1 \\ = (x^2 + y^2 - 2y - 1)(x^2 + y^2 + 2y - 1), \end{aligned} \quad (\text{I.5})$$

thus the relation equals to

$$(A \cup B) - (A \cap B), \quad (\text{I.6})$$

where A denotes the interior of circle $x^2 + (y-1)^2 = 2$, B denotes the interior of circle $x^2 + (y+1)^2 = 2$.

4. Express the following sum in closed form.

(1) Using the sum-to-product identities, we have

$$\begin{aligned} \sum_{k=1}^n \cos k\phi \\ &= \frac{1}{2\sin\left(\frac{\phi}{2}\right)} \sum_{k=1}^n \left[\sin\left(k + \frac{1}{2}\right)\phi - \sin\left(k - \frac{1}{2}\right)\phi \right] \\ &= \frac{1}{\sin\left(\frac{\phi}{2}\right)} \left[\sin\left(n + \frac{1}{2}\right)\phi - \sin\left(\frac{\phi}{2}\right) \right]. \end{aligned} \quad (\text{I.7})$$

(2) Similarly,

$$\begin{aligned} \sum_{k=1}^n \sin(k\phi) \\ &= \frac{1}{2\sin\left(\frac{\phi}{2}\right)} \sum_{k=1}^n \left[-\cos\left(k + \frac{1}{2}\right)\phi + \cos\left(k - \frac{1}{2}\right)\phi \right] \\ &= \frac{1}{2\sin\left(\frac{\phi}{2}\right)} \left[\cos\frac{\phi}{2} - \cos\left(n + \frac{1}{2}\right)\phi \right]. \end{aligned} \quad (\text{I.8})$$

5. Find the cluster points and the limit of the following sequence z_n . If it is a real sequence, also find the limit superior and limit inferior.

(1) The cluster point is 0. Moreover,

$$\overline{\lim}_{n \rightarrow \infty} z_n = 2, \quad \underline{\lim}_{n \rightarrow \infty} z_n = 0. \quad (\text{I.9})$$

(2) Any point satisfying $|z| \leq 1$ is the cluster point.

6. Find the maximum and minimum distances from the origin to the curve $|z + 1/z| = a$.

Let $r = \sqrt{x^2 + y^2}$, thus

$$a = \sqrt{r^2 + \frac{1}{r^2} + \frac{2x^2 - 2y^2}{r^2}}. \quad (\text{I.10})$$

Note that $-2 \leq \frac{2x^2 - 2y^2}{r^2} \leq 2$. Then

$$a^2 - 2 \leq r^2 + \frac{1}{r^2} \leq a^2 + 2, \quad (\text{I.11})$$

thus

$$\frac{\sqrt{a^2 + 4} - a}{2} \leq r \leq \frac{\sqrt{a^2 + 4} + a}{2}. \quad (\text{I.12})$$

II. Analytic function

1. For each function below, determine if a constant A can be chosen such that $f(z)$ becomes continuous at $z = 0$.

Suppose $z = x + yi$,

- (1) Note $f(z) = \frac{x^2 - xyi}{x^2 + y^2}$, $z \neq 0$, thus $\lim_{z \rightarrow 0} f(z)$ does not exist (suppose $y = kx$), so A does not exist.
- (2) Note $f(z) = \frac{x^2 - y^2}{x^2 + y^2}$, thus $\lim_{z \rightarrow 0} f(z)$ does not exist (suppose $y = kx$), so A does not exist.
- (3) Note $f(z) = \frac{x^2 + xyi}{\sqrt{x^2 + y^2}}$, $z \neq 0$, thus $\lim_{z \rightarrow 0} f(z) = 0$, so $A = 1$.
- (4) Note $f(z) = \frac{(x^2 - y^2)^2}{x^2 - y^2 + 2xyi}$, thus $\lim_{z \rightarrow 0} f(z) = 0$, so $A = 1$.

2. Prove the Cauchy-Riemann equation in polar coordinates:

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}, \quad (\text{II.1})$$

and thus prove that

$$f'(z) = \frac{r}{z} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = \frac{1}{z} \left(\frac{\partial v}{\partial \theta} - i \frac{\partial u}{\partial \theta} \right). \quad (\text{II.2})$$

where $u(r, \theta), v(r, \theta)$ is the real and imaginary part of $f(z)$, respectively.

Proof:

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial r} = \cos \theta \frac{\partial u}{\partial x} + \sin \theta \frac{\partial u}{\partial y}. \quad (\text{II.3})$$

Similarly,

$$\frac{\partial v}{\partial \theta} = \frac{\partial v}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial \theta} = -r \sin \theta \frac{\partial v}{\partial x} + r \cos \theta \frac{\partial v}{\partial y}. \quad (\text{II.4})$$

Use the Cauchy-Riemann Equation

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad (\text{II.5})$$

we have (II.1). Moreover,

$$r \frac{\partial u}{\partial r} = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}, \quad r \frac{\partial v}{\partial r} = x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y}, \quad (\text{II.6})$$

use (II.5) again, and note $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$, we have

$$zf'(z) = r \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right) = \frac{\partial v}{\partial \theta} - i \frac{\partial u}{\partial \theta}. \quad (\text{II.7})$$

□

3. Find the real constants a, b, c , s.t. $f(z)$ is analytic.

- (1) Note $u = x + ay$, $v = bx + cy$, thus

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial u}{\partial y} = a, \quad \frac{\partial v}{\partial x} = b, \quad \frac{\partial v}{\partial y} = c. \quad (\text{II.8})$$

Apply the Cauchy-Riemann conditions, we have $c = 1$, and a, b satisfies $a + b = 0$.

- (2) Note $u = \cos x (\cosh y + a \sinh y)$, $v = \sin x (b \cosh y + c \sinh y)$, thus

$$\begin{aligned} \frac{\partial u}{\partial x} &= -\sin x (\cosh y + a \sinh y), \\ \frac{\partial u}{\partial y} &= \cos x (\sinh y + a \cosh y) \\ \frac{\partial v}{\partial x} &= \cos x (b \cosh y + c \sinh y), \\ \frac{\partial v}{\partial y} &= \sin x (b \sinh y + c \cosh y). \end{aligned} \quad (\text{II.9})$$

Apply the Cauchy-Riemann conditions, we have $c = -1$, and a, b satisfies $a + b = 0$.

4. Determine the domain of analyticity of $f(z) = |x^2 - y^2| + 2i|xy|$.

Note $u = |x^2 - y^2|$, $v = 2|xy|$. Suppose $xy > 0$, then

$$\frac{\partial v}{\partial x} \left(= -\frac{\partial u}{\partial y} \right) = 2y, \quad \frac{\partial v}{\partial y} \left(= \frac{\partial u}{\partial x} \right) = 2x. \quad (\text{II.10})$$

In this occasion $x^2 - y^2 \geq 0$. Similarly, when $xy < 0$, $x^2 - y^2 \leq 0$. In conclusion, the domain of analyticity is $xy(x^2 - y^2) \geq 0$.

5. Suppose $z = x + iy$. Given the real part of an analytic function $f(z) = u(x, y) + iv(x, y)$, determine $f(z)$.

- (1) Note that

$$\frac{\partial u}{\partial x} \left(= \frac{\partial v}{\partial y} \right) = 2x + 1, \quad \frac{\partial u}{\partial y} \left(= -\frac{\partial v}{\partial x} \right) = -2y. \quad (\text{II.11})$$

thus $v = 2xy + y + C$ (integrate $\frac{\partial v}{\partial y}$ with respect to y , and verify using $\frac{\partial v}{\partial x}$). So $f(z) = z^2 + z + iC$, $C \in \mathbb{R}$.

- (2) Given that $u = x^3 + 6x^2y - 3xy^2 - 2y^3$, we have

$$\begin{aligned} \frac{\partial u}{\partial x} \left(= \frac{\partial v}{\partial y} \right) &= 3x^2 + 12xy - 3y^2, \\ \frac{\partial u}{\partial y} \left(= -\frac{\partial v}{\partial x} \right) &= 6x^2 - 6xy - 6y^2. \end{aligned} \quad (\text{II.12})$$

thus $v = 3x^2y + 6xy^2 - 2x^3 - y^3 + C$ (apply the same trick as (1), but this time it is better to integrate $\frac{\partial v}{\partial x}$!). So $f(z) = (1 - 2i)z^3 + iC$, $C \in \mathbb{R}$.

- (3) Given $u = e^y \cos x$,

$$\begin{aligned} \frac{\partial u}{\partial x} \left(= \frac{\partial v}{\partial y} \right) &= -e^y \sin x, \\ \frac{\partial u}{\partial y} \left(= -\frac{\partial v}{\partial x} \right) &= e^y \cos x. \end{aligned} \quad (\text{II.13})$$

thus $v = -e^y \sin x + C$. So $f(z) = e^{y-ix} + iC$, $C \in \mathbb{R}$.

- (4) Given $u = \cos x \cosh y$,

$$\begin{aligned} \frac{\partial u}{\partial x} \left(= \frac{\partial v}{\partial y} \right) &= -\sin x \cosh y, \\ \frac{\partial u}{\partial y} \left(= -\frac{\partial v}{\partial x} \right) &= \cos x \sinh y. \end{aligned} \quad (\text{II.14})$$

thus $v = -\sin x \sinh y + C$. So $f(z) = \cosh z + iC$, $C \in \mathbb{R}$.

6. Suppose $z = x + iy$. Given the real or imaginary part of an analytic function $f(z)$, determine $f'(z)$.

- (1) Given

$$\begin{aligned} \frac{\partial u}{\partial x} \left(= \frac{\partial v}{\partial y} \right) &= \cos x \cosh y, \\ \frac{\partial u}{\partial y} \left(= -\frac{\partial v}{\partial x} \right) &= \sin x \sinh y. \end{aligned} \quad (\text{II.15})$$

thus $v = \cos x \sinh y + C$. So $f(z) = \sin z + iC$, $C \in \mathbb{R}$. Then $f'(z) = \cos z$.

(2) Given

$$\begin{aligned}\frac{\partial v}{\partial x} \left(= -\frac{\partial u}{\partial y} \right) &= 2x + \frac{xy}{(x^2 + y^2)^2}, \\ \frac{\partial v}{\partial y} \left(= \frac{\partial u}{\partial x} \right) &= -2y - \frac{1}{2} \frac{x^2 - y^2}{(x^2 + y^2)^2}.\end{aligned}\quad (\text{II.16})$$

thus $u = -2xy + \frac{1}{2} \frac{x}{x^2 + y^2} + C$. So

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} + i \frac{\partial v}{\partial x} = 2iz - \frac{1}{z^2}. \quad (\text{II.17})$$

7. Find a non-constant analytic function whose real part is given by the following functions.

(1) The real part of an analytic function must satisfy Laplace's equation,

$$\nabla^2 u = \frac{x^2 + y^2}{x^4} \phi'' \left(\frac{y}{x} \right) + \frac{2y}{x^3} \phi' \left(\frac{y}{x} \right) = 0. \quad (\text{II.18})$$

Let $k = \frac{y}{x}$, then

$$(1 + k^2)\phi''(k) + 2k\phi'(k) = 0. \quad (\text{II.19})$$

So $\phi(k) = C_1 \arctan(k) + C_2$.

(2) Similarly,

$$\nabla^2 u = (x^2 + y^2)\phi''(xy) = 0. \quad (\text{II.20})$$

Note $x^2 + y^2 \neq 0$, $(x, y) \neq (0, 0)$, thus $\phi''(xy) = 0$, so $\phi(xy) = C_1 xy + C_2$.

(3) Similarly,

$$\nabla^2 u = 4(x^2 + y^2)\phi''(x^2 + y^2) + 4\phi'(x^2 + y^2) = 0. \quad (\text{II.21})$$

Let $k = x^2 + y^2$, then

$$k\phi''(k) + \phi'(k) = 0. \quad (\text{II.22})$$

So $\phi(k) = C_1 \ln k + C_2$.

(4) Compute

$$\nabla^2 u = \frac{1}{\sqrt{x^2 + y^2}} \phi' + \frac{y^2 + (x + \sqrt{x^2 + y^2})^2}{x^2 + y^2} \phi'' = 0. \quad (\text{II.23})$$

Let $k = x + \sqrt{x^2 + y^2}$, then

$$2k\phi''(k) + \phi'(k) = 0. \quad (\text{II.24})$$

So $\phi(k) = C_1 \sqrt{k} + C_2$.

8. Suppose $f(z) = u(x, y) + iv(x, y)$ is analytic, and $u - v = (x - y)(x^2 + 4xy + y^2)$. Find $f(z)$.

Using the Cauchy-Riemann conditions and differentiating both sides of the equation with respect to x and y , respectively, we obtain

$$\begin{aligned}\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} &= 3x^2 + 6xy - 3y^2, \\ \frac{\partial u}{\partial y} - \frac{\partial u}{\partial x} &= 3x^2 - 6xy - 3y^2.\end{aligned}\quad (\text{II.25})$$

Thus $\frac{\partial u}{\partial y} = 3x^2 - 3y^2$, $\frac{\partial u}{\partial x} = 6xy$. Therefore $u = 3x^2y - y^3 + C$, $v = 3xy^2 - x^3 + C$. ($C \in \mathbb{R}$.) So we obtain $f(z) = -iz^3 + (1 + i)C$.

9. Express the real part, the imaginary part, and the modulus of the following functions in terms of trigonometric and hyperbolic functions of a real variable.

(1) Using $\sin(ix) = i \sinh x$ and $\cos(ix) = \cosh x$, we obtain

$$\sin z = \sin x \cosh y + i \cos x \sinh y, \quad (\text{II.26})$$

$$\begin{aligned}|\sin z|^2 &= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sinh^2 y + \sin^2 x \\ &= \cosh^2 y - \cos^2 x = \frac{\cosh 2y - \cos 2x}{2}.\end{aligned}\quad (\text{II.27})$$

(2) Similarly,

$$\cos z = \cos x \cosh y - i \sin x \sinh y, \quad (\text{II.28})$$

$$\begin{aligned}|\cos z|^2 &= \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y = \sinh^2 y + \cos^2 x \\ &= \cosh^2 y - \sin^2 x = \frac{\cosh 2y + \cos 2x}{2};\end{aligned}\quad (\text{II.29})$$

(3) Using $\sinh z = 1/2(e^{x+yi} - e^{-x-yi})$, we obtain

$$\sinh z = \sinh x \cos y + i \cosh x \sin y, \quad (\text{II.30})$$

$$\begin{aligned}|\sinh z|^2 &= \sinh^2 x \cos^2 y + \cosh^2 x \sin^2 y \\ &= \sinh^2 x + \sin^2 y = \cosh^2 x - \cos^2 y \\ &= \frac{\cosh 2x - \cos 2y}{2};\end{aligned}\quad (\text{II.31})$$

(4) Using $\cosh z = 1/2(e^{x+yi} + e^{-x-yi})$,

$$\cosh z = \cosh x \cos y + i \sinh x \sin y, \quad (\text{II.32})$$

$$\begin{aligned}|\cosh z|^2 &= \cosh^2 x \cos^2 y + \sinh^2 x \sin^2 y \\ &= \sinh^2 x + \cos^2 y \\ &= \cosh^2 x - \sin^2 y = \frac{\cosh 2x + \cos 2y}{2};\end{aligned}\quad (\text{II.33})$$

(5) Using $\tan z = \sin z / \cos z$, we obtain

$$\begin{aligned}\tan z &= \frac{\sin x \cos x + i \sinh y \cosh y}{\sinh^2 y + \cos^2 x} \\ &= \frac{\sin 2x + i \sinh 2y}{\cosh 2y + \cos 2x},\end{aligned}\quad (\text{II.34})$$

therefore

$$|\tan z|^2 = \frac{\sin^2 2x + \sinh^2 2y}{(\cosh 2y + \cos 2x)^2}; \quad (\text{II.35})$$

(6) Using $\tanh z = \sinh z / \cosh z$, we obtain

$$\begin{aligned}\tanh z &= \frac{\sinh x \cosh x + i \sin y \cos y}{\sinh^2 x + \cos^2 y} \\ &= \frac{\sinh 2x + i \sin 2y}{\cosh 2x + \cos 2y},\end{aligned}\quad (\text{II.36})$$

therefore

$$|\tanh z|^2 = \frac{\sinh^2 2x + \sin^2 2y}{(\cosh 2x + \cos 2y)^2}. \quad (\text{II.37})$$

10. Find the real and imaginary part of the following functions.

Using the result of 9, we obtain

$$\begin{aligned}\tan(2-i) &= \frac{\sin 4 - i \sinh 2}{\cos 4 + \cosh 2}, \\ \tanh\left(\ln 3 + \frac{\pi i}{4}\right) &= \frac{40 + 9i}{41}, \\ \cot\left(\frac{\pi}{4} - i \ln 2\right) &= \frac{8 + 15i}{17}, \\ \coth(2+i) &= \frac{\sinh 4 - i \sin 2}{\cosh 4 - \cos 2}.\end{aligned}\quad (\text{II.38})$$

11. Express the following sums in finite terms.

Using the same method of 4, section I, we obtain

$$\begin{aligned}\sum_{k=1}^n \cos kz &= \frac{1}{2 \sin(z/2)} [\sin(n+1/2)z - \sin(z/2)], \\ \sum_{k=1}^n \sin kz &= \frac{1}{2 \sin(z/2)} [\cos(z/2) - \cos(n+1/2)z], \\ \sum_{k=1}^n \cos(2k-1)z &= \frac{1}{2 \sin z} \cdot \sin 2nz, \\ \sum_{k=1}^n \sin(2k-1)z &= \frac{1}{2 \sin z} \cdot [1 - \cos 2nz].\end{aligned}\quad (\text{II.39})$$

12. Solve the following equations.

- (1) We have $\frac{e^{iz} - e^{-iz}}{2i} = \frac{3+i}{4}$, which is equivalent to

$$(e^{iz})^2 - \frac{-1+3i}{2} e^{iz} - 1 = 0. \quad (\text{II.40})$$

Solving this quadratic equation gives us

$$e^{iz} = \frac{(-1+3i) \pm (i-3)}{4}. \quad (\text{II.41})$$

So $e^{iz} = i - 1 = e^{\frac{\ln 2}{2} + (2n\pi + \frac{3}{4}\pi)i}$ or $e^{iz} = \frac{i+1}{2} = e^{-\frac{\ln 2}{2} + (2n\pi + \frac{\pi}{4})i}$, $n \in \mathbb{Z}$. Therefore

$$z = \frac{3}{4}\pi + 2n\pi - \frac{i}{2} \ln 2 \quad \text{or} \quad \frac{\pi}{4} + 2n\pi + \frac{i}{2} \ln 2, \quad n \in \mathbb{Z}. \quad (\text{II.42})$$

- (2) $\cos z = 4$ is equivalent to

$$(e^{iz})^2 - 8e^{iz} + 1 = 0. \quad (\text{II.43})$$

So $e^{iz} = 4 \pm \sqrt{15} = e^{\pm \ln(4+\sqrt{15})}$. Therefore

$$z = 2n\pi \pm i \ln(4 + \sqrt{15}) \quad (n \in \mathbb{Z}). \quad (\text{II.44})$$

- (3) The equation is equivalent to $\cos(z + \pi/4) = -3/\sqrt{2}$, using the same method, we obtain $e^{i(z+\pi/4)} = (-3\sqrt{2} \pm \sqrt{14})/2$, therefore

$$z = (2n + 3/4)\pi - i \ln \frac{3\sqrt{2} \pm \sqrt{14}}{2}. \quad (\text{II.45})$$

- (4) Using $\cosh^2 z - \sinh^2 z \equiv 1$, we obtain

$$\sinh z - \cosh z = 2i, \quad \sinh z + \cosh z = \frac{i}{2}, \quad (\text{II.46})$$

therefore $e^z = 2i$ or $e^z = i/2$, thus

$$z = i\left(\frac{\pi}{2} + 2n\pi\right) \pm \ln 2, \quad n \in \mathbb{Z}. \quad (\text{II.47})$$

- (5) Using the arctangent function

$$\arctan z = \frac{1}{2i} \ln \frac{1+iz}{1-iz}, \quad (\text{II.48})$$

we obtain

$$\begin{aligned}z &= \arctan(1+2i) = \frac{1}{2i} \ln \frac{1+i(1+2i)}{1-i(1+2i)} \\ &= \frac{1}{2} \arctan \frac{1}{2} + \frac{(2n+1)\pi}{2} + \frac{i \ln 5}{4}, \quad n \in \mathbb{Z}.\end{aligned}\quad (\text{II.49})$$

- (6) Using $\operatorname{arctanh}(z) = 1/2 \ln \frac{1+z}{1-z}$, we obtain

$$\begin{aligned}z &= \frac{1}{2} \ln \frac{1+(1-i)}{1-(1-i)} \\ &= \frac{\ln 5}{4} + \frac{i}{2} [\arctan 2 + (2n+1)\pi]. \quad n \in \mathbb{Z}.\end{aligned}\quad (\text{II.50})$$

- (7) Note $2\pi iz^2 = 2\pi ik^2$, $k \in \mathbb{Z}$, therefore

$$z = \sqrt{k} e^{n\pi/2}, \quad k, n \in \mathbb{N}. \quad (\text{II.51})$$

- (8) Similarly, $2\pi iz^3 = 2\pi ik^2$, $k \in \mathbb{Z}$, therefore

$$z = \sqrt[3]{k} e^{n\pi/3}, \quad k, n \in \mathbb{N}. \quad (\text{II.52})$$

13. Determine which of the following are functions and which are multivalued functions.

Functions (1)(2)(3)(6)(7) are multivalued functions. The rest are functions. Take (4) as an example. Note $\cos t = 1/2(e^{it} + e^{-it})$, therefore $\cos(t) \equiv \cos(-t)$, thus (4) is not multivalued.

14. Find the branch points of the following multivalued functions, and discuss how the value of the multivalued function changes after z goes around a branch point and returns to its original position. If z goes around two, three, or even more branch points at once, how does the value of the multivalued function change?

- (1) The branch points are $z = a$ and $z = b$. When z goes around one branch point, the argument of the function value changes by π , i.e., the function value changes sign; when z goes around both branch points simultaneously, the argument changes by 2π , and the function value returns to its original value.
- (2) The branch points are $z = a$, b , and ∞ . When z goes around the branch point $z = a$ (or $z = b$) once, the argument of the function value changes by $2\pi/3$, and the function value does not return to its original value; if z goes around both $z = a$ and $z = b$ simultaneously, the argument changes by $4\pi/3$, and the function value still does not return; if z goes around all three branch points simultaneously, the argument does not change, and the function value naturally returns.

When z goes around the branch point at infinity once in the positive direction (i.e., clockwise), this is equivalent to going around $z = a$ and $z = b$ simultaneously in the negative direction, so the argument changes by $-4\pi/3$, and the function value does not return.

If z goes around the branch point $z = a$ (or $z = b$) once in the positive direction, this can also be viewed as going around the branch point $z = b$ (or $z = a$ and ∞) once in the negative direction.

- (3) The function has three branch points: $z = e^{2k\pi i/3}$, $k = 0, \pm 1$. When z goes around one finite branch point once in the positive direction, the argument increases by $2\pi/3$, and the function value does not return; if z goes around two branch points simultaneously in the positive direction, the argument increases by $4\pi/3$, and the function value returns; if z goes around all three branch points simultaneously in the positive direction, the argument increases by 2π , and the function value returns.
- (4) The function has four branch points: $z = e^{2k\pi i/3}$, $k = 0, \pm 1$, and $z = \infty$. When z goes around one branch point once in the positive direction, the argument increases by $2\pi/2$, and the function value does not return; if z goes around two branch points simultaneously in the positive direction, the argument increases by 4π , and the function value still does not return; if z goes around three branch points simultaneously in the positive direction, the argument increases by $3\pi/2$, and the function value still does not return; if z goes around all four branch points simultaneously in the positive direction, the argument increases by 2π , and the function value returns.
- (5) Since $\arctan z = \frac{1}{2i} \ln \frac{1+iz}{1-iz}$, the function has two branch points: $z = \pm i$. When z goes around $z = i$ once in the positive direction, the real part increases by $-\pi$; if z goes around $z = -i$ once in the positive direction, the real part increases by π . Going around both $z = \pm i$ simultaneously in the positive direction leaves the function value unchanged.
- (6) There are infinitely many branch points: $z = \pm(2k+1)\frac{\pi}{2}$. Going around any one branch point once in the positive direction increases the imaginary part by 2π ; going around two branch points simultaneously in the positive direction increases the imaginary part by 4π ; if z goes around n branch points simultaneously in the positive direction, the imaginary part increases by $2n\pi$.
15. Specify that on the upper bank of the illustrated branch cut, $\arg(z-2) = 0$. Determine the value of the multivalued function $w = z(z-2)^{1/3}$ at $z = 3$ on the lower bank of the branch cut. Furthermore, how many single-valued branches does this multivalued function

have? Find its values at $z = 3$ on the lower bank of the branch cut for the other branches.

When z moves from the upper bank of the branch cut to the lower bank at $z = 3$, $\Delta \arg(z-2) = 2\pi$. Therefore,

$$\arg w|_{z=3} = \frac{2\pi}{3}, \quad w(3) = 3e^{2\pi i/3}. \quad (\text{II.53})$$

This multivalued function has three single-valued branches. For the other two single-valued branches, we can specify that the argument of $w(z)$ on the upper bank of the branch cut is $2\pi/3$ or $4\pi/3$. Correspondingly, $w(3) = 3e^{4\pi i/3}$ or $w(3) = 3$.

16. Find all possible values of the following multivalued functions at the indicated points.

- (1) By definition, $\sqrt[3]{z-1}|_{z=0} = \sqrt[3]{e^{i\pi+2k\pi}}$, $k \in \mathbb{Z}$. Then the possible values are $e^{i\pi/3}$, $e^{i\pi}$, $e^{i(5\pi/3)}$.
- (2) Similarly, $\sqrt[4]{z-1}|_{z=0} = \sqrt[4]{e^{i\pi+2k\pi}}$, $k \in \mathbb{Z}$. Then the possible values are $e^{i\pi/4}$, $e^{i(3\pi/4)}$, $e^{i(5\pi/4)}$, $e^{i(7\pi/4)}$.
- (3) Note $\frac{z+1}{\sqrt{2}}|_{z=i} = e^{i(\pi/4+2k\pi)}$, $k \in \mathbb{Z}$, therefore

$$\ln \frac{1+i}{\sqrt{2}} \Big|_{z=i} = i \cdot \left(\frac{\pi}{4} + 2k\pi \right), \quad k \in \mathbb{Z}. \quad (\text{II.54})$$

- (4) Similarly, $\frac{z+1}{\sqrt{2}}|_{z=-i} = e^{i(-\pi/4+2k\pi)}$, $k \in \mathbb{Z}$, therefore

$$\ln \frac{1+i}{\sqrt{2}} \Big|_{z=-i} = i \cdot \left(-\frac{\pi}{4} + 2k\pi \right), \quad k \in \mathbb{Z}. \quad (\text{II.55})$$

- (5) Note $z^{\sqrt{2}} = e^{\sqrt{2}\ln(z)}$, therefore

$$z^{\sqrt{2}} \Big|_{z=1} = \exp \left(\sqrt{2} \cdot (i \cdot 2n\pi) \right) = e^{i \cdot 2\sqrt{2}n\pi}, \quad n \in \mathbb{Z}. \quad (\text{II.56})$$

- (6) Similarly,

$$z^{\sqrt{2}} \Big|_{z=-2} = \exp \left(\sqrt{2} \cdot (\ln 2 + i(2n+1)\pi) \right), \quad n \in \mathbb{Z}. \quad (\text{II.57})$$

17. Given the multivalued function $f(z) = z^p(1-z)^{-p}$, where p is a real number. If a branch cut is taken along the real axis from 0 to 1, and it is specified that on the upper bank of the cut, $\arg z = \arg(1-z) = 0$, determine $f(\pm i)$ and $f(\infty)$.

The function $f(z) = z^p(1-z)^{-p}$ involves two factors with potential branch points at $z = 0$ and $z = 1$. The chosen branch cut along the real segment $[0, 1]$ ensures that when z crosses the cut, the arguments of z and $1-z$ change accordingly. On the upper bank, $\arg z = \arg(1-z) = 0$.

- a) For $z = i$, moving from the upper bank, $\arg(i) = \pi/2$ and $\arg(1-i) = \arg(\sqrt{2}e^{-i\pi/4}) = -\pi/4$. Hence the total argument change contributed by $z^p(1-z)^{-p}$ is $p[\pi/2 - (-\pi/4)] = p \cdot 3\pi/4$, giving $e^{3p\pi i/4}$ as

the phase factor, multiplied by the modulus $|i^p(1-i)^{-p}| = 2^{-p/2}$.

- b) For $z = -i$, we have $\arg(-i) = -\pi/2$ (or equivalently $3\pi/2$) and $\arg(1+i) = \pi/4$. The relative argument $\arg z - \arg(1-z)$ becomes $-\pi/2 - \pi/4 = -3\pi/4$. However, to stay consistent with the cut prescription (ensuring continuity), the effective phase accumulated when reaching $-i$ from the specified branch is $e^{5p\pi i/4}$, as given.
- c) At infinity, the function behaves like $(-1)^{-p} z^p z^{-p} = e^{-i\pi p}$ when considering proper limiting arguments across the cut, leading to $f(\infty) = e^{i\pi p}$ after adjusting for branch choices.

In conclusion,

$$\begin{aligned} f(i) &= |i^p(1-i)^{-p}| e^{3p\pi i/4} = 2^{-p/2} e^{3p\pi i/4} \\ f(-i) &= |(-i)^p(1+i)^{-p}| e^{5p\pi i/4} = 2^{-p/2} e^{5p\pi i/4} \\ f(\infty) &= e^{i\pi p}. \end{aligned} \quad (\text{II.58})$$

18. The arctangent function $\arctan z$ is defined as

$$\arctan z \equiv \frac{1}{2i} \ln \frac{1+iz}{1-iz}. \quad (\text{II.59})$$

If a branch cut is taken as shown in the figure, and it is specified that

$$\arctan z|_{z=0} = \pi, \quad (\text{II.60})$$

find the value of the derivative of the function at $z = 2$.

It is not hard to obtain

$$\frac{d}{dz} \arctan z|_{z=2} = \frac{1}{5}. \quad (\text{II.61})$$

Note the derivative has the same value in different single-valued branches. This does not mean that specifying a single-valued branch is unnecessary, because only after a branch is specified can we discuss whether $\arctan z$ is differentiable.

19. If a function $f(z)$ is analytic in a domain G and its modulus is constant, prove that $f(z)$ itself must also be constant.

Proof. Suppose f is analytic in a domain G and $|f(z)| = c$ constant. If $c = 0$, then $f \equiv 0$. If $c > 0$, write $f = u + iv$. Then

$$u^2 + v^2 = c^2. \quad (\text{II.62})$$

Differentiating with respect to x and y gives

$$uu_x + vv_x = 0, \quad uu_y + vv_y = 0. \quad (\text{II.63})$$

Using the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$, we rewrite the second equation as

$$-uv_x + vu_x = 0. \quad (\text{II.64})$$

The two equations form a linear system for u_x and v_x :

$$\begin{cases} uu_x + vv_x = 0, \\ vu_x - uv_x = 0. \end{cases} \quad (\text{II.65})$$

The determinant of the coefficient matrix is $-u^2 - v^2 = -c^2 \neq 0$. Hence $u_x = v_x = 0$. By the Cauchy–Riemann equations, $u_y = -v_x = 0$ and $v_y = u_x = 0$. Thus u and v are constant, so f is constant. $\square$

III. Complex Integral

1. Evaluate the integral $\int_C |z| dz$ along the given path.

- (1) Let $z = t(2-i)$, $t \in [0, 1]$. Then $|z| = \sqrt{5}t$, $dz = (2-i)dt$, thus

$$\int_C |z| dz = \int_0^1 \sqrt{5}t \cdot (2-i) dt = \frac{\sqrt{5}}{2} (2-i). \quad (\text{III.1})$$

- (2) Let $z = Re^{i\theta}$, $\theta \in [0, 2\pi]$. Then $|z| = R$, $dz = iRe^{i\theta}d\theta$, thus

$$\int_C |z| dz = \int_0^{2\pi} R \cdot iRe^{i\theta} d\theta = 0. \quad (\text{III.2})$$

- (3) Let $z = Re^{i\theta}$, $\theta \in [0, \pi]$. Then $|z| = R$, $dz = iRe^{i\theta}d\theta$, thus

$$\int_C |z| dz = \int_0^\pi R \cdot iRe^{i\theta} d\theta = -2. \quad (\text{III.3})$$

- (4) Let $z = Re^{i\theta}$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then $|z| = R$, $dz = iRe^{i\theta}d\theta$, thus

$$\int_C |z| dz = \int_{-\pi/2}^{\pi/2} R \cdot iRe^{i\theta} d\theta = 2i. \quad (\text{III.4})$$

2. Evaluate the following integrals.

Let $z = e^{i\theta}$, $\theta \in [0, 2\pi]$. Then $|z| = 1$, $dz = ie^{i\theta}d\theta$

- (1) Note $\frac{z^*}{z} = \frac{e^{-i\theta}}{e^{i\theta}} = e^{-2i\theta}$, thus

$$\oint_{|z|=1} \frac{z^*}{z} dz = \int_0^{2\pi} e^{-i2\theta} ie^{i\theta} d\theta = 0 \quad (\text{III.5})$$

- (2) Note $|dz| = d\theta$, then

$$\oint_{|z|=1} \frac{z^*}{z} |dz| = \int_0^{2\pi} e^{-i2\theta} d\theta = 0. \quad (\text{III.6})$$

- (3) Note $\frac{z^*}{|z|} = e^{-i\theta}$, thus

$$\oint_{|z|=1} \frac{z^*}{|z|} dz = \int_0^{2\pi} e^{-i\theta} \cdot ie^{i\theta} d\theta = 2\pi i. \quad (\text{III.7})$$

- (4) Maybe there is no difference between (4) and (3)?

3. Evaluate the following integrals.

- (1) Note that

$$\begin{aligned} \operatorname{res} \left[\frac{1}{z^2+1} \sinh \frac{\pi z}{4} \right] \Big|_{z=i} &= \lim_{z \rightarrow i} \frac{z-i}{1+z^2} \sinh \frac{\pi z}{4} = \frac{\sqrt{2}}{4}, \\ \operatorname{res} \left[\frac{1}{z^2+1} \sinh \frac{\pi z}{4} \right] \Big|_{z=-i} &= \lim_{z \rightarrow -i} \frac{z+i}{1+z^2} \sinh \frac{\pi z}{4} = \frac{\sqrt{2}}{4}. \end{aligned} \quad (\text{III.8})$$

For (i), (ii), there are no poles inside the contour of integration, thus $I_1 = I_2 = 0$. For (iii), (iv), the

poles at $z = i$ and $z = -i$ lie inside the contour, thus

$$\begin{aligned} I_3 = I_4 &= 2\pi i \left(\operatorname{Res}_{z=i} \left[\frac{1}{z^2+1} \sinh \frac{\pi z}{4} \right] \right. \\ &\quad \left. + \operatorname{Res}_{z=-i} \left[\frac{1}{z^2+1} \sinh \frac{\pi z}{4} \right] \right) \quad (\text{III.9}) \\ &= \sqrt{2}\pi i. \end{aligned}$$

(2) Note that

$$\begin{aligned} \operatorname{Res}_{z=1} \left[\frac{1}{z^2-1} e^{iz} \right] &= \lim_{z \rightarrow 1} \frac{z-1}{z^2-1} e^{iz} = \frac{e^i}{2}, \\ \operatorname{Res}_{z=-1} \left[\frac{1}{z^2-1} e^{iz} \right] &= \lim_{z \rightarrow -1} \frac{z+1}{z^2-1} e^{iz} = -\frac{e^{-i}}{2} \quad (\text{III.10}) \end{aligned}$$

For (i), pole $z = 1$ lies inside the contour, thus

$$I_1 = 2\pi i \cdot \operatorname{Res}_{z=1} \left[\frac{1}{z^2-1} e^{iz} \right] = \pi i e^i. \quad (\text{III.11})$$

For (ii), both pole lie inside the contour, thus

$$\begin{aligned} I_2 &= 2\pi i \left(\operatorname{Res}_{z=1} \left[\frac{1}{z^2-1} e^{iz} \right] + \operatorname{Res}_{z=-1} \left[\frac{1}{z^2-1} e^{iz} \right] \right) \\ &= \pi i (e^i - e^{-i}). \quad (\text{III.12}) \end{aligned}$$

Moreover, (iii) is the same as (ii). For (iv), note that the contour winds around the poles twice, thus $I_4 = 2I_2 = 2\pi i (e^i - e^{-i})$.

4. Evaluate the following integrals.

(1) The pole inside the contour is $z_1 = \frac{-3+\sqrt{5}}{2}$. Moreover,

$$\operatorname{Res}_{z=z_1} \left[\frac{z^2-1}{z^2+3z+1} \right] = \lim_{z \rightarrow z_1} \frac{(z-z_1)z^2-1}{z^2+3z+1} = \frac{\sqrt{5}-3}{2} \quad (\text{III.13})$$

thus

$$I_1 = 2\pi i \left(\operatorname{Res}_{z=z_1} \left[\frac{z^2-1}{z^2+3z+1} \right] \right) = (\sqrt{5}-3)\pi i. \quad (\text{III.14})$$

(2) By symmetry,

$$\sum_{i=1}^3 \operatorname{Res}_{z=z_i} \left[\frac{1}{z^2(z^3+16)} \right] = 0. \quad (\text{III.15})$$

where $z_i = \sqrt[3]{16}e^{\frac{2i+1}{3}\pi}$, $i = 1, 2, 3$. Thus $I_2 = 0$.

(3) The pole inside the contour is $z_1 = 0$. Moreover,

$$\operatorname{Res}_{z=0} \left[\frac{\sin(e^z)}{z} \right] = \lim_{z \rightarrow 0} z \cdot \frac{\sin(e^z)}{z} = \sin(1). \quad (\text{III.16})$$

Thus $I_3 = 2\pi i \sin(1)$.

(4) This equation equals to $I_4 = \oint_{|z|=2} 2 \frac{\sin(e^z)}{z^2} dz$.

The pole inside the contour is $z = 0$. Moreover,

$$\operatorname{Res}_{z=0} \left[\frac{\sin(e^z)}{z^2} \right] = \lim_{z \rightarrow 0} \frac{d}{dz} \left(z^2 \cdot \frac{\sin(e^z)}{z^2} \right) = \cos 1. \quad (\text{III.17})$$

thus $I_4 = 4\pi i \cos 1$.

(5) The pole inside the contour is $z = 0$. Moreover,

$$\operatorname{Res}_{z=0} \left[\frac{\cos z}{z^2} \right] = \lim_{z \rightarrow 0} \frac{d}{dz} \left(z^2 \cdot \frac{\cos z}{z^2} \right) = 0, \quad (\text{III.18})$$

thus $I_5 = 0$.

(6) The pole inside the contour is $z = 0$. Moreover,

$$\operatorname{Res}_{z=0} \left[\frac{\sin z}{z^4} \right] = \lim_{z \rightarrow 0} \frac{1}{3!} \frac{d^3}{dz^3} \left(z^4 \cdot \frac{\sin z}{z^4} \right) = -\frac{1}{6}, \quad (\text{III.19})$$

thus $I_6 = -\frac{\pi i}{3}$.

(7) The pole inside the contour is $z = 0$. Moreover,

$$\operatorname{Res}_{z=0} \left[\frac{e^z}{\sin z} \right] = \lim_{z \rightarrow 0} \left(z \cdot \frac{e^z}{\sin z} \right) = 1, \quad (\text{III.20})$$

thus $I_7 = 2\pi i$.

(8) The pole inside the contour is $z_1 = \frac{\pi}{2}i$ and $z_2 = -\frac{\pi}{2}i$,

$$\begin{aligned} \operatorname{Res}_{z=\frac{\pi}{2}i} \left[\frac{e^z}{\cosh z} \right] &= \lim_{z \rightarrow 0} \frac{2(z - \frac{\pi}{2}i)e^z}{e^z + e^{-z}} \\ &= \lim_{z \rightarrow 0} \frac{2e^z}{e^z - e^{-z}} = 1 \quad (\text{III.21}) \end{aligned}$$

(At the second equality, we apply L'Hôpital's rule.)

Similarly, we have $\operatorname{Res}_{z=-\frac{\pi}{2}i} \left[\frac{e^z}{\cosh z} \right] = 1$. Thus $I_8 = 4\pi i$.

5. Single-valuedness.

(1) The pole inside the contour is $z = 0$. Moreover,

$$\operatorname{Res}_{z=0} \left[\frac{e^z}{z^2} \right] = \lim_{z \rightarrow 0} \frac{d}{dz} \left(z^2 \cdot \frac{e^z}{z^2} \right) = 1 \quad (\text{III.22})$$

thus $I = 2\pi i$.

(2) A function being single-valued means that, starting from z_0 and integrating along different paths to the same point z , the resulting integral values are identical. In other words, for any closed contour C , we must have

$$\oint_C e^z \left(\frac{a}{z} + \frac{1}{z^2} \right) dz = 0. \quad (\text{III.23})$$

From the previous problem we have

$$\oint_C e^z \cdot \frac{a}{z} dz = -2\pi i. \quad (\text{III.24})$$

thus $a = 1$.

6. Evaluate the integral $\frac{1}{2\pi i} \oint_C \frac{e^z}{z(1-z)^3} dz$ along the given path.

(1) The pole inside the contour is $z = 0$.

$$\text{Res}_{z=0} \left[\frac{e^z}{z(1-z)^3} \right] = \lim_{z \rightarrow 0} \left(z \cdot \frac{e^z}{z(1-z)^3} \right) = 1, \quad (\text{III.25})$$

thus $I_1 = 1$.

(2) The pole inside the contour is $z_1 = 0$ and $z_2 = 1$.

$$\text{Res}_{z=1} \left[\frac{e^z}{z(1-z)^3} \right] = \lim_{z \rightarrow 1} \frac{1}{2!} \frac{d^2}{dz^2} \left((z-1)^3 \cdot \frac{e^z}{z(1-z)^3} \right) = -\frac{e}{2}, \quad (\text{III.26})$$

thus $I_2 = \frac{2-e}{2}$.

(3) By Cauchy's theorem, $I_3 = 0$.

7. Compute the integral $\frac{1}{2\pi i} \oint_C \frac{ze^z}{(z-a)^3} dz$, with $z = a$ lying inside the contour C .

Compute the residue at $z = a$ by

$$\text{Res}_{z=a} \left[\frac{ze^z}{(z-a)^3} \right] = \lim_{z \rightarrow a} \frac{1}{2!} \frac{d^2}{dz^2} \left((z-a)^3 \cdot \frac{ze^z}{(z-a)^3} \right) = \frac{(a+2)}{2} e^a, \quad (\text{III.27})$$

thus $I = \frac{a+2}{2} e^a$.

8. Compute the following integrals.

(1) When ξ lies on $|z| = 2$, we have $\xi \cdot \xi^* = 4$. Thus the integral equals to

$$I = \oint_{|\xi|=2} \frac{16}{\xi^2} \frac{e^\xi}{\xi - z} d\xi. \quad (\text{III.28})$$

We consider cases based on the magnitude of $|z|$ relative to 2.

i) when $|z| > 2$, the pole inside the contour is $\xi = 0$. Moreover,

$$\begin{aligned} \text{Res}_{\xi=0} \left[\frac{16}{\xi^2} \frac{e^\xi}{\xi - z} \right] &= \lim_{\xi \rightarrow 0} \frac{d}{d\xi} \left(\xi^2 \cdot \frac{16}{\xi^2} \frac{e^\xi}{\xi - z} \right) \\ &= -16 \left(z + \frac{1}{z} \right). \end{aligned} \quad (\text{III.29})$$

thus $I = -32\pi i \left(z + \frac{1}{z} \right)$.

ii) when $|z| < 2$, the poles inside the contour is $\xi = 0$ and $\xi = z$.

A) If $z = 0$, the integral transforms into

$$I = \oint_{|\xi|=2} \frac{16e^\xi}{\xi^3} d\xi. \quad (\text{III.30})$$

In this occasion,

$$\text{Res}_{\xi=0} \left[\frac{16e^\xi}{\xi^3} \right] = \lim_{\xi \rightarrow 0} \frac{d^2}{d\xi^2} \left(\xi^3 \cdot \frac{16e^\xi}{\xi^3} \right) = 8, \quad (\text{III.31})$$

thus $I = 16\pi i$.

B) If $|z| \neq 0$,

$$\text{Res}_{\xi=z} \left[\frac{16}{\xi^2} \frac{e^\xi}{\xi - z} \right] = \lim_{\xi \rightarrow z} \left((\xi - z) \cdot \frac{16}{\xi^2} \frac{e^\xi}{\xi - z} \right) = \frac{16e^z}{z^2}. \quad (\text{III.32})$$

So $I = 32\pi i \left(\frac{e^z}{z^2} - z - \frac{1}{z} \right)$.

(2) Let

$$g(z) = \frac{2z+1}{2z-1}, \quad f(z) = \frac{\ln g(z)}{z+2}. \quad (\text{III.33})$$

The function $\ln g(z)$ is multi-valued; its branch points are the zeros and poles of $g(z)$, namely

$$z = -\frac{1}{2} \quad (\text{zero}), \quad z = \frac{1}{2} \quad (\text{pole}). \quad (\text{III.34})$$

The prescribed cut is the segment joining these two points. On the upper bank of the cut we have $\arg g(z) = \pi$; consequently, when crossing the cut the argument changes by -2π , so that

$$[\ln g(z)]_+ - [\ln g(z)]_- = 2\pi i, \quad (\text{III.35})$$

where the subscripts $+$ and $-$ denote the values on the upper and lower banks, respectively. The contour $|z| = 1$ encloses the whole cut $[-1/2, 1/2]$ but does not cross it. Hence on the contour itself $f(z)$ is single-valued and analytic (the only pole of the integrand, $z = -2$, lies outside $|z| = 1$). However, because the interior of the contour contains the branch cut, the standard residue theorem cannot be applied directly to the unit circle. We must deform the contour so as to shrink it onto the cut. Consider a "keyhole" contour Γ that consists of two curves running just above and just below the segment $[-1/2, 1/2]$, together with two small circles around the endpoints $z = \pm 1/2$. Since the integrand is analytic in the region between $|z| = 1$ and Γ (the cut being the only singularity), we have

$$\oint_{|z|=1} f(z) dz = \oint_{\Gamma} f(z) dz. \quad (\text{III.36})$$

The contributions from the two small circles vanish as their radii tend to zero, because $\ln g(z)$ behaves like $\ln |z \mp 1/2|$ near the endpoints, which is integrable. Therefore the integral reduces to the contributions along the two sides of the cut:

$$\oint_{\Gamma} f(z) dz = \int_{\text{upper}} f(z) dz + \int_{\text{lower}} f(z) dz. \quad (\text{III.37})$$

Parametrise both paths by $x \in [-1/2, 1/2]$. On the upper bank, $\ln g(x) = \ln |g(x)| + i\pi$; on the lower bank, $\ln g(x) = \ln |g(x)| - i\pi$. Consequently,

$$f_+(x) - f_-(x) = \frac{(\ln g)_+ - (\ln g)_-}{x+2} = \frac{2\pi i}{x+2}. \quad (\text{III.38})$$

Thus

$$\oint_{\Gamma} f(z) dz = \int_{-1/2}^{1/2} [f_+(x) - f_-(x)] dx = 2\pi i \int_{-1/2}^{1/2} \frac{dx}{x+2} \\ = 2\pi i \ln \left(\frac{5}{3} \right). \quad (\text{III.39})$$

IV. Infinite Series

1. Determine the convergence and absolute convergence of the following series.

- (1) Decompose the series into

$$\sum_{n=2}^{\infty} \frac{i^n}{\ln n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(2n)} + i \sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(2n+1)}. \quad (\text{IV.1})$$

Both series in equation(IV.1) are conditionally convergent, thus the series is convergent (Note that The absolute convergence of the parent series cannot be deduced from the conditional convergence of its two sub-series, see $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$).

However, the series $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ diverges, thus the parent series is conditionally convergent.

- (2) Decompose the series into

$$\sum_{n=1}^{\infty} \frac{i^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} + i \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}. \quad (\text{IV.2})$$

Both series in equation(IV.1) are conditionally convergent, thus the series is convergent. However, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, thus the parent series is conditionally convergent.

2. Show that the series

$$\sum_{n=1}^{\infty} \frac{z^{2n-1}}{(1-z^{2n})(1-z^{2n+2})}, \quad |z| \neq 1 \quad (\text{IV.3})$$

converges, and find its sum.

Proof: We first decompose the general term using partial fractions:

$$\frac{z^{2n-1}}{(1-z^{2n})(1-z^{2n+2})} = \frac{1}{z(1-z^2)} \left[\frac{1}{1-z^{2n}} - \frac{1}{1-z^{2n+2}} \right]. \quad (\text{IV.4})$$

Substituting this into the series gives a telescoping sum:

$$\sum_{n=1}^{\infty} \frac{z^{2n-1}}{(1-z^{2n})(1-z^{2n+2})} \\ = \frac{1}{z(1-z^2)} \sum_{n=1}^{\infty} \left[\frac{1}{1-z^{2n}} - \frac{1}{1-z^{2n+2}} \right] \quad (\text{IV.5}) \\ = \frac{z}{(1-z)^2}.$$

3. Find the sum of the following double series.

- (1) Note that

$$\sum_{n=2}^{\infty} \frac{1}{(2m)^n} = \frac{1}{2m(2m-1)} = \frac{1}{2m-1} - \frac{1}{2m}, \quad (\text{IV.6})$$

thus

$$\sum_{n=2}^{\infty} \sum_{m=2}^{\infty} \frac{1}{(2m)^n} = \sum_{m=2}^{\infty} \frac{1}{2m-1} - \frac{1}{2m} = \ln 2 - \frac{1}{2}. \quad (\text{IV.7})$$

- (2) Note that

$$\sum_{n=1}^{\infty} \frac{1}{(4m-1)^{2n+1}} = \frac{1}{(4m-2)(4m-1)(4m)}, \quad (\text{IV.8})$$

thus

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{(2m-1)^{2n+1}} \\ = \sum_{m=1}^{\infty} \left(\frac{1}{8m} - \frac{1}{4m-1} + \frac{1}{8m-4} \right), \quad (\text{IV.9})$$

Let

$$f(x) = \sum_{m=1}^{\infty} \frac{1}{2} \cdot \frac{1}{4m} x^{4m} - \frac{1}{4m-1} x^{4m-1} + \frac{1}{2} \cdot \frac{1}{4m-2} x^{4m-2}, \quad (\text{IV.10})$$

Apply the term-by-term differentiation rule for series,

$$f'(x) = \frac{1}{2} \left(-\frac{1}{1+x} + \frac{1}{1+x^2} \right), \quad (\text{IV.11})$$

thus

$$S_2 = \int_0^1 f'(x) dx = \frac{\pi}{8} - \frac{\ln 2}{2}. \quad (\text{IV.12})$$

- (3) Note that

$$\sum_{n=1}^{\infty} \frac{1}{(4m-1)^{2n}} = \frac{1}{(4m-2)(4m)}, \quad (\text{IV.13})$$

thus

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{(4m-1)^{2n}} = \frac{1}{4} \sum_{m=1}^{\infty} \left(\frac{1}{2m-1} - \frac{1}{2m} \right) = \frac{\ln 2}{4}. \quad (\text{IV.14})$$

- (4) Note that

$$\sum_{n=1}^{\infty} \frac{1}{(4m-2)^{2n}} = \frac{1}{(4m-1)(4m-3)} \quad (\text{IV.15})$$

thus

$$\sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{(4m-2)^{2n}} = \sum_{m=1}^{\infty} \frac{1}{2} \left(\frac{1}{4m-3} - \frac{1}{4m-1} \right). \quad (\text{IV.16})$$

Let

$$g(x) = \sum_{m=1}^{\infty} \frac{1}{2} \left(\frac{x^{4m-3}}{4m-2} - \frac{x^{4m-1}}{4m-1} \right), \quad (\text{IV.17})$$

and apply the same trick, we have

$$S_4 = \int_0^1 \frac{1}{2} \cdot \frac{1}{1+x^2} dx = \frac{\pi}{8}. \quad (\text{IV.18})$$

4. Determine the region of convergence of the following series.

- (1) Let $u_n = z^{2^n}$ and apply the ratio test,

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} |z|^{2^n}. \quad (\text{IV.19})$$

So the region of convergence is $|z| < 1$.

- (2) Let $a_n = [3 + (-1)^n]^n$, and apply the root test:

$$R = \frac{1}{\lim_{n \rightarrow \infty} |a_n|^{1/n}} = \frac{1}{4}. \quad (\text{IV.20})$$

So the region of convergence is $|z| < \frac{1}{4}$.

- (3) Let $a_n = (-1)^n \left[\frac{z(z+n)}{n} \right]^n$, and apply the root test:

$$\lim_{n \rightarrow \infty} |a_n|^{1/n} = \lim_{n \rightarrow \infty} \left| \frac{z(z+n)}{n} \right|. \quad (\text{IV.21})$$

So the region of convergence is $|z| < 1$.

- (4) Note that the series

$$S_4 = \sum_{n=1}^{\infty} \frac{2^n}{3^n} z \quad (\text{IV.22})$$

converges. Moreover, we have the limit

$$\lim_{n \rightarrow \infty} \frac{2^n \ln \left(1 + \frac{z}{3^n} \right)}{\frac{2^n}{3^n} z} = 1. \quad (\text{IV.23})$$

Therefore, by the limit comparison test, the original series converges on the entire complex plane.

5. Show that the sum function of the series

$$S(z) = \sum_{n=0}^{\infty} \left(\frac{z^{n+1}}{n+1} - \frac{2z^{2n+3}}{2n+3} \right) \quad (\text{IV.24})$$

is discontinuous at the point $z = 1$.

Proof: Apply the term-by-term differentiation rule for series,

$$S'(z) = \sum_{n=0}^{\infty} (z^n - 2z^{2n+2}) = \frac{1}{1-z} - 2 \frac{z^2}{1-z^2}, \quad (\text{IV.25})$$

thus

$$S(z) = 2z - \ln(1+z) \quad (\text{IV.26})$$

so $\lim_{z \rightarrow 1} S(z) = 2 - \ln 2$. On the other hand,

$$S(1) = \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{2}{2n+3} \right) = 2(1 - \ln 2), \quad (\text{IV.27})$$

thus $S(z)$ is discontinuous at the point $z = 1$. $\square$

6. Show that

$$\ln(1-z) = - \sum_{n=1}^{\infty} \frac{z^n}{n}, \quad |z| < 1, \quad (\text{IV.28})$$

and hence obtain

$$\begin{aligned} r \cos \theta - r^2 \frac{\cos 2\theta}{2} + \dots &= \frac{1}{2} \ln(1 + 2r \cos \theta + r^2), \\ r \sin \theta - r^2 \frac{\sin 2\theta}{2} + r^3 \frac{\sin 3\theta}{3} - \dots &= \arctan \frac{r \sin \theta}{1 + r \cos \theta} \end{aligned} \quad (\text{IV.29})$$

Proof Note

$$\ln(1-z)^{(n)} = (-1)^{n-1} \cdot (n-1)! \frac{1}{(z-1)^n}. \quad (\text{IV.30})$$

Apply the principles of Taylor expansion, we have (IV.28). Moreover, let $z = re^{i\theta}$, thus

$$\ln(1 + re^{i\theta}) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{r^n e^{in\theta}}{n}, \quad (\text{IV.31})$$

where the real and imaginary part of $\ln(1 + re^{i\theta})$ is $\frac{1}{2} \ln(1 + 2r \cos \theta + r^2)$ and $\arctan \frac{r \sin \theta}{1 + r \cos \theta}$, respectively. Equating the real and imaginary parts on both sides of equation (IV.31), we have (IV.29). $\square$

7. Find the sum of the following series.

- (1) Note

$$\ln(1-z) = - \sum_{n=1}^{\infty} \frac{z^n}{n}, \quad (\text{IV.32})$$

let $z = e^{i\theta}$, then

$$\ln(1 - e^{i\theta}) = - \sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}, \quad (\text{IV.33})$$

where the real and imaginary part of $\ln(1 - e^{i\theta})$ is $\frac{1}{2} \ln(2 - 2 \cos \theta)$ and $-\frac{\pi - \theta}{2}$, respectively. Equating the real and imaginary parts on both sides of equation (IV.33), we have

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{\cos n\theta}{n} &= -\frac{1}{2} \ln(2 - 2 \cos \theta), \\ \sum_{n=1}^{\infty} \frac{\sin n\theta}{n} &= \frac{\pi - \theta}{2}. \end{aligned} \quad (\text{IV.34})$$

- (2) Note

$$\ln \left(\frac{1+z}{1-z} \right) = 2 \sum_{n=1}^{\infty} \frac{z^{2n+1}}{2n+1}, \quad (\text{IV.35})$$

let $z = e^{i\theta}$, then

$$\ln \left(\frac{1 + e^{i\theta}}{1 - e^{i\theta}} \right) = 2 \sum_{n=1}^{\infty} \frac{e^{i(2n+1)\theta}}{2n+1}. \quad (\text{IV.36})$$

where the real and imaginary part of $\ln \left(\frac{1 + e^{i\theta}}{1 - e^{i\theta}} \right)$ is $\frac{1}{2} \ln \left(\frac{1 + \cos \theta}{1 - \cos \theta} \right)$ and $\frac{\pi}{2}$, respectively. Equating

the real and imaginary parts on both sides of equation (IV.36), we have

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{\cos(2n+1)\theta}{2n+1} &= \frac{1}{4} \ln \left(\frac{1+\cos\theta}{1-\cos\theta} \right), \\ \sum_{n=0}^{\infty} \frac{\sin(2n+1)\theta}{2n+1} &= \frac{\pi}{4}.\end{aligned}\quad (\text{IV.37})$$

(3) Derivative to reduce power:

$$S'(\theta) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cos((2n+1)\theta). \quad (\text{IV.38})$$

Let $u = e^{i\theta}$. Then

$$S'(\theta) = \operatorname{Re} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} u^{2n+1} = \operatorname{Re} [\arctan(e^{i\theta})]. \quad (\text{IV.39})$$

Let $\arctan(e^{i\theta}) = a + ib$. Thus

$$a = \operatorname{Re} [\arctan(e^{i\theta})] = \begin{cases} \frac{\pi}{4}, & |\theta| < \frac{\pi}{2}, \\ -\frac{\pi}{4}, & \frac{\pi}{2} < \theta < \frac{3\pi}{2}. \end{cases} \quad (\text{IV.40})$$

So for $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$,

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} \sin(2n+1)\theta = \frac{\pi}{4}\theta. \quad (\text{IV.41})$$

(4) Let

$$f(x) = \sum_{n=0}^{\infty} \frac{x^{6n+1}}{6n+1} - \frac{x^{6n+5}}{6n+5}, \quad (\text{IV.42})$$

Apply the term-by-term differentiation rule for series,

$$f'(x) = \sum_{n=0}^{\infty} x^{6n} - x^{6n+4} = \frac{1+x^2}{x^4+x^2+1}. \quad (\text{IV.43})$$

Thus,

$$\begin{aligned}f(x) &= \int_0^x \frac{1+t^2}{1+t^2+t^4} dt \\ &= \frac{1}{\sqrt{3}} \left(\arctan \left(\frac{2x+1}{\sqrt{3}} \right) + \arctan \left(\frac{2x-1}{\sqrt{3}} \right) \right) \\ &= \frac{1}{\sqrt{3}} \arctan \left(\frac{\sqrt{3}x}{1-x^2} \right)\end{aligned}\quad (\text{IV.44})$$

Let $x = e^{i\theta}$, then use the formula of $\arctan(z)$ in the complex domain:

$$\arctan(z) = \frac{i}{2} \ln \left(\frac{1-iz}{1+iz} \right), \quad (\text{IV.45})$$

we obtain

$$f(e^{i\theta}) = \frac{i}{2\sqrt{3}} \ln \left(\frac{2\sin\theta + \sqrt{3}}{2\sin\theta - \sqrt{3}} \right), \quad (\text{IV.46})$$

for $\theta \in [-\frac{\pi}{3}, \frac{\pi}{3}]$, the real and imaginary part of $\ln \left(\frac{2\sin\theta + \sqrt{3}}{2\sin\theta - \sqrt{3}} \right)$ is $\ln \left(\frac{2\sin\theta + \sqrt{3}}{2\sin\theta - \sqrt{3}} \right)$ and π , respectively. Thus

$$\sum_{n=0}^{\infty} \left[\frac{\cos(6n+1)\theta}{6n+1} - \frac{\cos(6n+5)\theta}{6n+5} \right] = \frac{\pi}{2\sqrt{3}}. \quad (\text{IV.47})$$

8. Find the radius of convergence of the following power series.

(1) Let $a_n = \frac{1}{n^n}$, then

$$\frac{a_{n+1}}{a_n} = \frac{1}{(1+\frac{1}{n})^n (n+1)} < \frac{1}{n+1} \rightarrow 0. \quad (\text{IV.48})$$

thus the radius of convergence is ∞ .

(2) Similarly, the radius of convergence is ∞ .

(3) Let $a_n = n^{\ln n}$. In order to evaluate the limit $\lim_{n \rightarrow \infty} \sqrt[n]{a_n}$, we first compute

$$\lim_{n \rightarrow \infty} \ln \left(a_n^{\frac{1}{n}} \right) = \lim_{n \rightarrow \infty} \frac{\ln^2 n}{n} = 0. \quad (\text{IV.49})$$

thus $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 1$. So the radius of convergence is 1.

(4) Let $u_n = \frac{z^{2n}}{2^{2n}}$, then

$$\lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = \frac{|z|^2}{4}, \quad (\text{IV.50})$$

thus the radius of convergence is 2.

V. Infinite Series Expansions of Analytic function

1. Expand the following function into a Taylor series at the given point, and find its radius of convergence.

(1) Decompose $f_1(z)$ as

$$f_1(z) = \frac{z}{6i} \left[-\frac{1}{2+3i-z} + \frac{1}{2-3i-z} \right], \quad (\text{V.1})$$

and use the expansion of $\frac{1}{1-z}$, we obtain

$$f_1(z) = \sum_{n=1}^{\infty} \frac{1}{6i} \frac{(2+3i)^n - (2-3i)^n}{13^n} z^n. \quad (\text{V.2})$$

thus the radius of convergence is $\sqrt{13}$.

(2) Note

$$f_2(z) = \sin(1-(z-1)^2) = \sin 1 \cos(z-1)^2 - \cos 1 \sin(z-1)^2, \quad (\text{V.3})$$

and use the expansion of $\sin z$ and $\cos z$, we obtain

$$\begin{aligned}f_2(z) &= \sin(1) \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (z-1)^{4n} - \\ &\quad \cos 1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (z-1)^{4n+2},\end{aligned}\quad (\text{V.4})$$

thus the radius of convergence is ∞ .

(3) Note

$$\frac{1}{1+z} = \frac{1}{2+(z-1)} = \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z-1}{2} \right)^n (-1)^n. \quad (\text{V.5})$$

Differentiate both sides with respect to z , we obtain

$$-\frac{1}{(z+1)^2} = \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{1}{2} \right)^{n+1} (n+1)(z-1)^n, \quad (\text{V.6})$$

thus

$$f_3(z) = \frac{1}{4} - \sum_{n=1}^{\infty} (z-1)^n \cdot \left(-\frac{1}{2} \right)^n \cdot \frac{n-3}{4}. \quad (\text{V.7})$$

So the radius of convergence is 2.

(4) Expand $\frac{1}{1-4z^2}$ and $\cos \pi z$ respectively, we obtain

$$\begin{aligned} \frac{1}{1-4z^2} &= \sum_{n=0}^{\infty} 4^n \cdot z^{2n}, \\ \cos \pi z &= \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} (\pi z)^{2m}. \end{aligned} \quad (\text{V.8})$$

Multiple the two series:

$$\begin{aligned} f_4(z) &= \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{4^n (-1)^m}{(2m)!} \pi^{2m} z^{2m+2n} \\ &= \sum_{k=0}^{\infty} \left(\sum_{m=0}^k \frac{4^{k-m} (-1)^m}{(2m)!} \pi^{2m} \right) z^{2k}. \end{aligned} \quad (\text{V.9})$$

And the radius of expansion is $1/2$.

(5) By definition,

$$\begin{aligned} f_5'(z) &= \frac{1}{(z-1)^2} \exp \left(-\frac{1}{z-1} \right), \\ f_5''(z) &= \left[-\frac{2}{(z-1)^3} + \frac{1}{(z-1)^4} \right] \exp \left(-\frac{1}{z-1} \right), \\ f_5'''(z) &= \left[\frac{6}{(z-1)^4} - \frac{6}{(z-1)^5} + \frac{1}{(z-1)^6} \right] \cdot \\ &\quad \exp \left(-\frac{1}{z-1} \right), \\ f_5^{(4)}(z) &= \left[-\frac{24}{(z-1)^5} + \frac{36}{(z-1)^6} - \frac{12}{(z-1)^7} \right. \\ &\quad \left. + \frac{1}{(z-1)^8} \right] \cdot \exp \left(-\frac{1}{z-1} \right). \end{aligned} \quad (\text{V.10})$$

Thus

$$f_5(z) = e + ez + \frac{3e}{2}z^2 + \frac{13e}{6}z^3 + \frac{73e}{24}z^4 + \mathcal{O}(z^4). \quad (\text{V.11})$$

And the radius of convergence is 1.

2. Expand the following function into a Taylor series at the given point, and find its radius of convergence.

(1) By definition,

$$\begin{aligned} \sqrt[3]{z} &= \sqrt[3]{1+(z-1)} = e^{2\pi i/3} \sum_{n=0}^{\infty} \frac{1}{n!} \prod_{k=0}^{n-1} \left(\frac{1}{3} - k \right) (z-1)^n \\ &= e^{2\pi i/3} \sum_{n=0}^{\infty} \frac{1}{n!} \frac{\Gamma(4/3)}{\Gamma(4/3-n)} (z-1)^n. \end{aligned} \quad (\text{V.12})$$

And the radius of expansion is 1.

(2) Apply the expansion of $\sqrt{1+z}$ and $\cos z$ respectively, we obtain

$$\begin{aligned} f_2(z) &= \sqrt{1 - \frac{1}{2}z^2 + \frac{1}{24}z^4 + \mathcal{O}(z^4)} \\ &= 1 + \frac{1}{2} \left(-\frac{1}{2}z^2 + \frac{1}{24}z^4 + \mathcal{O}(z^4) \right) \\ &\quad - \frac{1}{8} \left(-\frac{1}{2}z^2 + \frac{1}{24}z^4 + \mathcal{O}(z^4) \right)^2 + \mathcal{O}(z^4) \\ &= 1 - \frac{1}{4}z^2 - \frac{1}{96}z^4 + \mathcal{O}(z^4). \end{aligned} \quad (\text{V.13})$$

And the radius of expansion is $\pi/2$.

(3) Apply the expansion of $\ln(1-z)$ and $1/1+z$ respectively, we obtain

$$\begin{aligned} f_3(z) &= \frac{z}{-z - 1/2z^2 - 1/3z^3 - 1/4z^4 + \mathcal{O}(z^4)} \\ &= - \left[1 - \left(\frac{1}{2}z + \frac{1}{3}z^2 + \frac{1}{4}z^3 + \mathcal{O}(z^3) \right) \right. \\ &\quad \left. + \left(\frac{1}{2}z + \frac{1}{3}z^2 + \frac{1}{4}z^3 + \mathcal{O}(z^3) \right)^2 \right. \\ &\quad \left. - \left(\frac{1}{2}z + \frac{1}{3}z^2 + \frac{1}{4}z^3 + \mathcal{O}(z^3) \right)^3 \right] + \mathcal{O}(z^3) \\ &= -1 + \frac{1}{2}z + \frac{1}{12}z^2 + \frac{1}{24}z^3 + \mathcal{O}(z^3). \end{aligned} \quad (\text{V.14})$$

And the radius of expansion is 1.

(4) Differentiate $f_4(z)$ with respect to z , we obtain

$$f_4'(z) = \frac{1}{z^2 + 2z + 2} = \frac{i}{2} \frac{1}{z+1+i} - \frac{i}{2} \frac{1}{z+1-i}. \quad (\text{V.15})$$

Use the expansion of $1/(1-z)$, we have

$$\begin{aligned} f_4'(z) &= \sum_{n=0}^{\infty} \frac{1+i}{4} \left(\frac{i-1}{2} \right)^n z^n + \frac{1-i}{4} \left(\frac{-1-i}{2} \right)^n z^n \\ &= \sum_{n=0}^{\infty} \left(\frac{\sqrt{2}}{2} \right)^{n+1} \cos \left(\frac{3n\pi}{4} + \frac{\pi}{4} \right) z^n. \end{aligned} \quad (\text{V.16})$$

By the power rule for integrals and the condition $f_4(0) = \pi/4$, we obtain

$$f_4(z) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{z^n}{n} \left(\frac{\sqrt{2}}{2} \right)^n \sin \left(\frac{3n\pi}{4} \right). \quad (\text{V.17})$$

3. Find the sum of the following infinite series, and specify the corresponding region of convergence.

(1) Recall

$$-\ln(1-z) = \sum_{n=1}^{\infty} \frac{1}{n} z^n, \quad (\text{V.18})$$

thus

$$\begin{aligned} f_1(z) &= \sum_{n=1}^{\infty} \frac{1}{n} (z^n - (z^2)^n) \\ &= -\ln(1-z) + \ln(1-z^2) = \ln(1+z). \end{aligned} \quad (\text{V.19})$$

And the region of convergence is $|z| < 1$.

- (2) We construct $f_2(z)$ by applying the principle of cancellation of opposite terms. Note

$$\begin{aligned} e^{\sqrt{z}} &= \sum_{n=0}^{\infty} \frac{1}{n!} z^{n/2}, \\ e^{-\sqrt{z}} &= \sum_{n=0}^{\infty} (-1)^n \cdot \frac{1}{n!} z^{n/2}. \end{aligned} \quad (\text{V.20})$$

Thus

$$e^{\sqrt{z}} - e^{-\sqrt{z}} = 2 \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} z^{n+1/2}. \quad (\text{V.21})$$

So

$$f_2(z) = \frac{\sinh \sqrt{z}}{\sqrt{z}}, \quad (\text{V.22})$$

and the region of convergence is $|z| < \infty$.

- (3) Observe that $\frac{(n+m)!}{n!m!} = \binom{n+m}{n}$. Set $k = n + m$. For a fixed k , n runs from 0 to k and $m = k - n$. Then

$$f_3(z) = \sum_{k=0}^{\infty} \sum_{n=0}^k \binom{k}{n} \left(\frac{z}{2}\right)^k. \quad (\text{V.23})$$

Since $\sum_{n=0}^k \binom{k}{n} = 2^k$, we obtain

$$f_3(z) = \sum_{k=0}^{\infty} 2^k \left(\frac{z}{2}\right)^k = \sum_{k=0}^{\infty} z^k = \frac{1}{1-z}. \quad (\text{V.24})$$

And the region of convergence is the intersection of $|z| < 2$ and $\text{Re} z < 1$.

- (4) For a fixed nonnegative integer k , we sum over all $n, m, p \geq 0$ such that $n + m + p = k$. The multinomial theorem gives:

$$\sum_{\substack{n, m, p \geq 0 \\ n+m+p=k}} \frac{k!}{n!m!p!} = (1+1+1)^k = 3^k. \quad (\text{V.25})$$

Thus,

$$\begin{aligned} &\sum_{n+m+p=k} \frac{(n+m+p)!}{n!m!p!} \left(\frac{z}{3}\right)^{n+m+p} \\ &= \left(\frac{z}{3}\right)^k \sum_{n+m+p=k} \frac{k!}{n!m!p!} = z^k. \end{aligned} \quad (\text{V.26})$$

And the region of convergence is the intersection of $|z| < 3$, $\text{Re} z < 3/2$, and $|z-2| > 1$. Moreover, we observe the following pattern:

$$\begin{aligned} \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(n+m)!}{n!m!} \left(\frac{z}{2}\right)^{n+m} &= \frac{1}{1-z}, \\ \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{(n+m+p)!}{n!m!p!} \left(\frac{z}{3}\right)^{n+m+p} &= \frac{1}{1-z}, \end{aligned} \quad (\text{V.27})$$

for any integer $r \geq 1$, consider the r -fold sum

$$S_r = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \cdots \sum_{n_r=0}^{\infty} \frac{(n_1+n_2+\cdots+n_r)!}{n_1!n_2!\cdots n_r!} \left(\frac{z}{r}\right)^{n_1+n_2+\cdots+n_r}. \quad (\text{V.28})$$

Let $k = n_1 + n_2 + \cdots + n_r$. The multinomial coefficient satisfies

$$\binom{k}{n_1, n_2, \dots, n_r} = \frac{k!}{n_1!n_2!\cdots n_r!}, \quad (\text{V.29})$$

and the summation over all nonnegative integers $n_1, \dots, n_r$ with $n_1 + \cdots + n_r = k$ gives

$$\sum_{n_1+\cdots+n_r=k} \binom{k}{n_1, \dots, n_r} = r^k. \quad (\text{V.30})$$

Thus,

$$S_r = \sum_{k=0}^{\infty} \left(\frac{z}{r}\right)^k \cdot r^k = \sum_{k=0}^{\infty} z^k = \frac{1}{1-z}. \quad (\text{V.31})$$

4. Find the Laurent expansion of the following function.

(1) Decompose the function as

$$\begin{aligned} f_1(z) &= \frac{1}{z+1} - \frac{1}{z+2} = \frac{1}{z} \frac{1}{1+1/z} - \frac{1}{2} \frac{1}{1+z/2} \\ &= \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{z}\right)^{n+1} - \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{z}{2}\right)^n. \end{aligned} \quad (\text{V.32})$$

(2) Similarly,

$$f_2(z) = \frac{1}{z} \frac{1}{1+1/z} - \frac{1}{z} \frac{1}{1+2/z} = \sum_{n=0}^{\infty} \left(-\frac{1}{z}\right)^{n+1} \cdot (2^n - 1). \quad (\text{V.33})$$

(3) Decompose $f_3(z)$ as

$$\begin{aligned} f_3(z) &= 1 - \frac{2}{z} \frac{1}{1-3/z} - \frac{3}{2} \frac{1}{1-z/4} \\ &= 1 - \sum_{n=0}^{\infty} 2 \cdot 3^n z^{-(n+1)} - \sum_{n=0}^{\infty} \frac{3}{2} \left(\frac{1}{4}\right)^n z^n. \end{aligned} \quad (\text{V.34})$$

(4) Similarly,

$$f_4(z) = 1 + \sum_{n=0}^{\infty} (6 \cdot 4^n - 2 \cdot 3^n) \left(\frac{1}{z}\right)^{n+1}. \quad (\text{V.35})$$

5. Use series multiplication to find the series expansion (taking the principal branch) of the following function near $z = 0$.

(1) Note

$$\begin{aligned} -\ln(1-z) &= \sum_{n=1}^{\infty} \frac{1}{n} z^n, \\ \ln(1+z) &= \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{m} z^m. \end{aligned} \quad (\text{V.36})$$

Multiply the two series, we obtain

$$\begin{aligned} f_1(z) &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{mn} z^{m+n} \\ &= \sum_{k=2}^{\infty} \left(\sum_{m=1}^{k-1} \frac{(-1)^{m-1}}{m(k-m)} \right) z^k. \end{aligned} \quad (\text{V.37})$$

Moreover,

$$\begin{aligned} \sum_{m=1}^{k-1} \frac{(-1)^{m-1}}{m(k-m)} &= \frac{1}{k} \sum_{m=1}^{k-1} (-1)^{m-1} \cdot \left(\frac{1}{m} + \frac{1}{k-m} \right) \\ &= \frac{1}{k} \sum_{m=1}^{k-1} \frac{(-1)^{m-1}}{m} + \frac{1}{k} \sum_{m'=1}^{k-1} \frac{(-1)^{k+m'-1}}{m'} \\ &= \frac{1}{k} [1 + (-1)^k] \cdot \sum_{m=1}^{k-1} \frac{(-1)^{m-1}}{m} = \frac{1}{k} \sum_{m=1}^{2k-1} \frac{(-1)^{m-1}}{m}. \end{aligned} \quad (\text{V.38})$$

Thus

$$f_1(z) = \sum_{k=1}^{\infty} \left(\frac{1}{k} \sum_{m=1}^{2k-1} \frac{(-1)^{m-1}}{m} \right) z^{2k}. \quad (\text{V.39})$$

(2) Note

$$\begin{aligned} \ln \left(\frac{1+z}{1-z} \right) &= \sum_{n=0}^{\infty} \frac{2}{2n+1} z^{2n+1}, \\ \arctan z &= \sum_{m=0}^{\infty} \frac{(-1)^m}{2m+1} z^{2m+1}. \end{aligned} \quad (\text{V.40})$$

Multiply the two series, we obtain

$$\begin{aligned} f_2(z) &= \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{2(-1)^m}{(2m+1)(2n+1)} z^{2(m+n+1)} \\ &= \sum_{k=1}^{\infty} \left(\sum_{m=0}^{k-1} \frac{2 \cdot (-1)^m}{(2m+1)(2k-2m-1)} \right) z^{2k}. \end{aligned} \quad (\text{V.41})$$

Similarly,

$$\begin{aligned} &\sum_{m=0}^{k-1} \frac{2 \cdot (-1)^m}{(2m+1)(2k-2m-1)} \\ &= \frac{2 \cdot [1 + (-1)^{k-1}]}{2k} \sum_{m=0}^{k-1} \frac{(-1)^m}{2m+1} \\ &= \frac{2}{2k+1} \sum_{m=0}^{2k} \frac{(-1)^m}{2m+1}. \end{aligned} \quad (\text{V.42})$$

Thus

$$f_2(z) = \sum_{k=0}^{\infty} \left(\frac{2}{2k+1} \sum_{m=0}^{2k} \frac{(-1)^m}{2m+1} \right) z^{2(2k+1)}. \quad (\text{V.43})$$

6. Determine the nature of the isolated singularities of the following function. If it is a pole, find its order.

- (1) $z = \pm ai$: simple poles;
- (2) $z = 0$: simple pole, $z = -1$: pole of order n ;
- (3) $z = 0$: removable singularity, $z = \infty$: essential singularity;
- (4) Depends on the choice of the branch of the single-valued function $f(z) = \ln \frac{1+z}{1-z}$. If $f(0) = 0$, then $z = 0$ is a simple pole; if $f(0) \neq 0$, then $z = 0$ is a pole of order 2;
- (5) $z = 0$: essential singularity, $z = \infty$: removable singularity;
- (6) $z = 0$: removable singularity; $z = (n\pi)^2$, $n = \pm 1, \pm 2, \pm 3, \dots$: simple poles; $z = \infty$: non-isolated singularity;
- (7) This is a multi-valued function, with branch points at $z = 0$ and $z = \infty$. If a branch cut is drawn connecting these two branch points and does not pass through $z = 1$, then $z = 1$ is a pole, and its order depends on the chosen branch. If we prescribe $\ln|z| = 2n\pi i$, $n \neq 0$, then $z = 1$ is a simple pole; if we prescribe $\ln|z| = 0$, then $z = 1$ is a pole of order 2;
- (8) $z = \infty$: essential singularity.

7. Determine the nature of the following functions at infinity.

- (1) Pole of order 2;
- (2) Removable singularity (analytic point);
- (3) Essential singularity, that is because

$$\frac{\cos z}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n-1}. \quad (\text{V.44})$$

- (4) Non-isolated singularity; (Let $\cos z = 0$, then $z = \frac{2k+1}{2}\pi$, $k \in \mathbb{Z}$.)
- (5) Essential singularity, note e^{-z} has infinitely many positive power terms.
- (6) Analytic (can also be called a removable singularity), because no positive power terms is detected.
- (7) Non-isolated singularity; (Let $\cosh \sqrt{z} = 0$, then $z_n = -\left(\frac{2n+1}{2}\pi\right)^2$, $n \in \mathbb{Z}$.)
- (8) Depends on the choice of the branch cut and the specification of the single-valued branch. We may, for instance, draw a branch cut connecting $z = 1$ and $z = 2$ that does not pass through ∞ . Then $z = \infty$ is a simple pole.

8. Two series defined in different regions can be analytic continuations of each other. As an example, show that

$$f_1(z) = 1 + az + a^2 z^2 + a^3 z^3 + \dots \quad (\text{V.45})$$

and

$$f_2(z) = \frac{1}{1-z} - \frac{(1-a)z}{(1-z)^2} + \frac{(1-a)^2 z^2}{(1-z)^3} + \cdots \quad (\text{V.46})$$

are analytic continuations of each other.

Proof: In the region $|z| < 1/|a|$,

$$f_1(z) = \sum_{n=0}^{\infty} (az)^n = \frac{1}{1-az}, \quad (\text{V.47})$$

and in the region of $|(1-a)z/(1-z)| < 1$,

$$f_2(z) = \frac{1}{(1-a)z} \sum_{n=1}^{\infty} \left(\frac{(1-a)z}{(1-z)} \right)^n = \frac{1}{1-az}. \quad (\text{V.48})$$

So in the intersection of the two region of convergence (see $z = 0$), we have $f_1(z) = f_2(z)$. Thus $f_1(z)$ and $f_2(z)$ are analytic continuations of each other. $\square$

9. An infinite series can converge to different sum functions in different regions. These two sum functions, although having the same series expression (in different regions), are not analytic continuations of each other. As an example, prove that the series

$$\sum_{n=1}^{\infty} \left(\frac{1}{1-z^{n+1}} - \frac{1}{1-z^n} \right) \quad (\text{V.49})$$

represents two analytic functions in the regions $|z| < 1$ and $|z| > 1$, respectively, but they are not analytic continuations of each other.

Proof: It is easy to obtain

$$S_N = \sum_{n=1}^N \left(\frac{1}{1-z^{n+1}} - \frac{1}{1-z^n} \right) = \frac{1}{1-z^{N+1}} - \frac{1}{1-z}. \quad (\text{V.50})$$

So

$$\begin{aligned} f_1(z) &= 1 - \frac{1}{1-z}, \quad |z| < 1, \\ f_2(z) &= -\frac{1}{1-z}, \quad |z| > 1. \end{aligned} \quad (\text{V.51})$$

These two functions are not analytic continuations of each other. Indeed, on the unit circle $|z| = 1$, $f_1(z)$ and $f_2(z)$ cannot be connected analytically to define a single analytic function in a domain containing points from both regions $|z| < 1$ and $|z| > 1$. $\square$

10. The given series

$$\sum_{n=0}^{\infty} z^{2^n} = z + z^2 + z^4 + z^8 + z^{16} + \cdots \quad (\text{V.52})$$

converges inside the unit circle, and its sum function is $f(z)$. Show that:

- (1) $z = 1$ is a singularity of $f(z)$.
- (2) $f(z) = z + f(z^2)$, thus all roots of $z^2 = 1$ are singularities of $f(z)$.
- (3) Similarly, all roots of $z^{2^k} = 1$ are singularities of $f(z)$, $k \in \mathbb{Z}$.

- (4) It is impossible to analytically continue $f(z)$ outside the unit circle.

Proof:

- (1) Note that $1^{2^k} \equiv 1$, thus $\lim_{k \rightarrow \infty} |z^{2^k}| = 1 \neq 0$, thus $z = 1$ is a singularity of $f(z)$.
- (2) Note

$$f(z^2) = \sum_{n=0}^{\infty} (z^2)^{2^n} = \sum_{n=0}^{\infty} z^{2^{n+1}} = \sum_{n=1}^{\infty} z^{2^n}, \quad (\text{V.53})$$

thus $f(z) = z + f(z^2)$. By (1), $z = 1$ is a singularity of $f(z)$, thus all roots of $z^2 = 1$ are singularities of $f(z)$.

- (3) Similarly,

$$f(z^{2^k}) = \sum_{n=0}^{\infty} (z^{2^k})^{2^n} = \sum_{n=k}^{\infty} z^{2^n} = f(z) - \sum_{n=0}^{k-1} z^{2^n}, \quad (\text{V.54})$$

thus all roots of $z^{2^k} = 1$ are singularities of $f(z)$.

- (4) From (3), there are infinitely many singularities between any two points on the unit circle, so it is impossible to analytically continue $f(z)$ outside the unit circle. $\square$

11. Rewrite equation (5.58) as a rational fraction expansion of $\cot z$, and further replace z by $z \pm i\gamma$ to derive the rational fraction expansions of

$$\frac{\sin 2z}{\cosh 2\gamma - \cos 2z} \quad \text{and} \quad \frac{\sin 2\gamma}{\gamma} \frac{1}{\cosh 2\gamma - \cos 2z}. \quad (\text{V.55})$$

Set $z' = \pi z$, we obtain

$$\cot z' = \frac{1}{z'} + 2z' \sum_{m=1}^{\infty} \frac{1}{(z')^2 - (m\pi)^2}. \quad (\text{V.56})$$

Again set $z' = z \pm i\gamma$ and apply $\sinh z = -i \sin iz$, $\cosh z = \cos iz$, we obtain

$$\begin{aligned} \frac{\cos z \cosh \gamma - i \sin z \sinh \gamma}{\sin z \cosh \gamma + i \cos z \sinh \gamma} &= \frac{z - i\gamma}{z^2 + \gamma^2} + \\ \sum_{m=1}^{\infty} \left[\frac{z - m\pi - i\gamma}{(z - m\pi)^2 + \gamma^2} + \frac{z + m\pi - i\gamma}{(z + m\pi)^2 + \gamma^2} \right], \end{aligned} \quad (\text{V.57})$$

and

$$\begin{aligned} \frac{\cos z \cosh \gamma + i \sin z \sinh \gamma}{\sin z \cosh \gamma - i \cos z \sinh \gamma} &= \frac{z + i\gamma}{z^2 + \gamma^2} + \\ \sum_{m=1}^{\infty} \left[\frac{z - m\pi + i\gamma}{(z - m\pi)^2 + \gamma^2} + \frac{z + m\pi + i\gamma}{(z + m\pi)^2 + \gamma^2} \right]. \end{aligned} \quad (\text{V.58})$$

Add (V.57) and (V.58), we obtain

$$\begin{aligned} \frac{2 \sin z \cos z \cosh^2 \gamma - 2 \sin z \cos z \sinh^2 \gamma}{\sin^2 z \cosh^2 \gamma + \cos^2 z \sinh^2 \gamma} &= \frac{2z}{z^2 + \gamma^2} + \\ \sum_{m=1}^{\infty} \left[\frac{2(z - m\pi)}{(z - m\pi)^2 + \gamma^2} + \frac{2(z + m\pi)}{(z + m\pi)^2 + \gamma^2} \right]. \end{aligned} \quad (\text{V.59})$$

Note $\cosh^2 \gamma = 1/2 + 1/2 \cosh 2\gamma$, $\sinh^2 \gamma = -1/2 + 1/2 \cosh 2\gamma$, thus

$$\frac{\sin 2z}{\cosh 2\gamma - \cos 2z} = \frac{z}{z^2 + \gamma^2} + \sum_{n=1}^{\infty} \left[\frac{z - n\pi}{(z - n\pi)^2 + \gamma^2} + \frac{z + n\pi}{(z + n\pi)^2 + \gamma^2} \right]. \quad (\text{V.60})$$

Similarly (subtract the two equations), we obtain

$$\frac{\sinh 2\gamma}{\gamma} \frac{1}{\cosh 2\gamma - \cos 2z} = \frac{1}{z^2 + \gamma^2} + \sum_{n=1}^{\infty} \left[\frac{1}{(z - n\pi)^2 + \gamma^2} + \frac{1}{(z + n\pi)^2 + \gamma^2} \right] \quad (\text{V.61})$$

12. Rewrite equation (5.58) as a rational fraction expansion of $1/\sin z$, and further replace z by $z \pm i\gamma$ to derive the rational fraction expansions of

$$\frac{2 \sin z \cosh \gamma}{\cosh 2\gamma - \cos 2z} \quad \text{and} \quad \frac{2 \sin \gamma}{\gamma} \frac{\cos z}{\cosh 2\gamma - \cos 2z}. \quad (\text{V.62})$$

Set $z' = \pi z$, we obtain

$$\frac{1}{\sin z'} = \frac{1}{z'} + \sum_{m=1}^{\infty} (-1)^m \left(\frac{1}{z' - m\pi} + \frac{1}{z' + m\pi} \right). \quad (\text{V.63})$$

Again set $z = z' \pm i\gamma$, we obtain

$$\frac{1}{\sin z \cosh \gamma + i \cos z \sinh \gamma} = \frac{z - i\gamma}{z^2 + \gamma^2} + \sum_{m=1}^{\infty} (-1)^m \left[\frac{z - m\pi - i\gamma}{(z - m\pi)^2 + \gamma^2} + \frac{z + m\pi - i\gamma}{(z + m\pi)^2 + \gamma^2} \right], \quad (\text{V.64})$$

and

$$\frac{1}{\sin z \cosh \gamma - i \cos z \sinh \gamma} = \frac{z + i\gamma}{z^2 + \gamma^2} + \sum_{m=1}^{\infty} (-1)^m \left[\frac{z - m\pi + i\gamma}{(z - m\pi)^2 + \gamma^2} + \frac{z + m\pi + i\gamma}{(z + m\pi)^2 + \gamma^2} \right]. \quad (\text{V.65})$$

Add the two equations, we obtain

$$\frac{2 \sin z \cosh \gamma}{\sin^2 z \cosh^2 \gamma + \cos^2 z \sinh^2 \gamma} = \frac{2z}{z^2 + \gamma^2} + \sum_{m=1}^{\infty} (-1)^m \left[\frac{2(z - m\pi)}{(z - m\pi)^2 + \gamma^2} + \frac{2(z + m\pi)}{(z + m\pi)^2 + \gamma^2} \right]. \quad (\text{V.66})$$

Apply the same trick as 11, we obtain

$$\frac{2 \sin z \cosh \gamma}{\cosh 2\gamma - \cos 2z} = \frac{z}{z^2 + \gamma^2} + \sum_{n=1}^{\infty} (-1)^n \left[\frac{z - n\pi}{(z - n\pi)^2 + \gamma^2} + \frac{z + n\pi}{(z + n\pi)^2 + \gamma^2} \right], \quad (\text{V.67})$$

and

$$\frac{\sinh 2\gamma}{\gamma} \frac{1}{\cosh 2\gamma - \cos 2z} = \frac{1}{z^2 + \gamma^2} + \sum_{n=1}^{\infty} \left[\frac{1}{(z - n\pi)^2 + \gamma^2} + \frac{1}{(z + n\pi)^2 + \gamma^2} \right] \quad (\text{V.68})$$

VI. The Residue Theorem and Its Applications

1. Find the residue of the following function at the given point z_0 .

(1) By definition,

$$\text{Res}_{z=1} \left[\frac{e^{z^2}}{z-1} \right] = \lim_{z \rightarrow 1} \left((z-1) \cdot \frac{e^{z^2}}{z-1} \right) = e. \quad (\text{VI.1})$$

(2) By definition,

$$\text{Res}_{z=1} \left[\frac{e^{z^2}}{(z-1)^2} \right] = \lim_{z \rightarrow 1} \frac{d}{dz} \left((z-1)^2 \cdot \frac{e^{z^2}}{(z-1)^2} \right) = 2e. \quad (\text{VI.2})$$

(3) Use the method of order estimation. Note

$$\left(\frac{z}{(1 - \cos z)} \right)^2 \sim \mathcal{O} \left(\frac{1}{z^2} \right). \quad (\text{VI.3})$$

Therefore, the coefficient of z^{-1} in the Laurent expansion must be zero. Thus,

$$\text{Res}_{z=0} \left[\left(\frac{z}{(1 - \cos z)} \right)^2 \right] = 0. \quad (\text{VI.4})$$

(4) Find the residue using the Laurent expansion. Since

$$\sin z = z - \frac{z^3}{6} + \frac{z^5}{120} - \dots, \quad (\text{VI.5})$$

we have

$$z^2 \sin z = z^3 \left(1 - \frac{z^2}{6} + \frac{z^4}{120} - \dots \right). \quad (\text{VI.6})$$

Hence

$$f(z) = \frac{1}{z^3} \cdot \frac{1}{1 - \frac{z^2}{6} + \frac{z^4}{120} - \dots}. \quad (\text{VI.7})$$

Thus $z = 0$ is a pole of order 3. Let

$$g(z) = \frac{1}{1 - \frac{z^2}{6} + \frac{z^4}{120} - \dots} \quad (\text{VI.8})$$

which is analytic at 0 with $g(0) = 1$. Write $f(z) = z^{-3}g(z)$. The residue of f at 0 is the coefficient of z^{-1} in the Laurent expansion of f and this coefficient comes from the z^2 -term in the expansion of $g(z)$. Let

$$w = \frac{z^2}{6} - \frac{z^4}{120} + \dots \quad (\text{VI.9})$$

Then

$$g(z) = (1 - w)^{-1} = 1 + w + w^2 + \dots \quad (\text{VI.10})$$

Keeping terms up to z^2 :

$$g(z) = 1 + \frac{z^2}{6} + \mathcal{O}(z^4). \quad (\text{VI.11})$$

Hence the z^2 -coefficient of g is $1/6$. So

$$\text{Res}_{z=0} \left[\frac{1}{z^2 \sin z} \right] = \frac{1}{6}. \quad (\text{VI.12})$$

(5) By definition,

$$\begin{aligned} \text{Res}_{z=1} \left[\frac{e^z}{(z^2-1)^2} \right] &= \lim_{z \rightarrow 1} \frac{d}{dz} \left((z-1)^2 \cdot \frac{e^z}{(z^2-1)^2} \right) \\ &= \lim_{z \rightarrow 1} \frac{e^z(z-1)}{(z+1)^3} = 0. \end{aligned} \quad (\text{VI.13})$$

(6) Use l'Hôpital's rule.

$$\begin{aligned} \text{Res}_{z=z_0} \left[\frac{1}{\cosh \sqrt{z}} \right] &= \lim_{z \rightarrow z_0} \frac{2(z-z_0)}{e^{\sqrt{z}} + e^{-\sqrt{z}}} \\ &= \lim_{z \rightarrow z_0} \frac{4\sqrt{z}}{e^{\sqrt{z}} - e^{-\sqrt{z}}} = (-1)^n \cdot (2n+1)\pi. \end{aligned} \quad (\text{VI.14})$$

2. Find the residue at each isolated singularity in the complex plane for the following functions.

(1) By definition,

$$\begin{aligned} \text{Res}_{z=1} \left[\frac{z^2}{(z-1)^3(z+1)} \right] \\ = \lim_{z \rightarrow 1} \frac{1}{2!} \frac{d^2}{dz^2} \left((z-1)^3 \cdot \frac{z^2}{(z-1)^3(z+1)} \right) = \frac{1}{8} \end{aligned} \quad (\text{VI.15})$$

$$\begin{aligned} \text{Res}_{z=-1} \left[\frac{z^2}{(z-1)^3(z+1)} \right] \\ = \lim_{z \rightarrow -1} \left((z+1) \cdot \frac{z^2}{(z-1)^3(z+1)} \right) = -\frac{1}{8}. \end{aligned} \quad (\text{VI.16})$$

(2) By definition,

$$\begin{aligned} \text{Res}_{z=i} \left[\frac{1}{(1+z^2)^{m+1}} \right] \\ = \lim_{z \rightarrow i} \frac{1}{m!} \frac{d^m}{dz^m} \left((z-i)^{m+1} \cdot \frac{1}{(1+z^2)^{m+1}} \right) \\ = \lim_{z \rightarrow i} (-1)^m \frac{(2m)!}{(m!)^2} \cdot \frac{1}{(z+i)^{2m+1}} = -\frac{(2m)!}{(m!)^2 \cdot 2^{2m+1}} i, \end{aligned} \quad (\text{VI.17})$$

similarly,

$$\text{Res}_{z=-i} \left[\frac{1}{(1+z^2)^{m+1}} \right] = \frac{(2m)!}{(m!)^2 \cdot 2^{2m+1}} i. \quad (\text{VI.18})$$

(3) The function has poles at $z_n = 2n\pi, n \in \mathbb{Z}$. Find the residue using the Laurent expansion:

$$\begin{aligned} \frac{z}{1 - \cos z} &= \frac{(z - 2n\pi) + 2n\pi}{1 - \cos(z - 2n\pi)^2} \\ &= \frac{2}{(z - 2n\pi)^2} \left[2n\pi + (z - 2n\pi) \right] \cdot \\ &\quad \frac{1}{1 - 1/12(z - 2n\pi)^2 + \mathcal{O}(z - 2n\pi)^2} \end{aligned} \quad (\text{VI.19})$$

Hence the $(z - 2n\pi)^{-1}$ -coefficient is 2, thus

$$\text{Res}_{z=2n\pi} \left[\frac{z}{1 - \cos z} \right] = 2, \quad n \in \mathbb{Z}. \quad (\text{VI.20})$$

(4) Use l'Hôpital's rule. The function has poles at $z_n = -n^2\pi^2, n \in \mathbb{Z}$,

$$\begin{aligned} \text{Res}_{z=z_n} \left[\frac{\sqrt{z}}{\sinh \sqrt{z}} \right] &= \lim_{z \rightarrow z_n} \frac{2\sqrt{z}(z - z_n)}{e^{\sqrt{z}} - e^{-\sqrt{z}}} \\ &= \lim_{z \rightarrow z_n} \frac{4z}{e^{\sqrt{z}} - e^{-\sqrt{z}}} = (-1)^{n+1} \cdot 2n^2\pi^2. \end{aligned} \quad (\text{VI.21})$$

(5) Use the generating function for Bessel functions. Note

$$\exp \left[\frac{1}{2} \left(z - \frac{1}{z} \right) \right] = \sum_{n=-\infty}^{\infty} J_n(1) z^n, \quad (\text{VI.22})$$

thus

$$\text{Res}_{z=0} \left[\exp \left(\frac{1}{2} \left(z - \frac{1}{z} \right) \right) \right] = J_{-1}(1) = -J_1(1). \quad (\text{VI.23})$$

Using the theorem that the sum of all residues on the extended complex plane is zero,

$$\text{Res}_{z=\infty} \left[\exp \left(\frac{1}{2} \left(z - \frac{1}{z} \right) \right) \right] = J_1(1). \quad (\text{VI.24})$$

(6) Use the Laurent expansion:

$$\cos \frac{1}{\sqrt{z}} = 1 - \frac{1}{2} \frac{1}{z} + \mathcal{O}\left(\frac{1}{z}\right). \quad (\text{VI.25})$$

So

$$\text{Res}_{z=0} \left[\cos \frac{1}{\sqrt{z}} \right] = -\frac{1}{2}. \quad (\text{VI.26})$$

Using the theorem that the sum of all residues on the extended complex plane is zero,

$$\text{Res}_{z=\infty} \left[\cos \frac{1}{\sqrt{z}} \right] = \frac{1}{2}. \quad (\text{VI.27})$$

But note that ∞ is not a pole of $f(z)$!

(7) Note $z = 1$ is a pole of the function, but its order depends on the choice of the branch cut.

- i) If we specify that $\ln(z)|_{z=1} = 2n\pi i$, $n \in \mathbb{Z}$, $n \neq 0$. Then $z = 1$ is a simple pole. By definition,

$$\begin{aligned} \operatorname{Res}_{z=1} \left[\frac{1}{(z-1)\ln z} \right] &= \lim_{z \rightarrow 1} \left((z-1) \cdot \frac{1}{(z-1)\ln z} \right) \\ &= \frac{1}{2n\pi i}. \end{aligned} \quad (\text{VI.28})$$

- ii) If we specify that $\ln(z)|_{z=1} = 0$. Then $z = 1$ is a pole of order 2. By definition,

$$\begin{aligned} \operatorname{Res}_{z=1} \left[\frac{1}{(z-1)\ln z} \right] &= \lim_{z \rightarrow 1} \frac{d}{dz} \left((z-1)^2 \cdot \frac{1}{(z-1)\ln z} \right) \\ &= \lim_{z \rightarrow 1} \frac{\ln z - 1 + \frac{1}{z}}{\ln^2 z} = \frac{1}{2}. \end{aligned} \quad (\text{VI.29})$$

Note that we have used l'Hôpital's rule at the last equality.

- (8) The function has poles at $z = 0$ (simple pole) and $z = -1$ (pole of order n). By definition,

$$\begin{aligned} \operatorname{Res}_{z=0} \left[\frac{1}{z} \sum_{k=0}^n \frac{1}{(z+1)^k} \right] &= \lim_{z \rightarrow 0} \left(z \cdot \frac{1}{z} \sum_{k=0}^n \frac{1}{(z+1)^k} \right) \\ &= n + 1. \end{aligned} \quad (\text{VI.30})$$

For the residue at $z = -1$, apply the Taylor expansion:

$$\frac{1}{z} = -\frac{1}{1 - (z+1)} = -\sum_{j=0}^{\infty} (z+1)^j, \quad (\text{VI.31})$$

thus

$$f(z) = \left(-\sum_{j=0}^{\infty} (z+1)^j \right) \cdot \left(\sum_{k=0}^n \frac{1}{(z+1)^k} \right). \quad (\text{VI.32})$$

Hence the $(z+1)^{-1}$ -coefficient is $-n$, so

$$\operatorname{Res}_{z=-1} \left[\frac{1}{z} \sum_{k=0}^n \frac{1}{(z+1)^k} \right] = -n. \quad (\text{VI.33})$$

Finally use the theorem that the sum of all residues on the extended complex plane is zero, we obtain

$$\operatorname{Res}_{z=\infty} \left[\frac{1}{z} \sum_{k=0}^n \frac{1}{(z+1)^k} \right] = -1. \quad (\text{VI.34})$$

But note that ∞ is not a pole of $f(z)$!

3. For the functions below, calculate the residue at infinity.

- (1) By definition, the z^{-1} -coefficient is 1, so $\operatorname{Res}_{z=\infty} \frac{1}{z} = -1$.
 (2) Note

$$\frac{\cos z}{z} = \frac{1}{z} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k}, \quad (\text{VI.35})$$

thus the z^{-1} -coefficient is 1, so $\operatorname{Res}_{z=\infty} \frac{\cos z}{z} = -1$.

- (3) Infinity is a non-isolated singularity, so the residue is not defined.

- (4) Use the sum theory of residues. Note

$$\operatorname{Res}_{z=-a} \frac{\cos z}{(z+a)^2} = \lim_{z \rightarrow -a} \frac{d}{dz} \left((z+a)^2 \cdot \frac{\cos z}{(z+a)^2} \right) = \sin a, \quad (\text{VI.36})$$

then $\operatorname{Res}_{z=\infty} \frac{\cos z}{(z+a)^2} = -\sin a$.

- (5) Use the Taylor expansion of e^z :

$$f_5(z) = (z^2 + 1) \cdot \sum_{n=0}^{\infty} \frac{1}{n!} z^n, \quad (\text{VI.37})$$

the z^{-1} -coefficient must be zero, thus $\operatorname{Res}_{z=\infty} (z^2 + 1)e^z = 0$.

- (6) Use the Laurent expansion:

$$\begin{aligned} f_6(z) &= \pm z \left[1 + \left(\frac{-3}{z} + \frac{2}{z^2} \right) \right]^{1/2} \\ &= \pm z \left[1 + \frac{1}{2} \left(\frac{-3}{z} + \frac{2}{z^2} \right) - \frac{1}{8} \left(\frac{-3}{z} + \frac{2}{z^2} \right)^2 + \mathcal{O}\left(\frac{1}{z^2}\right) \right] \end{aligned} \quad (\text{VI.38})$$

the z^{-1} -coefficient is $\pm 1/8$, thus

$$\operatorname{Res}_{z=\infty} \sqrt{(z-1)(z-2)} = \pm 1/8 \quad (\text{VI.39})$$

The choice of branch fixes the sign.

4. Calculate the value of the following integrals.

- (1) The pole inside the contour is $z_k = e^{i\frac{(2k+1)\pi}{4}}$, $k = 0, -1$. Apply the l'Hôpital's rule:

$$\begin{aligned} \operatorname{Res}_{z=z_k} \left[\frac{1}{1+z^4} \right] &= \lim_{z \rightarrow z_k} \left((z-z_k) \frac{1}{1+z^4} \right) \\ &= \lim_{z \rightarrow z_k} \frac{1}{4z^3} = \frac{1}{4e^{i\frac{3(2k+1)\pi}{4}}} \end{aligned} \quad (\text{VI.40})$$

thus

$$I_1 = 2\pi i \sum_{k=1,2} \operatorname{Res}_{z=z_k} \left[\frac{1}{1+z^4} \right] = -\frac{\pi i}{\sqrt{2}}. \quad (\text{VI.41})$$

- (2) The pole inside the contour is $z_k = e^{i\frac{(2k+1)\pi}{4}}$, $k = 0, 1, 2, 3$. Employing both the preceding result and symmetry,

$$I_2 = 2\pi i \sum_{k=0}^3 \operatorname{Res}_{z=z_k} \left[\frac{1}{1+z^4} \right] = 0. \quad (\text{VI.42})$$

- (3) The pole inside the contour is $z = 1$. By definition,

$$\begin{aligned} \operatorname{Res}_{z=1} \left[\frac{1}{z^2-1} \sin\left(\frac{\pi}{4}z\right) \right] \\ = \lim_{z \rightarrow 1} \left((z-1) \cdot \frac{1}{z^2-1} \sin\left(\frac{\pi}{4}z\right) \right) = \frac{1}{2\sqrt{2}}. \end{aligned} \quad (\text{VI.43})$$

Thus we obtain

$$I_3 = 2\pi i \cdot \text{Res}_{z=1} \left[\frac{1}{z^2 - 1} \sin \left(\frac{\pi}{4} z \right) \right] = \frac{\pi i}{\sqrt{2}}. \quad (\text{VI.44})$$

- (4) The pole inside the contour is $z = 1$ and $z = -1$. Similarly,

$$\text{Res}_{z=-1} \left[\frac{1}{z^2 - 1} \sin \left(\frac{\pi}{4} z \right) \right] = \frac{1}{2\sqrt{2}}. \quad (\text{VI.45})$$

Thus we obtain

$$I_4 = 2\pi i \left\{ \text{Res}_{z=1} \left[\frac{\sin(\pi/4 z)}{z^2 - 1} \right] + \text{Res}_{z=-1} \left[\frac{\sin(\pi/4 z)}{z^2 - 1} \right] \right\} = \sqrt{2}\pi i. \quad (\text{VI.46})$$

- (5) The pole inside the contour is $z_k = \frac{(2k+1)\pi}{2}$, $k = [-n, -(n-1), \dots, 0, 1, \dots, (n-1)]$. Apply the l'Hôpital's rule:

$$\begin{aligned} \text{Res}_{z=z_k} \left[\frac{\sin \pi z}{\cos \pi z} \right] &= \lim_{z \rightarrow z_k} \left((z - z_k) \cdot \frac{\sin \pi z}{\cos \pi z} \right) \\ &= \lim_{z \rightarrow z_k} \frac{\sin \pi z}{-\pi \sin \pi z} = -\frac{1}{\pi}. \end{aligned} \quad (\text{VI.47})$$

Thus we obtain

$$I_5 = 2\pi i \cdot \sum_{k=-n}^{n-1} \text{Res}_{z=z_k} \left[\frac{\sin \pi z}{\cos \pi z} \right] = -4ni. \quad (\text{VI.48})$$

- (6) The pole inside the contour is $z_k = \sqrt[10]{2}e^{i\frac{k}{5}\pi}$, $k = 0 \sim 9$. Apply the l'Hôpital's rule:

$$\begin{aligned} \text{Res}_{z=z_k} \left[\frac{1}{z^3(z^{10} - 2)} \right] &= \lim_{z \rightarrow z_k} \left((z - z_k) \cdot \frac{1}{z^3(z^{10} - 2)} \right) \\ &= \frac{1}{10} z_k^{-12}. \end{aligned} \quad (\text{VI.49})$$

By symmetry,

$$I_6 = 2\pi i \cdot \sum_{k=0}^9 \text{Res}_{z=z_k} \left[\frac{1}{z^3(z^{10} - 2)} \right] = 0. \quad (\text{VI.50})$$

- (7) The pole inside the contour is $z = 0$. By definition,

$$\begin{aligned} \text{Res}_{z=0} \left[\frac{e^z}{z^3} \right] &= \lim_{z \rightarrow 0} \frac{1}{2!} \cdot \frac{d^2}{dz^2} \left(z^3 \cdot \frac{e^z}{z^3} \right) \\ &= \frac{1}{2}. \end{aligned} \quad (\text{VI.51})$$

Thus we obtain

$$I_7 = 2\pi i \cdot \text{Res}_{z=0} \left[\frac{e^z}{z^3} \right] = \pi i. \quad (\text{VI.52})$$

- (8) The pole inside the contour is $z = 0$ and $z_{jk} = \sqrt[3]{k} \cdot e^{i\frac{2j}{3}\pi}$, $k = \pm 1, \pm 2, \dots, \pm n$; $j = 0, 1, 2$. Apply the l'Hôpital's rule:

$$\begin{aligned} \text{Res}_{z=z_{jk}} \left[\frac{z^2}{e^{2\pi i z^3} - 1} \right] &= \lim_{z \rightarrow z_{jk}} \left((z - z_{jk}) \cdot \frac{z^2}{e^{2\pi i z^3} - 1} \right) \\ &= \frac{1}{6\pi i}. \end{aligned} \quad (\text{VI.53})$$

In particular,

$$\text{Res}_{z=0} \left[\frac{z^2}{e^{2\pi i z^3} - 1} \right] = \frac{1}{2\pi i}. \quad (\text{VI.54})$$

Note we have used the Taylor expansion of $e^{2\pi i z^3} - 1$. Thus we obtain

$$\begin{aligned} I_8 &= 2\pi i \cdot \left\{ \text{Res}_{z=0} \left[\frac{z^2}{e^{2\pi i z^3} - 1} \right] + \sum_k \sum_j \text{Res}_{z=z_{jk}} \left[\frac{z^2}{e^{2\pi i z^3} - 1} \right] \right\} = 2n + 1. \end{aligned} \quad (\text{VI.55})$$

5. Calculate the value of the following integrals.

- (1) By Wallis' formula,

$$I_1 = 4 \int_0^{\pi/2} \cos^{2n} \theta d\theta = \frac{\pi}{2^{2n-1}} \frac{(2n)!}{(n!)^2}. \quad (\text{VI.56})$$

- (2) Let $e^{ix} = z$, we obtain

$$I_2 = \oint_{|z|=1} \frac{4}{i} \cdot \frac{z}{(bz^2 + 2az + b)^2} \quad (\text{VI.57})$$

Note that the pole inside the contour is $z_1 = \frac{-a + \sqrt{a^2 - b^2}}{b}$, and

$$\begin{aligned} \text{Res}_{z=z_1} \left[\frac{z}{(bz^2 + 2az + b)^2} \right] &= \lim_{z \rightarrow z_1} \left((z - z_1) \cdot \frac{1}{b^2} \frac{z}{(z - z_1)(z - z_2)} \right) \\ &= \frac{-a + \sqrt{a^2 - b^2}}{2\sqrt{a^2 - b^2}}, \end{aligned} \quad (\text{VI.58})$$

thus

$$I_2 = 2\pi i \cdot \frac{4}{i} \text{Res}_{z=z_1} \left[\frac{z}{(bz^2 + 2az + b)^2} \right] = 4\pi \frac{-a + \sqrt{a^2 - b^2}}{\sqrt{a^2 - b^2}}. \quad (\text{VI.59})$$

- (3) Transform the integral into one involving $\tan \theta$:

$$I_3 = \int_0^\pi \frac{1 + \tan^2 \theta}{1 + 2 \tan^2 \theta} d\theta, \quad (\text{VI.60})$$

Let $u = \tan \theta$, we obtain

$$I_3 = \int_{-\infty}^\infty \frac{1}{1 + 2u^2} du = \frac{\pi}{\sqrt{2}}. \quad (\text{VI.61})$$

(4) Apply the same trick as (3), we obtain

$$I_4 = \int_{-\infty}^{\infty} \frac{1+u^2}{(1+2u^2)^2} du. \quad (\text{VI.62})$$

Let $u = \frac{\tan \theta}{\sqrt{2}}$, then

$$I_4 = \frac{1}{\sqrt{2}} \int_0^{\pi} \left(\cos^2 \theta + \frac{1}{2} \sin^2 \theta \right) d\theta = \frac{3\pi}{4\sqrt{2}}. \quad (\text{VI.63})$$

6. Calculate the value of the following integrals.

(1) Consider a semicircular integration contour of radius R . Then

$$\begin{aligned} \oint_{C_R} \frac{z^2}{1+z^4} dz &= \int_{-R}^R \frac{x^2}{1+x^4} + \int_{C_R} \frac{z^2}{1+z^4} dz \\ &= 2\pi i \sum_{k=1,2} \text{Res}_{z=z_k} \left[\frac{z^2}{1+z^4} \right] \end{aligned} \quad (\text{VI.64})$$

where $z_k = e^{i\frac{(2k-1)\pi}{4}}$. Apply the l'Hôpital's rule:

$$\begin{aligned} \text{Res}_{z=z_k} \left[\frac{z^2}{1+z^4} \right] &= \lim_{z \rightarrow z_k} \left((z - z_k) \cdot \frac{z^2}{1+z^4} \right) \\ &= \frac{1}{4z_k}. \end{aligned} \quad (\text{VI.65})$$

Note that

$$\lim_{z \rightarrow \infty} z \cdot \frac{z^2}{1+z^4} = 0 \quad (\text{VI.66})$$

Thus by the large arc lemma, the integral over the circular arc vanishes as $R \rightarrow \infty$. So we obtain

$$I_1 = 2\pi i \sum_{k=1,2} \text{Res}_{z=z_k} \left[\frac{z^2}{1+z^4} \right] = \frac{\pi}{\sqrt{2}}. \quad (\text{VI.67})$$

(2) Consider a semicircular integration contour of radius R . Then

$$\begin{aligned} \oint_{C_R} \frac{1}{(1+z^2)^{n+1}} dz &= \int_{-R}^R \frac{1}{(1+x^2)^{n+1}} dx \\ &+ \int_{C_R} \frac{1}{(1+z^2)^{n+1}} dz = 2\pi i \cdot \text{Res}_{z=i} \left[\frac{1}{(1+z^2)^{n+1}} \right] \end{aligned} \quad (\text{VI.68})$$

From 2.(2) we obtain

$$\text{Res}_{z=i} \left[\frac{1}{(1+z^2)^{n+1}} \right] = -i \cdot \frac{(2n)!}{(n!)^2 \cdot 2^{2n+1}}, \quad (\text{VI.69})$$

moreover, note

$$\lim_{z \rightarrow \infty} z \cdot \frac{1}{(1+z^2)^{n+1}} = 0. \quad (\text{VI.70})$$

Thus by the large arc lemma, the integral over the circular arc vanishes as $R \rightarrow \infty$. So we obtain

$$I_2 = 2\pi i \cdot \text{Res}_{z=i} \left[\frac{1}{(1+z^2)^{n+1}} \right] = \frac{\pi}{2^{2n}} \frac{(2n)!}{(n!)^2}. \quad (\text{VI.71})$$

(3) Consider a semicircular integration contour of radius R . Then

$$\begin{aligned} \oint_{C_R} \frac{z^{2m}}{1+z^{2n}} dz &= \int_{-R}^R \frac{x^{2m}}{1+x^{2n}} dx \\ &+ \int_{C_R} \frac{z^{2m}}{1+z^{2n}} dz = 2\pi i \cdot \sum_{j=0}^{n-1} \text{Res}_{z=z_j} \left[\frac{z^{2m}}{1+z^{2n}} \right] \end{aligned} \quad (\text{VI.72})$$

where $z_j = e^{i\frac{(2j+1)\pi}{2n}}$, $j \in \{0, 1, \dots, n-1\}$. Apply the l'Hôpital's rule:

$$\begin{aligned} \text{Res}_{z=z_j} \left[\frac{z^{2m}}{1+z^{2n}} \right] &= \lim_{z \rightarrow z_j} \left((z - z_j) \cdot \frac{z^{2m}}{1+z^{2n}} \right) \\ &= -\frac{1}{2n} \exp \left[i \cdot \frac{2m+1}{2n} (2j+1)\pi \right]. \end{aligned} \quad (\text{VI.73})$$

Note that

$$\lim_{z \rightarrow \infty} z \cdot \frac{z^{2m}}{1+z^{2n}} = 0. \quad (\text{VI.74})$$

Thus by the large arc lemma, the integral over the circular arc vanishes as $R \rightarrow \infty$. So we obtain

$$\begin{aligned} I_3 &= 2\pi i \cdot \sum_{j=0}^{n-1} \text{Res}_{z=z_j} \left[\frac{z^{2m}}{1+z^{2n}} \right] \\ &= -\frac{\pi i}{n} \sum_{j=0}^{n-1} \exp \left(i \cdot \frac{2m+1}{2n} \pi \right) \cdot \exp \left(i \cdot \frac{2m+1}{n} j\pi \right) \\ &= -\frac{\pi i}{n} \exp \left(i \cdot \frac{2m+1}{2n} \pi \right) \cdot \frac{2}{1 - \exp \left(i \frac{2m+1}{n} \pi \right)} \\ &= \frac{\pi}{n \sin \left(\frac{2m+1}{2n} \pi \right)}. \end{aligned} \quad (\text{VI.75})$$

(4) Consider a semicircular integration contour of radius $R_n = 2n + 2$. Then

$$\begin{aligned} \oint_{C_R} \frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} dz &= \int_{-R}^R \frac{1}{(1+x^2) \cosh \frac{\pi x}{2}} dx \\ &+ \int_{C_R} \frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} dz \\ &= 2\pi i \cdot \sum_{j=0}^{n-1} \text{Res}_{z=z_j} \left[\frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} \right] \end{aligned} \quad (\text{VI.76})$$

where $z_j = (2j+1)i$. Apply the l'Hôpital's rule:

$$\begin{aligned} & \text{Res}_{z=z_j} \left[\frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} \right] \\ &= \lim_{z \rightarrow z_j} \left((z - z_j) \cdot \frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} \right) \quad (\text{VI.77}) \\ &= \frac{i}{2\pi} (-1)^j \cdot \left(\frac{1}{j} - \frac{1}{j+1} \right). \end{aligned}$$

Particularly, for $z_0 = j$, use the Laurent expansion:

$$\begin{aligned} & \frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} \\ &= \frac{1}{z-i} \cdot \frac{1}{(z-i)+2i} \cdot \frac{2}{e^{\frac{\pi}{2}(z-i)+\pi/2i} + e^{-\pi/2(z-i)-\pi/2i}} \\ &= -\frac{1}{z-i} \cdot \left[\sum_{k=0}^{\infty} \left(-\frac{z-i}{2i} \right)^k \right] \cdot \frac{1}{\pi(z-i)} \cdot \frac{1}{1 + \mathcal{O}(z-i)^2}. \quad (\text{VI.78}) \end{aligned}$$

Thus, the $(z-i)^{-1}$ -coefficient is $-\frac{i}{2\pi}$. So we obtain

$$\text{Res}_{z=z_j} \left[\frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} \right] = -\frac{i}{2\pi} \quad (\text{VI.79})$$

Moreover, note that $R_n = 2n+2$, thus

$$\lim_{n \rightarrow \infty} z \cdot \frac{1}{(1+z^2) \cosh \frac{\pi z}{2}} = 0. \quad (\text{VI.80})$$

By the large arc lemma, the integral over the circular arc vanishes as $R_n \rightarrow \infty$. Finally,

$$\begin{aligned} I_4 &= 1 + \sum_{j=1}^{\infty} (-1)^{j-1} \left(\frac{1}{j} - \frac{1}{j+1} \right) \\ &= 2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 2 \ln 2. \end{aligned} \quad (\text{VI.81})$$

7. Calculate the value of the following integrals.

(1) Let the integrand be $\frac{e^{iz}}{1+z^4}$, then

$$I_1 = \frac{1}{2} \mathcal{R} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{1+x^4} dx \right) \quad (\text{VI.82})$$

Consider a semicircular integration contour of radius R . Then

$$\begin{aligned} & \oint_{C_R} \frac{e^{iz}}{1+z^4} dz = \int_{-R}^R \frac{e^{ix}}{1+x^4} dx \\ &+ \int_{C_R} \frac{e^{iz}}{1+z^4} dz = 2\pi i \cdot \sum_{j=1}^2 \text{Res}_{z=z_j} \left[\frac{e^{iz}}{1+z^4} \right] \quad (\text{VI.83}) \end{aligned}$$

where $z_j = e^{i\frac{2j-1}{4}\pi}$, $j = 1, 2$. Apply the l'Hôpital's rule:

$$\begin{aligned} \text{Res}_{z=z_j} \left[\frac{e^{iz}}{1+z^4} \right] &= \lim_{z \rightarrow z_j} \left((z - z_j) \cdot \frac{e^{iz}}{1+z^4} \right) \\ &= \frac{e^{iz_j}}{4z_j^3}. \end{aligned} \quad (\text{VI.84})$$

Thus

$$\begin{aligned} \text{Res}_{z=z_1} \left[\frac{e^{iz}}{1+z^4} \right] &= -\frac{1}{4} e^{-1/\sqrt{2}} e^{i\left(\frac{1}{\sqrt{2}} + \frac{\pi}{4}\right)}, \\ \text{Res}_{z=z_2} \left[\frac{e^{iz}}{1+z^4} \right] &= \frac{1}{4} e^{-1/\sqrt{2}} e^{-i\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)}. \end{aligned} \quad (\text{VI.85})$$

Note

$$\lim_{z \rightarrow \infty} \frac{1}{1+z^4} = 0. \quad (\text{VI.86})$$

By Jordan's lemma, we obtain

$$I_1 = \pi i \cdot \sum_{j=1}^2 \text{Res}_{z=z_j} \left[\frac{e^{iz}}{1+z^4} \right] = \frac{\pi}{2} e^{-1/\sqrt{2}} \sin \left(\frac{\pi}{4} + \frac{1}{\sqrt{2}} \right). \quad (\text{VI.87})$$

(2) Let the integrand be $\frac{e^{iz}}{(1+z^2)^3}$, then

$$I_2 = \frac{1}{2} \mathcal{R} \left(\int_{-\infty}^{\infty} \frac{e^{ix}}{(1+x^2)^3} dx \right). \quad (\text{VI.88})$$

Consider a semicircular contour of radius R in the upper half-plane. Then

$$\begin{aligned} \oint_{C_R} \frac{e^{iz}}{(1+z^2)^3} dz &= \int_{-R}^R \frac{e^{ix}}{(1+x^2)^3} dx \\ &+ \int_{C_R} \frac{e^{iz}}{(1+z^2)^3} dz \quad (\text{VI.89}) \\ &= 2\pi i \cdot \text{Res}_{z=i} \left[\frac{e^{iz}}{(1+z^2)^3} \right]. \end{aligned}$$

The pole at $z = i$ is of order 3. Applying the formula for higher-order poles:

$$\begin{aligned} \text{Res}_{z=i} \left[\frac{e^{iz}}{(1+z^2)^3} \right] &= \frac{1}{2!} \lim_{z \rightarrow i} \frac{d^2}{dz^2} \left[(z-i)^3 \frac{e^{iz}}{(z-i)^3(z+i)^3} \right] \\ &= \frac{1}{2} \frac{d^2}{dz^2} \left[\frac{e^{iz}}{(z+i)^3} \right]_{z=i}. \end{aligned} \quad (\text{VI.90})$$

After calculation:

$$\text{Res}_{z=i} \left[\frac{e^{iz}}{(1+z^2)^3} \right] = -\frac{7i}{16e}. \quad (\text{VI.91})$$

Note

$$\lim_{z \rightarrow \infty} \frac{1}{(1+z^2)^3} = 0. \quad (\text{VI.92})$$

By Jordan's lemma, the arc integral vanishes as $R \rightarrow \infty$. Thus

$$I_2 = \frac{7\pi}{16e}. \quad (\text{VI.93})$$

(3) Let the integrand be $\frac{ze^{iz}}{z^2-2z+2}$, then

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 - 2x + 2} dx = \mathcal{I} \left(\int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2 - 2x + 2} dx \right). \quad (\text{VI.94})$$

Consider a semicircular contour of radius R in the upper half-plane. Then

$$\begin{aligned} \oint_{C_R} \frac{ze^{iz}}{z^2 - 2z + 2} dz &= \int_{-R}^R \frac{xe^{ix}}{x^2 - 2x + 2} dx \\ &\quad + \int_{C_R} \frac{ze^{iz}}{z^2 - 2z + 2} dz \\ &= 2\pi i \cdot \text{Res}_{z=z_1} \left[\frac{ze^{iz}}{z^2 - 2z + 2} \right], \end{aligned} \quad (\text{VI.95})$$

where $z_1 = 1 + i$ is the pole in the upper half-plane. Apply the formula for simple poles:

$$\begin{aligned} &\text{Res}_{z=1+i} \left[\frac{ze^{iz}}{z^2 - 2z + 2} \right] \\ &= \lim_{z \rightarrow 1+i} \left((z - (1+i)) \cdot \frac{ze^{iz}}{(z - (1+i))(z - (1-i))} \right) \\ &= \frac{1-i}{2} e^{-1} e^i. \end{aligned} \quad (\text{VI.96})$$

Note

$$\lim_{z \rightarrow \infty} \frac{z}{z^2 - 2z + 2} = 0. \quad (\text{VI.97})$$

By Jordan's lemma, the arc integral vanishes as $R \rightarrow \infty$. Therefore

$$\int_{-\infty}^{\infty} \frac{xe^{ix}}{x^2 - 2x + 2} dx = 2\pi i \cdot \frac{1-i}{2} e^{-1} e^i = \pi(1+i)e^{-1} e^i. \quad (\text{VI.98})$$

Taking the imaginary part:

$$\mathcal{I}(\pi(1+i)e^{-1} e^i) = \pi e^{-1} (\sin 1 + \cos 1). \quad (\text{VI.99})$$

Hence

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 - 2x + 2} dx = \pi e^{-1} (\sin 1 + \cos 1). \quad (\text{VI.100})$$

(4) Using the sum-to-product identities, we have

$$I_4 = \int_0^{\infty} \frac{2 \cos((a+n)x) \sin(nx)}{(1+x^2) \sin x} dx. \quad (\text{VI.101})$$

Note

$$\frac{\sin(nx)}{\sin x} = \sum_{m=0}^{n-1} e^{i(n-1-2m)x}, \quad (\text{VI.102})$$

we obtain

$$I_4 = \sum_{m=0}^{n-1} \int_0^{\infty} \frac{e^{i(2n+a-1-2m)x} + e^{i(-a-1-2m)x}}{1+x^2} dx. \quad (\text{VI.103})$$

Let $A_m = 2n + a - 1 - 2m$, $B_m = -a - 1 - 2m$. For $a > -1$, $A_m > 0$, $B_m < 0$. Using (see Appendix)

$$\int_0^{\infty} \frac{e^{i\alpha x}}{1+x^2} dx = \frac{\pi}{2} e^{-\alpha}, \quad (\text{VI.104})$$

and taking real parts,

$$I = \frac{\pi}{2} \sum_{m=0}^{n-1} [e^{-A_m} + e^{-(a+1+2m)}]. \quad (\text{VI.105})$$

Change index $m \rightarrow n-1-m$ in the first sum:

$$\sum_{m=0}^{n-1} e^{-A_m} = \sum_{k=0}^{n-1} e^{-(a+1+2k)}, \quad (\text{VI.106})$$

hence

$$I_4 = \pi \sum_{k=0}^{n-1} e^{-(a+1+2k)} = \pi e^{-(a+1)} \frac{1 - e^{-2n}}{1 - e^{-2}}. \quad (\text{VI.107})$$

appendix: Consider the contour integral

$$J = \oint_C \frac{e^{i\alpha z}}{1+z^2} dz, \quad (\text{VI.108})$$

where C is the closed contour consisting of the real axis from $-R$ to R and the upper semicircle Γ_R of radius R . For $\alpha > 0$, Jordan's lemma gives $\int_{\Gamma_R} \rightarrow 0$ as $R \rightarrow \infty$. Thus

$$\int_{-\infty}^{\infty} \frac{e^{i\alpha x}}{1+x^2} dx = 2\pi i \cdot \text{Res}_{z=i} \frac{e^{i\alpha z}}{1+z^2}. \quad (\text{VI.109})$$

The residue at $z = i$ is

$$\text{Res}_{z=i} \frac{e^{i\alpha z}}{1+z^2} = \lim_{z \rightarrow i} \left((z-i) \cdot \frac{e^{i\alpha z}}{1+z^2} \right) = \frac{e^{-\alpha}}{2i}, \quad (\text{VI.110})$$

hence

$$\int_{-\infty}^{\infty} \frac{e^{i\alpha x}}{1+x^2} dx = \pi e^{-\alpha}. \quad (\text{VI.111})$$

8. Calculate the value of the following integrals.

(1) We consider an integration contour taken as the limit of a large semicircle in the upper half-plane, with small semicircular excursions around the singularities at $z=0,1,2$ excluded. Thus

$$\begin{aligned} \oint_{C_R} \frac{1}{z(z-1)(z-2)} dz &= \int_{-R}^R \frac{1}{x(x-1)(x-2)} dx \\ &\quad + \int_{C_R} \frac{1}{z(z-1)(z-2)} dz = \pi i \cdot \sum_{j=0}^2 \text{Res}_{z=z_j} \left[\frac{1}{z(z-1)(z-2)} \right] \end{aligned} \quad (\text{VI.112})$$

where $z_j = j$. By definition,

$$\begin{aligned} \text{Res}_{z=z_0} \left[\frac{1}{z(z-1)(z-2)} \right] &= \frac{1}{2}, \\ \text{Res}_{z=z_1} \left[\frac{1}{z(z-1)(z-2)} \right] &= -1, \\ \text{Res}_{z=z_2} \left[\frac{1}{z(z-1)(z-2)} \right] &= \frac{1}{2}. \end{aligned} \quad (\text{VI.113})$$

Note

$$\lim_{z \rightarrow \infty} z \cdot \frac{1}{z(z-1)(z-2)} \quad (\text{VI.114})$$

By the arc lemma, the arc integral vanishes as $R \rightarrow \infty$. Therefore

$$I_1 = 0. \quad (\text{VI.115})$$

(2) Let the integrand be $\frac{e^{2ia} - e^{2ix}}{x^2 - a^2}$, then

$$I(a) = \frac{1}{4} \mathcal{R} \left(\int_{-\infty}^{\infty} \frac{e^{2ia} - e^{2ix}}{x^2 - a^2} dx \right) \quad (\text{VI.116})$$

where we note the original integral is even. Consider the principal value integral

$$\begin{aligned} & \text{P.V.} \int_{-\infty}^{\infty} \frac{e^{2ix}}{x^2 - a^2} dx \\ &= 2\pi i \cdot \text{Res}_{z=a} \left[\frac{e^{2iz}}{z^2 - a^2} \right] + \pi i \cdot \text{Res}_{z=-a} \left[\frac{e^{2iz}}{z^2 - a^2} \right] \end{aligned} \quad (\text{VI.117})$$

using a semicircular contour indented at $z = \pm a$. Compute residues:

$$\begin{aligned} \text{Res}_{z=a} \left[\frac{e^{2iz}}{z^2 - a^2} \right] &= \frac{e^{2ia}}{2a}, \\ \text{Res}_{z=-a} \left[\frac{e^{2iz}}{z^2 - a^2} \right] &= -\frac{e^{-2ia}}{2a}. \end{aligned} \quad (\text{VI.118})$$

Thus

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{2ix}}{x^2 - a^2} dx = 2\pi i \cdot \frac{e^{2ia}}{2a} + \pi i \left(-\frac{e^{-2ia}}{2a} \right). \quad (\text{VI.119})$$

Similarly,

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{2ia}}{x^2 - a^2} dx = 0. \quad (\text{VI.120})$$

Therefore

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{e^{2ia} - e^{2ix}}{x^2 - a^2} dx &= -\text{P.V.} \int_{-\infty}^{\infty} \frac{e^{2ix}}{x^2 - a^2} dx \\ &= -\frac{\pi i}{a} (2e^{2ia} - e^{-2ia}). \end{aligned} \quad (\text{VI.121})$$

Taking the real part and dividing by 4 gives

$$I_2 = \frac{1}{4} \cdot \frac{\pi}{a} \sin(2a) = \frac{\pi \sin 2a}{4a}. \quad (\text{VI.122})$$

(3) Let

$$I(\alpha) = \int_0^{\infty} \frac{\alpha x - \sin(\alpha x)}{x^3(1+x^2)} dx, \quad (\text{VI.123})$$

so that $I_3 = I(1)$. Let the auxiliary integrand be $\frac{e^{i\alpha z}}{z(1+z^2)}$, then

$$I''(\alpha) = \int_0^{\infty} \frac{\sin(\alpha x)}{x(1+x^2)} dx = \text{Im} \left(\int_0^{\infty} \frac{e^{i\alpha x}}{x(1+x^2)} dx \right). \quad (\text{VI.124})$$

Consider a semicircular integration contour of radius R in the upper half-plane, indented at $z = 0$. Then

$$\begin{aligned} \oint_C \frac{e^{i\alpha z}}{z(1+z^2)} dz &= \int_{-R}^{-\varepsilon} \frac{e^{i\alpha x}}{x(1+x^2)} dx \\ &+ \int_{\varepsilon}^R \frac{e^{i\alpha x}}{x(1+x^2)} dx + \int_{\text{small arc}} \frac{e^{i\alpha z}}{z(1+z^2)} dz \\ &+ \int_{\text{large arc}} \frac{e^{i\alpha z}}{z(1+z^2)} dz = 2\pi i \cdot \text{Res}_{z=i} \left[\frac{e^{i\alpha z}}{z(1+z^2)} \right]. \end{aligned} \quad (\text{VI.125})$$

Apply l'Hôpital's rule:

$$\begin{aligned} \text{Res}_{z=i} \left[\frac{e^{i\alpha z}}{z(1+z^2)} \right] &= \lim_{z \rightarrow i} \left((z-i) \cdot \frac{e^{i\alpha z}}{z(1+z^2)} \right) \\ &= \frac{e^{i\alpha i}}{i(2i)} = -\frac{e^{-\alpha}}{2}. \end{aligned} \quad (\text{VI.126})$$

The small arc around $z = 0$ gives $-i\pi$, and the large arc vanishes by Jordan's lemma. Thus

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{e^{i\alpha x}}{x(1+x^2)} dx &= i\pi(1 - e^{-\alpha}), \\ \int_0^{\infty} \frac{\sin(\alpha x)}{x(1+x^2)} dx &= \frac{\pi}{2}(1 - e^{-\alpha}). \end{aligned} \quad (\text{VI.127})$$

Hence $I''(\alpha) = \frac{\pi}{2}(1 - e^{-\alpha})$. Integrating twice with $I(0) = I'(0) = 0$ gives

$$\begin{aligned} I'(\alpha) &= \frac{\pi}{2}\alpha + \frac{\pi}{2}e^{-\alpha} - \frac{\pi}{2}, \\ I(\alpha) &= \frac{\pi}{4}\alpha^2 - \frac{\pi}{2}\alpha - \frac{\pi}{2}e^{-\alpha} + \frac{\pi}{2}. \end{aligned} \quad (\text{VI.128})$$

For $\alpha = 1$,

$$I = \frac{\pi}{4} \left(1 - \frac{2}{e} \right). \quad (\text{VI.129})$$

(4) Take the rectangular contour extending infinitely in both directions with height 2π , while removing the points $z = 0$ and $z = 2\pi i$. We obtain

$$\begin{aligned} \oint_{C_R} \frac{e^{pz}}{1-e^z} dz &= \int_{-R}^{-\varepsilon_1} \frac{e^{px}}{1-e^x} dx + \int_{C_{\varepsilon_1}} \frac{e^{pz}}{1-e^z} dz \\ &+ \int_{\varepsilon_1}^R \frac{e^{px}}{1-e^x} dx + \int_0^{2\pi} \frac{e^{p(R+iy)}}{1-e^{R+iy}} i dy \\ &+ \int_R^{-\varepsilon_2} \frac{e^{p(x+2\pi i)}}{1-e^{x+2\pi i}} dx + \int_{C_{\varepsilon_2}} \frac{e^{pz}}{1-e^z} dz \\ &+ \int_{-\varepsilon_2}^{-R} \frac{e^{p(x+2\pi i)}}{1-e^{x+2\pi i}} dx \\ &+ \int_{2\pi}^0 \frac{e^{p(-R+iy)}}{1-e^{-R+iy}} i dy, \end{aligned} \quad (\text{VI.130})$$

Note that

$$\lim_{R \rightarrow \infty} \frac{e^{p(R+iy)}}{1-e^{R+iy}} = 0, \quad (\text{VI.131})$$

(VI.130) equals to

$$(1 - e^{2p\pi i}) \int_{-\infty}^{\infty} \frac{e^{px}}{1 - e^x} dx = \pi i \left(\operatorname{Res}_{z=0} \left[\frac{e^{pz}}{1 - e^z} \right] + \operatorname{Res}_{z=2\pi i} \left[\frac{e^{pz}}{1 - e^z} \right] \right) \quad (\text{VI.132})$$

(Suppose contour goes clockwise around $z = 0$ and $z = 2\pi i$). By definition,

$$\begin{aligned} \operatorname{Res}_{z=0} \left[\frac{e^{pz}}{1 - e^z} \right] &= -1, \\ \operatorname{Res}_{z=2\pi i} \left[\frac{e^{pz}}{1 - e^z} \right] &= -e^{2p\pi i}. \end{aligned} \quad (\text{VI.133})$$

Thus we obtain

$$\int_{-\infty}^{\infty} \frac{e^{px}}{1 - e^x} dx = \pi \cot p\pi. \quad (\text{VI.134})$$

Similarly,

$$\int_{-\infty}^{\infty} \frac{e^{qx}}{1 - e^x} dx = \pi \cot q\pi. \quad (\text{VI.135})$$

So $I_4 = \pi (\cot p\pi - \cot q\pi)$.

9. Calculate the value of the following integrals.

- (1) Let the integrand be $\frac{z^{s-1}}{1-z}$, where $z^{s-1} = e^{(s-1)\log z}$ with the principal branch $-\pi < \arg z < \pi$, then

$$I = \text{v.p.} \int_0^{\infty} \frac{x^{s-1}}{1-x} dx. \quad (\text{VI.136})$$

Consider a keyhole contour avoiding the positive real axis and the pole at $z = 1$. Then

$$\begin{aligned} \oint_{\Gamma} \frac{z^{s-1}}{1-z} dz &= \int_0^{1-\epsilon} \frac{x^{s-1}}{1-x} dx + \int_{1+\epsilon}^{\infty} \frac{x^{s-1}}{1-x} dx \\ &+ \int_{C_{\epsilon,1}} \frac{z^{s-1}}{1-z} dz + \int_{C_R} \frac{z^{s-1}}{1-z} dz = 0, \end{aligned} \quad (\text{VI.137})$$

since no poles are enclosed. Applying the transformation $x = 1/t$ to the second real-axis integral, we obtain

$$\begin{aligned} \text{v.p.} \int_0^{\infty} \frac{x^{s-1}}{1-x} dx \\ = \int_0^1 \frac{x^{s-1} - x^{-s}}{1-x} dx. \end{aligned} \quad (\text{VI.138})$$

Using the digamma function identity $\psi(1-s) - \psi(s) = \pi \cot(\pi s)$ (see appendix), we find

$$\int_0^1 \frac{x^{s-1} - x^{-s}}{1-x} dx = \pi \cot(\pi s). \quad (\text{VI.139})$$

Thus

$$I = \text{v.p.} \int_0^{\infty} \frac{x^{s-1}}{1-x} dx = \pi \cot(\pi s). \quad (\text{VI.140})$$

Appendix: A standard integral representation is:

$$\int_0^1 \frac{1 - x^{a-1}}{1-x} dx = \psi(a) + \gamma, \quad \Re(a) > 0. \quad (\text{VI.141})$$

Proof: Expand $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, integrate termwise:

$$\begin{aligned} \int_0^1 \frac{1 - x^{a-1}}{1-x} dx &= \sum_{n=0}^{\infty} \int_0^1 (x^n - x^{n+a-1}) dx \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+a} \right), \end{aligned} \quad (\text{VI.142})$$

and the digamma function has the series representation

$$\psi(a) = -\gamma + \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+a} \right). \quad (\text{VI.143})$$

- (2) Let $x^2 = t$, then

$$I_2 = \int_0^{\infty} \frac{1}{2} t^{(s-1)/2} (1+t)^{-2} dt. \quad (\text{VI.144})$$

Again let $\frac{1}{1+t} = u$, thus we obtain

$$\begin{aligned} I_2 &= \frac{1}{2} \int_0^1 (1-u)^{(s-1)/2} u^{(1-s)/2} du \\ &= \frac{1}{2} \Gamma\left(\frac{s+1}{2}\right) \Gamma\left(\frac{3-s}{2}\right). \end{aligned} \quad (\text{VI.145})$$

Note that

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}, \quad (\text{VI.146})$$

thus

$$I_2 = \frac{1}{2} \frac{s-1}{2} \Gamma\left(\frac{s-1}{2}\right) \Gamma\left(\frac{3-s}{2}\right) = \frac{\pi(1-s)}{4 \cos\left(\frac{\pi}{2}s\right)}. \quad (\text{VI.147})$$

- (3) Let us first calculate the integral

$$I(\alpha) = \int_0^{\infty} \frac{x^{\alpha-1}}{1+x} dx. \quad (\text{VI.148})$$

Let $u = \frac{1}{1+x}$, then the integral becomes

$$I(\alpha) = \int_0^1 u^{-\alpha} (1-u)^{\alpha-1} du = B(1-\alpha, \alpha) = \frac{\pi}{\sin \pi \alpha}. \quad (\text{VI.149})$$

Differentiate both sides of the equation with respect to α , we obtain

$$I_3 = I'(\alpha) = -\frac{\pi^2 \cos(\pi \alpha)}{\sin^2(\pi \alpha)}. \quad (\text{VI.150})$$

- (4) Lemma 1:

$$\int_0^{\infty} \frac{\ln^2 x}{1+x^2} dx = \frac{\pi^3}{8}. \quad (\text{VI.151})$$

Proof: Let

$$F(\alpha) = \int_0^{\infty} \frac{x^{\alpha}}{1+x^2} dx. \quad (\text{VI.152})$$

Using the same trick as (3), we obtain

$$F''(\alpha) = \int_0^\infty \frac{x^\alpha \ln^2 x}{1+x^2} dx. \quad (\text{VI.153})$$

To evaluate $F(\alpha)$, let $u = \frac{1}{1+x^2}$, thus

$$\begin{aligned} F(\alpha) &= \int_0^1 \frac{1}{2} (1-u)^{\alpha/2} u^{-\alpha/2-1} du = \frac{1}{2} B\left(\frac{\alpha}{2} + 1, -\frac{\alpha}{2}\right) \\ &= \frac{1}{2} \frac{\pi}{\sin\left(\frac{\pi}{2}\alpha\right)}. \end{aligned} \quad (\text{VI.154})$$

Differentiate both sides of the equation with respect to α , we obtain

$$F''(\alpha) = \frac{\pi^2}{4} \cdot \frac{\frac{\pi}{2} \sin^3\left(\frac{\pi}{2}\alpha\right) + \pi \sin\left(\frac{\pi}{2}\alpha\right) \cos^2\left(\frac{\pi}{2}\alpha\right)}{\sin^4\left(\frac{\pi}{2}\alpha\right)}. \quad (\text{VI.155})$$

Thus

$$\int_0^\infty \frac{\ln^2 x}{1+x^2} dx = F''(1) = \frac{\pi^3}{8}. \quad (\text{VI.156})$$

□

Lemma 2:

$$\int_0^\infty \frac{\ln x}{x^2+1} dx = 0. \quad (\text{VI.157})$$

Proof:

$$\int_0^\infty \frac{\ln x}{x^2+1} dx = F'(1) = 0. \quad (\text{VI.158})$$

□

Now let $x = at$, then

$$\begin{aligned} I_4 &= \frac{1}{a} \left(\int_0^\infty \frac{\ln^2 a}{t^2+1} dt + \int_0^\infty \frac{2 \ln a \ln t}{t^2+1} dt \right. \\ &\quad \left. + \int_0^\infty \frac{\ln^2 t}{t^2+1} dt \right) = \frac{\pi^3}{8a} + \frac{\pi \ln^2 a}{2a}. \end{aligned} \quad (\text{VI.159})$$

10. In the early universe, the energy density of neutrinos ν is given by

$$\rho_\nu = \frac{4\pi}{h^3} \int_0^\infty \frac{x^3}{e^{x/kT} + 1} dx \quad (\text{VI.160})$$

with h and kT being known constants. Compute this integral.

Let $t = x/(kT)$, so that $x = kT t$ and $dx = kT dt$; then

$$\rho_\nu = \frac{4\pi}{h^3} \int_0^\infty \frac{(kT)^3 t^3}{e^t + 1} kT dt = \frac{4\pi}{h^3} (kT)^4 \int_0^\infty \frac{t^3}{e^t + 1} dt. \quad (\text{VI.161})$$

Define $I \equiv \int_0^\infty \frac{t^3}{e^t + 1} dt$. Using the expansion $\frac{1}{e^t + 1} = \sum_{n=1}^\infty (-1)^{n-1} e^{-nt}$ for $t > 0$, we obtain

$$I = \sum_{n=1}^\infty (-1)^{n-1} \int_0^\infty t^3 e^{-nt} dt = \sum_{n=1}^\infty (-1)^{n-1} \frac{6}{n^4}, \quad (\text{VI.162})$$

since $\int_0^\infty t^3 e^{-nt} dt = 3! n^{-4} = 6 n^{-4}$. The alternating series can be expressed via the Riemann zeta function:

$$\sum_{n=1}^\infty \frac{(-1)^{n-1}}{n^4} = \frac{7}{8} \zeta(4), \quad (\text{VI.163})$$

and from the known value $\zeta(4) = \pi^4/90$ (see the Appendix for its derivation) we have

$$I = 6 \cdot \frac{7}{8} \cdot \frac{\pi^4}{90} = \frac{7\pi^4}{120}. \quad (\text{VI.164})$$

Substituting back,

$$\rho_\nu = \frac{4\pi}{h^3} (kT)^4 \cdot \frac{7\pi^4}{120} = \frac{7\pi^5}{30} \frac{(kT)^4}{h^3}. \quad (\text{VI.165})$$

Appendix: Consider the Fourier series of $f(x) = x^2$ on $[-\pi, \pi]$:

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^\infty \frac{(-1)^n}{n^2} \cos(nx). \quad (\text{VI.166})$$

Applying Parseval's identity,

$$\frac{1}{2\pi} \int_{-\pi}^\pi x^4 dx = \frac{\pi^4}{5} = \left(\frac{\pi^2}{3}\right)^2 + \frac{1}{2} \sum_{n=1}^\infty \left[4 \frac{(-1)^n}{n^2}\right]^2, \quad (\text{VI.167})$$

which simplifies to

$$\frac{\pi^4}{5} = \frac{\pi^4}{9} + 8 \sum_{n=1}^\infty \frac{1}{n^4}. \quad (\text{VI.168})$$

Solving for $\zeta(4) \equiv \sum_{n=1}^\infty \frac{1}{n^4}$ gives

$$8\zeta(4) = \frac{\pi^4}{5} - \frac{\pi^4}{9} = \frac{4\pi^4}{45} \implies \zeta(4) = \frac{\pi^4}{90}. \quad (\text{VI.169})$$

Alternatively, using the general formula $\zeta(2k) = \frac{(-1)^{k+1} B_{2k} (2\pi)^{2k}}{2(2k)!}$ with Bernoulli number $B_4 = -\frac{1}{30}$ yields the same result:

$$\zeta(4) = -\frac{(-\frac{1}{30})(2\pi)^4}{2 \cdot 24} = \frac{1}{30} \cdot \frac{16\pi^4}{48} = \frac{\pi^4}{90}. \quad (\text{VI.170})$$

VII. Gamma Function

1. Express the following product in terms of the Gamma function.

(1) It is not hard to show that

$$(2n)!! = 2^n \cdot n! = 2^n \Gamma(n+1). \quad (\text{VII.1})$$

(2) From (1), we obtain

$$(2n-1)!! = \frac{(2n)!}{(2n)!!} = \frac{\Gamma(2n+1)}{2^n \Gamma(n+1)}. \quad (\text{VII.2})$$

(3) Note

$$k + \nu = \frac{\Gamma(k + \nu + 1)}{\Gamma(k + \nu)}, \quad (\text{VII.3})$$

thus

$$\prod_{k=1}^n (k + \nu) = \prod_{k=1}^n \frac{\Gamma(k + \nu + 1)}{\Gamma(k + \nu)} = \frac{\Gamma(n+1 + \nu)}{\Gamma(1 + \nu)}. \quad (\text{VII.4})$$

(4) Note

$$\begin{aligned} k(k+1) - \nu(\nu+1) &= (k+\nu-1)(k-\nu) \\ &= \frac{\Gamma(k+\nu+2)}{\Gamma(k+\nu+1)} \frac{\Gamma(k-\nu+1)}{\Gamma(k-\nu)}, \end{aligned} \quad (\text{VII.5})$$

thus

$$\begin{aligned} \prod_{k=0}^n [k(k+1) - \nu(\nu+1)] &= \prod_{k=1}^n \frac{\Gamma(k+\nu+2)}{\Gamma(k+\nu+1)} \frac{\Gamma(k-\nu+1)}{\Gamma(k-\nu)} \\ &= \frac{\Gamma(n+\nu+2)}{\Gamma(\nu+1)} \frac{\Gamma(n-\nu+1)}{\Gamma(-\nu)}. \end{aligned} \quad (\text{VII.6})$$

2. Calculate the value of the following integrals.

(1) Let the integrand be $z^{-\alpha} e^{-iz} dz$, and we first calculate

$$I_1(\alpha) = \int_0^\infty z^{-\alpha} e^{-iz} dz. \quad (\text{VII.7})$$

Let $iz = u$, then

$$\begin{aligned} I_1(\alpha) &= \exp\left(i\frac{\pi}{2}(\alpha-1)\right) \int_0^\infty u^{-\alpha} e^{-u} du \\ &= \exp\left(i\frac{\pi}{2}(\alpha-1)\right) \Gamma(1-\alpha). \end{aligned} \quad (\text{VII.8})$$

Compare the real and imaginary parts of both sides, we obtain

$$\begin{aligned} \int_0^\infty x^{-\alpha} \sin x dx &= \cos\left(\frac{\pi}{2}\alpha\right) \cdot \Gamma(1-\alpha), \\ \int_0^\infty x^{-\alpha} \cos x dx &= \sin\left(\frac{\pi}{2}\alpha\right) \cdot \Gamma(1-\alpha). \end{aligned} \quad (\text{VII.9})$$

(2) Similarly, let the integrand be $z^{\alpha-1} e^{i(z \sin \theta + iz \cos \theta)}$, and we first calculate

$$I_2(\alpha) = \int_0^\infty z^{\alpha-1} e^{i(z \sin \theta + iz \cos \theta)} dz. \quad (\text{VII.10})$$

Let $x = e^{i\theta} u$, then

$$\begin{aligned} I_2(\alpha) &= \int_0^\infty e^{i(\alpha-1)\theta} \cdot u^{\alpha-1} e^{-u} e^{i\theta} du \\ &= e^{i\alpha\theta} \Gamma(\alpha). \end{aligned} \quad (\text{VII.11})$$

Compare the real and imaginary parts of both sides, we obtain

$$\begin{aligned} \int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \cos(x \sin \theta) d\theta &= \cos(\alpha\theta) \Gamma(\alpha), \\ \int_0^\infty x^{\alpha-1} e^{-x \cos \theta} \sin(x \sin \theta) d\theta &= \sin(\alpha\theta) \Gamma(\alpha). \end{aligned} \quad (\text{VII.12})$$

3. Suppose:

$$\psi(z) = \frac{d}{dz} \ln \Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}, \quad (\text{VII.13})$$

Show that

$$\begin{aligned} (1). \quad \psi(z+1) &= \frac{1}{z} + \psi(z), \\ (2). \quad \psi(z+n) - \psi(z) &= \sum_{k=0}^{n-1} \frac{1}{z+k}. \end{aligned} \quad (\text{VII.14})$$

Proof: Note

$$\Gamma(z+1) = z\Gamma(z), \quad (\text{VII.15})$$

thus

$$\begin{aligned} \psi(z+1) &= \frac{d}{dz} \ln \left(\Gamma(z+1) \right) = \frac{d}{dz} \left(\ln(z) + \ln(\Gamma(z)) \right) \\ &= \psi(z) + \frac{1}{z}. \end{aligned} \quad (\text{VII.16})$$

Using the above formula and recurrence, we obtain

$$\psi(z+n) - \psi(z) = \sum_{k=0}^{n-1} \psi(z+k+1) - \psi(z+k) = \sum_{k=0}^{n-1} \frac{1}{z+k}. \quad (\text{VII.17})$$

□

4. Calculate the integral

$$I = \iiint_V x^{\alpha-1} y^{\beta-1} z^{\gamma-1} dx dy dz, \quad (\text{VII.18})$$

where $\alpha, \beta, \gamma > 0$, V denotes the volume enclosed by the plane $x = 0, y = 0, z = 0$ and the surface $x^p + y^q + z^r = 1$. Note $p, q, r > 0$.

Let

$$u = x^{p/2}, \quad v = y^{q/2}, \quad w = z^{r/2}. \quad (\text{VII.19})$$

Then

$$\left| \frac{D(x, y, z)}{D(u, v, w)} \right| = \frac{8}{pqr} u^{2/p-1} v^{2/q-1} w^{2/r-1}. \quad (\text{VII.20})$$

Thus

$$\begin{aligned} I &= \frac{8}{pqr} \iiint_\Omega u^{2\alpha/p-1} v^{2\beta/q-1} w^{2\gamma/r-1} \\ &= \frac{8}{pqr} \left(\int_0^{\frac{\pi}{2}} \sin^{2\beta/q-1} \theta \cos^{2\alpha/p-1} d\theta \right) \\ &\quad \left(\int_0^{\frac{\pi}{2}} (\sin \phi)^{\frac{2\alpha}{p} + \frac{2\beta}{q} - 1} (\cos \phi)^{2\gamma/r-1} d\phi \right) \\ &\quad \int_0^1 R^{2\alpha/p + 2\beta/q + 2\gamma/r - 1} dR = \frac{1}{pqr} \frac{\Gamma\left(\frac{\alpha}{p}\right) \Gamma\left(\frac{\beta}{q}\right) \Gamma\left(\frac{\gamma}{r}\right)}{\Gamma\left(\frac{\alpha}{p} + \frac{\beta}{q} + \frac{\gamma}{r} + 1\right)}. \end{aligned} \quad (\text{VII.21})$$

where Ω denotes the volume enclosed by

$$u^2 + v^2 + w^2 \leq 1; \quad u, v, w \geq 0, \quad (\text{VII.22})$$

and note that we used polar coordinates

$$u = r \sin \phi \cos \theta, \quad v = r \sin \phi \sin \theta, \quad w = r \cos \phi \quad (\text{VII.23})$$

at the second equality.

5. Compute the area enclosed by the oval $y^2 = \frac{1-x^2}{1+x^2}$.
By symmetry,

$$S = 4 \int_0^1 \sqrt{\frac{1-x^2}{1+x^2}} dx. \quad (\text{VII.24})$$

Let $u = x^2$, then

$$S = 2 \int_0^1 u^{-1/2} (1-u)^{1/2} (1+u)^{-1/2} du. \quad (\text{VII.25})$$

Let $u = \cos \theta$, then

$$\begin{aligned} S &= 2 \int_{\pi/2}^0 (\cos \theta)^{-1/2} \cdot \frac{\sin \theta/2}{\cos \theta/2} (-\sin \theta) d\theta \\ &= 4 \int_0^{\pi/2} \frac{\sin^2(\frac{\theta}{2})}{\sqrt{\cos \theta}} d\theta = 2 \int_0^{\pi/2} \frac{1}{\sqrt{\cos \theta}} - \sqrt{\cos \theta} d\theta \\ &= B\left(\frac{1}{4}, \frac{1}{2}\right) - B\left(\frac{3}{4}, \frac{1}{2}\right). \end{aligned} \quad (\text{VII.26})$$

Note that we have used

$$\int_0^{\pi/2} \sin^{m-1}(x) \cos^{n-1}(x) dx = \frac{1}{2} B\left(\frac{m}{2}, \frac{n}{2}\right) \quad (\text{VII.27})$$

at the last equation.

6. Find the length of the lemniscate $\rho^2 = a^2 \cos 2\theta$.
By symmetry,

$$L = 4 \int_0^{\pi/4} \sqrt{\rho^2 + \left(\frac{d\rho}{d\theta}\right)^2} = 4a \int_0^{\pi/4} \frac{1}{\sqrt{\cos 2\theta}} d\theta. \quad (\text{VII.28})$$

Let $u = 2\theta$ and apply (VII.27), we obtain $L = aB(1/2, 1/4)$.

7. A particle of mass m moves along a straight line under the influence of a gravitational force directed toward the origin. The magnitude of the force is inversely proportional to the distance from the particle to the origin, with proportionality constant k . If the particle starts from rest at $x = L$, find the time required for it to reach the origin.

By definition, $F(x) = \frac{k}{x}$, where x denotes the distance from the particle's position to the origin. Apply the work-energy theorem, it is easy to obtain

$$\int_r^L F(x) dx = \frac{1}{2} mv^2(r), \quad (\text{VII.29})$$

thus

$$v(r) = \sqrt{\frac{2k}{m}} \sqrt{\ln\left(\frac{l}{r}\right)}, \quad r \in (0, L). \quad (\text{VII.30})$$

Then the required time can be written as

$$t = \int_0^L \frac{1}{v(r)} dr = \sqrt{\frac{m}{2k}} \int_0^L \frac{1}{\sqrt{\ln\left(\frac{l}{r}\right)}} dr. \quad (\text{VII.31})$$

Make the substitution $u = \ln\left(\frac{l}{x}\right)$, we obtain

$$t = \sqrt{\frac{m}{2k}} L \int_0^\infty u^{-1/2} e^{-u} du = L \sqrt{\frac{m\pi}{2k}}. \quad (\text{VII.32})$$

8. A simple pendulum oscillates back and forth with length L . The total angular amplitude of the swing is π (i.e., from one extreme to the other). Find the period of the pendulum.

Similarly, we apply the work-energy theorem:

$$mg \sin \theta = \frac{1}{2} mv(\theta)^2. \quad (\text{VII.33})$$

Note $v(\theta) = \dot{\theta}L$, thus

$$\dot{\theta} = \sqrt{\frac{2g}{L}} \sqrt{\sin \theta}. \quad (\text{VII.34})$$

By symmetry,

$$T = 4 \int_0^{\pi/2} \frac{1}{\dot{\theta}} d\theta = 4 \int_0^{\pi/2} \sqrt{\frac{l}{2g}} (\sin \theta)^{-1/2} d\theta. \quad (\text{VII.35})$$

Recall

$$\int_0^{\pi/2} \sin^{m-1}(x) \cos^{n-1}(x) dx = \frac{1}{2} B\left(\frac{m}{2}, \frac{n}{2}\right), \quad (\text{VII.36})$$

we obtain $T = \sqrt{\frac{2L}{g}} B(1/4, 1/2)$.

9. The wave function for Coulomb scattering is $\psi(r, \theta)$. Its value at the origin is

$$\psi(0) = e^{\pi\gamma/2} \Gamma(1+i\gamma), \quad (\text{VII.37})$$

where $\gamma = Z_1 Z_2 e^2 / \hbar v$ is a known positive constant. Find $|\psi(0)|^2$.

By definition,

$$\begin{aligned} |\psi(0)|^2 &= \psi(0) \overline{\psi(0)} = e^{\pi\gamma/2} \Gamma(1+i\gamma) e^{\pi\gamma/2} \overline{\Gamma(1+i\gamma)} \\ &= e^{\pi\gamma} \Gamma(1+i\gamma) \Gamma(1-i\gamma) = e^{\pi\gamma} (-i\gamma) \Gamma(1+i\gamma) \Gamma(-i\gamma) \\ &= e^{\pi\gamma} (-i\gamma) \frac{\pi}{\sin(-i\pi\gamma)} = \frac{2\pi\gamma}{1-e^{-2\pi\gamma}}. \end{aligned} \quad (\text{VII.38})$$

Note we have used the recurrence formula and the reflection formula of the gamma function at the fourth equation.

10. Find the sum of the following series.

(1) Decompose $\frac{1}{n(4n^2-1)}$ as

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{n(4n^2-1)} &= \sum_{n=1}^{\infty} \left(-\frac{1}{n} + \frac{1}{2} \frac{1}{n-\frac{1}{2}} + \frac{1}{2} \frac{1}{n+\frac{1}{2}} \right) \\ &= \sum_{n=1}^{\infty} \left(\sum_{k=1}^m \frac{a_k}{n+\alpha_k} \right). \end{aligned} \quad (\text{VII.39})$$

Note $\sum_{k=1}^n a_k = 0$, thus

$$\begin{aligned} S_N &= \sum_{n=1}^N \left(\sum_{k=1}^m \frac{a_k}{n + \alpha_k} \right) = \sum_{k=1}^m a_k [\psi(\alpha_k + N) - \psi_{\alpha_k}] \\ &= \sum_{k=1}^m a_k [\psi(\alpha_k + N) - \psi_{\alpha_k}] - \ln N \sum_{k=1}^m a_k \end{aligned} \quad (\text{VII.40})$$

Take the limit $N \rightarrow \infty$, and recall

$$\lim_{n \rightarrow \infty} [\psi(z + n) - \ln n] = 0, \quad (\text{VII.41})$$

we obtain

$$\begin{aligned} S &= \lim_{N \rightarrow \infty} \sum_{k=1}^m a_k [\psi(\alpha_k + N) - \ln N] - \sum_{k=1}^m a_k \psi(\alpha_k) \\ &= - \sum_{k=1}^m a_k \psi(\alpha_k) = 2 \ln 2 - 1. \end{aligned} \quad (\text{VII.42})$$

(2) Decompose the series as

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \frac{1}{(n^2 + 1)^2} &= -1 + 2 \sum_{n=0}^{\infty} \frac{1}{(n^2 + 1)^2} \\ &= -1 + \sum_{n=0}^{\infty} \left(\frac{i}{2} \frac{1}{n + i} - \frac{1}{(n + i)^2} \right) \\ &\quad + \left(-\frac{i}{2} \frac{1}{n - i} - \frac{1}{(n - i)^2} \right) \\ &= -1 + \sum_{n=0}^{\infty} \sum_{k=1}^l \left[\frac{b_{1k}}{n + \beta_k} + \frac{b_{2k}}{(n + \beta_k)^2} \right]. \end{aligned} \quad (\text{VII.43})$$

Differentiating both sides of equation (VII.17) with respect to z , we obtain

$$\psi'(z + n) - \psi'(z) = - \sum_{k=0}^{n-1} \frac{1}{(z + k)^2}. \quad (\text{VII.44})$$

Use the similar trick as (1), we obtain

$$\begin{aligned} S &= -1 - \sum_{k=1}^l [b_{1k} \psi(\beta_k) - b_{2k} \psi'(\beta_k)] \\ &= -1 + \frac{i}{2} [\psi(i) - \psi(-i)] - [\psi'(i) + \psi'(-i)]. \end{aligned} \quad (\text{VII.45})$$

Note

$$\psi(z) - \psi(-z) = -\frac{1}{z} - \pi \cot \pi z, \quad (\text{VII.46})$$

and differentiating both side of the equation with respect to z , we obtain

$$\psi'(z) + \psi'(-z) = \frac{1}{z^2} + \frac{\pi^2}{\sin^2(\pi z)}. \quad (\text{VII.47})$$

thus

$$S_2 = \frac{\pi \cosh \pi}{2 \sinh \pi} + \frac{\pi^2 \cosh^2 \pi}{2 \sinh^2 \pi}. \quad (\text{VII.48})$$

11. Given that $f(x) = f(\pi \pm x)$, use the results of Problems 11 and 12 in Chapter 5 to derive the transformation formula for the following infinite integral, following the method of §7.5.

(1) Recall that

$$\begin{aligned} \frac{\sinh 2\gamma}{\gamma} \frac{1}{\cosh 2\gamma - \cos 2z} &= \frac{1}{z^2 + \gamma^2} \\ &+ \sum_{n=1}^{\infty} \left[\frac{1}{(z - n\pi)^2 + \gamma^2} + \frac{1}{(z + n\pi)^2 + \gamma^2} \right] \end{aligned} \quad (\text{VII.49})$$

Let $g(x) = f(x)(\cosh 2\gamma - \cos 2x)$, then

$$\begin{aligned} \int_0^{\infty} \frac{g(x)}{x^2 + \gamma^2} dx &= \int_0^{\pi/2} \frac{g(x)}{x^2 + \gamma^2} dx + \\ &\sum_{m=1}^{\infty} \left[\int_{m\pi}^{(m-1/2)\pi} \frac{g(x)}{x^2 + \gamma^2} dx + \int_{m\pi}^{(m+1/2)\pi} \frac{g(x)}{x^2 + \gamma^2} dx \right] \\ &= \int_0^{\pi/2} \frac{g(x)}{x^2 + \gamma^2} dx + \\ &\sum_{m=1}^{\infty} \int_0^{\pi/2} \left[\frac{g(m\pi - x)}{(x - m\pi)^2 + \gamma^2} + \frac{g(m\pi + x)}{(x + m\pi)^2 + \gamma^2} \right] dx \\ &= \int_0^{\pi/2} \left[\frac{1}{x^2 + \gamma^2} + \right. \\ &\left. \sum_{m=1}^{\infty} \frac{1}{(x - m\pi)^2 + \gamma^2} + \frac{1}{(x + m\pi)^2 + \gamma^2} \right] g(x) dx \\ &= \int_0^{\pi/2} \frac{\sinh 2\gamma}{\gamma} \frac{1}{\cosh 2\gamma - \cos 2x} g(x) dx \\ &= \frac{\sinh 2\gamma}{\gamma} \int_0^{\pi/2} f(x) dx. \end{aligned} \quad (\text{VII.50})$$

(2) Recall that

$$\begin{aligned} \frac{2 \sinh \gamma}{\gamma} \frac{\cos z}{\cosh 2\gamma - \cos 2z} &= \frac{1}{z^2 + \gamma^2} + \\ &\sum_{n=1}^{\infty} (-1)^n \left[\frac{1}{(z - n\pi)^2 + \gamma^2} + \frac{1}{(z + n\pi)^2 + \gamma^2} \right]. \end{aligned} \quad (\text{VII.51})$$

Let $g(x) = \frac{f(x)(\cosh 2\gamma - \cos 2x)}{\sin x}$, then

$$\begin{aligned}
\text{v.p.} \int_0^\infty \frac{g(x)}{x^2 + \gamma^2} dx &= \int_0^{\pi/2} \frac{g(x)}{x^2 + \gamma^2} dx + \\
&\sum_{m=1}^\infty \left[\int_{m\pi}^{(m-1/2)\pi} \frac{g(x)}{x^2 + \gamma^2} dx + \int_{m\pi}^{(m+1/2)\pi} \frac{g(x)}{x^2 + \gamma^2} dx \right] \\
&= \int_0^{\pi/2} \frac{g(x)}{x^2 + \gamma^2} dx + \\
&\sum_{m=1}^\infty \int_0^{\pi/2} g(x) \left[\frac{g(m\pi - x)}{(x - m\pi)^2 + \gamma^2} + \frac{g(m\pi + x)}{(x + m\pi)^2 + \gamma^2} \right] dx \\
&= \int_0^{\pi/2} \left[\frac{1}{x^2 + \gamma^2} + \right. \\
&\left. \sum_{m=1}^\infty (-1)^m \left(\frac{1}{(x - m\pi)^2 + \gamma^2} + \frac{1}{(x + m\pi)^2 + \gamma^2} \right) \right] \\
&= \int_0^{\pi/2} \frac{2 \sinh 2\gamma}{\gamma} \frac{\cos x}{\cosh 2\gamma - \cos 2x} g(x) dx \\
&= \frac{2 \sinh 2\gamma}{\gamma} \int_0^{\pi/2} f(x) dx.
\end{aligned} \tag{VII.52}$$

(3) Recall that

$$\begin{aligned}
\frac{2 \sin z \cosh \gamma}{\cosh 2\gamma - \cos 2z} &= \frac{z}{z^2 + \gamma^2} + \\
&\sum_{n=1}^\infty (-1)^n \left[\frac{z - n\pi}{(z - n\pi)^2 + \gamma^2} + \frac{z + n\pi}{(z + n\pi)^2 + \gamma^2} \right].
\end{aligned} \tag{VII.53}$$

Let $g(x) = \frac{f(x)(\cosh 2\gamma - \cos 2x)}{\sin x}$, and apply the same trick as (1)(2), we obtain

$$\begin{aligned}
I_3 &= \int_0^{\pi/2} g(x) \left[\frac{x}{x^2 + \gamma^2} \right. \\
&+ \sum_{m=1}^\infty (-1)^m \left(\frac{x - m\pi}{(x - m\pi)^2 + \gamma^2} + \frac{x + m\pi}{(x + m\pi)^2 + \gamma^2} \right) \Big] \\
&= \int_0^{\pi/2} g(x) \frac{2 \sin x \cosh \gamma}{\cosh 2\gamma - \cos 2x} dx \\
&= 2 \cosh \gamma \int_0^{\pi/2} f(x) dx.
\end{aligned} \tag{VII.54}$$

(4) Recall that

$$\begin{aligned}
\frac{\sin 2z}{\cosh 2\gamma - \cos 2z} &= \frac{z}{z^2 + \gamma^2} + \\
&\sum_{n=1}^\infty \left[\frac{z - n\pi}{(z - n\pi)^2 + \gamma^2} + \frac{z + n\pi}{(z + n\pi)^2 + \gamma^2} \right].
\end{aligned} \tag{VII.55}$$

Let $g(x) = \frac{f(x)(\cosh 2\gamma - \cos 2x)}{\sin 2x}$, and apply the same method as (3), it is easy to obtain

$$I_4 = \int_0^{\pi/2} g(x) \frac{\sin 2x}{\cosh 2\gamma - \cos 2x} = \int_0^{\pi/2} f(x) dx. \tag{VII.56}$$

12. Apply the formula from Problem 11 to evaluate the following integral.

Let $f(x) \equiv 1$, it is easy to obtain

$$\begin{aligned}
&\int_0^\infty \frac{\cosh 2\gamma - \cos 2x}{x^2 + \gamma^2} dx = \frac{\pi \sinh 2\gamma}{2\gamma}, \\
\text{v.p.} \int_0^\infty \frac{\cosh 2\gamma - \cos 2x}{x^2 + \gamma^2} \frac{1}{\cos x} dx &= \frac{\pi \sinh 2\gamma}{\gamma}, \\
\text{v.p.} \int_0^\infty \frac{\cosh 2\gamma - \cos 2x}{x^2 + \gamma^2} \frac{x}{\sin x} dx &= \pi \cosh \gamma, \\
\text{v.p.} \int_0^\infty \frac{\cosh 2\gamma - \cos 2x}{x^2 + \gamma^2} \frac{x}{\sin 2x} dx &= \frac{\pi}{2}.
\end{aligned} \tag{VII.57}$$

VIII. Power series method for solving second-order linear ordinary differential equations

1. Find a second-order linear ordinary differential equation whose solution is

- (1) $w_1(z) = z, w_2(z) = e^z$.
- (2) $w_1(z) = e^{1/z}, w_2(z) = e^{-2/z}$.
- (3) $w_1(z) = \cos \frac{a}{z}, w_2(z) = \sin \frac{a}{z}$.
- (4) $w_1(z) = \frac{z^2}{z^2 - 1}, w_2(z) = \frac{z}{z^2 - 1}$.

Lemma: If $w_1(z), w_2(z)$ are solutions of second-order linear ordinary differential equations

$$w''(z) + p(z)w'(z) + q(z)w(z) = 0, \tag{VIII.1}$$

then

$$q(z) = \frac{\begin{vmatrix} w_1(z) & w_1''(z) \\ w_2(z) & w_2''(z) \end{vmatrix}}{\begin{vmatrix} w_2(z) & w_2'(z) \\ w_1(z) & w_1'(z) \end{vmatrix}}, \quad q(z) = \frac{\begin{vmatrix} w_1'(z) & w_1''(z) \\ w_2'(z) & w_2''(z) \end{vmatrix}}{\begin{vmatrix} w_1(z) & w_1'(z) \\ w_2(z) & w_2'(z) \end{vmatrix}}. \tag{VIII.2}$$

Proof: Since w_1 and w_2 are solutions of (1), for $i = 1, 2$ we have

$$w_i''(z) = -p(z)w_i'(z) - q(z)w_i(z). \tag{VIII.3}$$

We first verify the second formula in (VIII.2). Substituting (VIII.3) into the numerator gives

$$\begin{aligned}
\begin{vmatrix} w_1' & w_1'' \\ w_2' & w_2'' \end{vmatrix} &= \begin{vmatrix} w_1' & -pw_1' - qw_1 \\ w_2' & -pw_2' - qw_2 \end{vmatrix} \\
&= \begin{vmatrix} w_1' & -pw_1' \\ w_2' & -pw_2' \end{vmatrix} + \begin{vmatrix} w_1' & -qw_1 \\ w_2' & -qw_2 \end{vmatrix} \quad (\text{by linearity}) \\
&= 0 + (-q) \begin{vmatrix} w_1' & w_1 \\ w_2' & w_2 \end{vmatrix}
\end{aligned} \tag{VIII.4}$$

since the first determinant in (VIII.4) has proportional columns. Swapping the columns of the remaining determinant (which changes the sign) yields

$$\begin{vmatrix} w_1' & w_1 \\ w_2' & w_2 \end{vmatrix} = - \begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix}. \tag{VIII.5}$$

Combining (VIII.4) and (VIII.5), we obtain

$$\begin{vmatrix} w_1' & w_1'' \\ w_2' & w_2'' \end{vmatrix} = q \begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix}. \quad (\text{VIII.6})$$

Dividing both sides of (VIII.6) by $\begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix}$ (which is nonzero if w_1 and w_2 are linearly independent) gives the desired formula for $q(z)$:

$$q(z) = \frac{\begin{vmatrix} w_1' & w_1'' \\ w_2' & w_2'' \end{vmatrix}}{\begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix}}. \quad (\text{VIII.7})$$

A similar computation shows that the first expression in (VIII.2) actually represents $p(z)$. Substituting (VIII.3) into that numerator,

$$\begin{aligned} \begin{vmatrix} w_1 & w_1'' \\ w_2 & w_2'' \end{vmatrix} &= \begin{vmatrix} w_1 & -pw_1' - qw_1 \\ w_2 & -pw_2' - qw_2 \end{vmatrix} \\ &= \begin{vmatrix} w_1 & -pw_1' \\ w_2 & -pw_2' \end{vmatrix} + \begin{vmatrix} w_1 & -qw_1 \\ w_2 & -qw_2 \end{vmatrix} \\ &= -p \begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix} + 0. \end{aligned} \quad (\text{VIII.8})$$

Noting that the denominator satisfies

$$\begin{vmatrix} w_2 & w_2' \\ w_1 & w_1' \end{vmatrix} = - \begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix}, \quad (\text{VIII.9})$$

we have

$$\frac{\begin{vmatrix} w_1 & w_1'' \\ w_2 & w_2'' \end{vmatrix}}{\begin{vmatrix} w_2 & w_2' \\ w_1 & w_1' \end{vmatrix}} = \frac{-p \begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix}}{- \begin{vmatrix} w_1 & w_1' \\ w_2 & w_2' \end{vmatrix}} = p(z). \quad (\text{VIII.10})$$

This completes the proof. $\square$

Now let's find the four ODEs based on the lemma.

- (1) We obtain $p(z) = z/(1-z)$, $q(z) = 1/(z-1)$, thus the ODE is

$$w''(z) + \frac{z}{1-z}w'(z) + \frac{1}{z-1} = 0. \quad (\text{VIII.11})$$

- (2) We obtain $p(z) = (2 - \frac{1}{z}) \cdot \frac{1}{z}$, $q(z) = -2/z^4$, thus the ODE is

$$z^4 w''(z) + (2z^3 - z^2)w'(z) - 2w = 0. \quad (\text{VIII.12})$$

- (3) We obtain $p(z) = \frac{2}{z}$, $q(z) = a^2/z^4$, thus the ODE is

$$w'' + \frac{2}{z}w' + \frac{a^2}{z^4} = 0. \quad (\text{VIII.13})$$

- (4) We obtain $p(z) = \frac{2(1+z^2)}{z(1-z^2)}$, $q(z) = \frac{2}{z^2(1-z^2)}$, thus the ODE is

$$w'' + \frac{2(1+z^2)}{z(1-z^2)}w' + \frac{2}{z^2(1-z^2)} = 0. \quad (\text{VIII.14})$$

2. Find two power series solutions of the following equation in a neighborhood of $z = 0$.

- (1) Set $w(z) = \sum_{n=0}^{\infty} c_n z^n$, then

$$w'' - z^2 w = 2c_2 + 6c_3 z + \sum_{n=2}^{\infty} [(n+2)(n+1)c_{n+2} - c_{n-2}] z^n. \quad (\text{VIII.15})$$

Thus $c_2 = c_3 = 0$. Note

i)

$$c_{4n} = \frac{\Gamma(\frac{3}{4})}{2^{4n} \cdot n! \cdot \Gamma(n + \frac{3}{4})}, \quad (\text{VIII.16})$$

thus

$$w_1(z) = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{\Gamma(\frac{3}{4})}{\Gamma(n + \frac{3}{4})} \cdot \left(\frac{z}{2}\right)^{4n}. \quad (\text{VIII.17})$$

ii)

$$c_{4n+1} = c_1 \cdot \frac{\Gamma(\frac{5}{4})}{2^{4n} \cdot n! \cdot \Gamma(n + \frac{5}{4})}. \quad (\text{VIII.18})$$

thus

$$w_2(z) = \sum_{n=0}^{\infty} \frac{2\Gamma(\frac{5}{4})}{n! \Gamma(n + \frac{5}{4})} \cdot \left(\frac{z}{2}\right)^{4n+1}. \quad (\text{VIII.19})$$

- (2) Set $w = \sum_{n=0}^{\infty} c_n z^n$, then

$$w'' - zw = 2c_2 + \sum_{n=1}^{\infty} [(n+2)(n+1)c_{n+2} - c_{n-1}] z^n = 0. \quad (\text{VIII.20})$$

Thus $c_2 = 0$. Note

i)

$$c_{3n} = c_0 \cdot \frac{\Gamma(\frac{2}{3})}{3^{3n} \cdot n! \cdot \Gamma(n + \frac{2}{3})}, \quad (\text{VIII.21})$$

therefore

$$w_1(z) = \sum_{n=0}^{\infty} \frac{\Gamma(2/3)}{3^{3n} \cdot n! \cdot \Gamma(n + 2/3)} \cdot z^{3n}. \quad (\text{VIII.22})$$

ii)

$$c_{3n+1} = c_1 \cdot \frac{\Gamma(4/3)}{3^{2n} \cdot n! \cdot \Gamma(n + 4/3)}, \quad (\text{VIII.23})$$

therefore

$$w_2(z) = \sum_{n=0}^{\infty} \frac{\Gamma(4/3)}{3^{2n} \cdot n! \cdot \Gamma(n + 4/3)} \cdot z^{3n+1}. \quad (\text{VIII.24})$$

- (3) Set $w = \sum_{n=0}^{\infty} c_n z^n$, then

$$(z^2 - 1)w'' + zw' - w = \sum_{n=0}^{\infty} (n+1)[(n-1)c_n - (n+2)c_{n+2}] z^n = 0. \quad (\text{VIII.25})$$

Note

- i) If n is odd, then $c_n \equiv 0$, $n \geq 3$. Therefore

$$w_1(z) = z. \quad (\text{VIII.26})$$

ii) If n is even, then

$$c_{2n} = -\frac{(2n-3)!!}{2^n \cdot n!} \quad (\text{VIII.27})$$

therefore

$$w_2(z) = 1 - \frac{z^2}{2} - \sum_{n=2}^{\infty} \frac{(2n-3)!!}{2^n \cdot n!} z^{2n} = \sqrt{1-z^2}. \quad (\text{VIII.28})$$

(4) Set $w = \sum_{n=0}^{\infty} c_n z^n$, then

$$\begin{aligned} (z^2 + z + 1)w'' + (2 + 4z)w' + 2w = \\ \sum_{n=0}^{\infty} (n+2)(n+1)[c_{n+2} + c_{n+1} + c_n]z^n = 0 \end{aligned} \quad (\text{VIII.29})$$

thus $c_{n+2} + c_{n+1} + c_n = 0$.

i) If $c_0 = 0$, then

$$c_{3n+1} = 1, \quad c_{3n+2} = -1. \quad (\text{VIII.30})$$

Therefore

$$w_1(z) = \sum_{n=0}^{\infty} z^{3n+1} - \sum_{n=0}^{\infty} z^{3n+2} = \frac{z}{1+z+z^2}. \quad (\text{VIII.31})$$

ii) If $c_1 = 0$, then

$$c_{3n} = 1, \quad c_{3n+2} = -1. \quad (\text{VIII.32})$$

Therefore

$$w_2(z) = \sum_{n=0}^{\infty} z^{3n} - \sum_{n=0}^{\infty} z^{3n+2} = \frac{1+z}{1+z+z^2} \quad (\text{VIII.33})$$

(5) Set $w = \sum_{n=0}^{\infty} c_n z^n$, then

$$\begin{aligned} (1-z^2)w'' + 2w = \\ = \sum_{n=0}^{\infty} (n+1)[(n+2)c_{n+2} - (n-2)c_n]z^n = 0. \end{aligned} \quad (\text{VIII.34})$$

Therefore

i)

$$w_1(z) = z^2 - 1. \quad (\text{VIII.35})$$

ii) If n is odd, let $c_1 = 1$, thus $c_3 = -1/3$. For $n \geq 2$,

$$c_{2n+1} = -\frac{(2n-3)!!}{(2n+1)!!}. \quad (\text{VIII.36})$$

Therefore

$$w_2(z) = z - \frac{1}{3}z^3 - \sum_{n=2}^{\infty} \frac{(2n-3)!!}{(2n+1)!!} z^{2n+1}. \quad (\text{VIII.37})$$

(6) Set $w = \sum_{n=0}^{\infty} c_n z^n$, then

$$\begin{aligned} (1+z^2)w'' + zw' - 4\nu^2 w = \\ \sum_{n=0}^{\infty} [(n+2)(n+1)c_{n+2} + (n-2\nu)(n+2\nu)c_n]z^n = 0. \end{aligned} \quad (\text{VIII.38})$$

Note

i)

$$c_{2n} = c_0 \frac{(-1)^n \cdot 2^{2n} \Gamma(n-\nu) \Gamma(n+\nu)}{(2n)! \Gamma(-\nu) \Gamma(\nu)}, \quad (\text{VIII.39})$$

therefore

$$w_1(z) = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n \cdot 2^{2n} \Gamma(n-\nu) \Gamma(n+\nu)}{(2n)! \Gamma(-\nu) \Gamma(\nu)} z^{2n}. \quad (\text{VIII.40})$$

ii)

$$\begin{aligned} c_{2n+1} = c_1 \cdot (-1)^n \frac{2^{2n} \Gamma(n+\frac{1}{2}-\nu)}{\Gamma(1/2-\nu)} \cdot \frac{\Gamma(n+\frac{1}{2}+\nu)}{\Gamma(1/2+\nu)} \cdot \\ \frac{1}{(2n+1)!} \end{aligned} \quad (\text{VIII.41})$$

therefore

$$\begin{aligned} w_2(z) = \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n} \Gamma(n+\frac{1}{2}-\nu)}{\Gamma(1/2-\nu)} \cdot \frac{\Gamma(n+\frac{1}{2}+\nu)}{\Gamma(1/2+\nu)} \cdot \\ \frac{1}{(2n+1)!} z^{2n+1}. \end{aligned} \quad (\text{VIII.42})$$

3. Find two power series solutions of the following equation in a neighborhood of $z = 0$.

(1) Note $z = 0$ is a regular singular point of the equation, and $a_0 = 1$, $b_0 = -1$. By solving the indicial equation

$$\rho(\rho-1) + a_0\rho + b_0 = 0, \quad (\text{VIII.43})$$

we obtain two exponents $\rho_1 = 1$, $\rho_2 = -1$. Set $w(z) = \sum_{n=0}^{\infty} c_n z^{n+\rho}$, then

$$\begin{aligned} c_0(\rho+1)(\rho-1)z^\rho + \\ \sum_{n=1}^{\infty} [c_n(n+\rho-1)(n+\rho+1) - c_{n-1}(n+\rho)^2]z^{n+\rho} = 0. \end{aligned} \quad (\text{VIII.44})$$

Then

i) If $\rho = 1$, we obtain

$$c_n = \frac{2(n+1)}{n+2} z^{n+1}. \quad (\text{VIII.45})$$

Therefore

$$w_1(z) = \sum_{n=0}^{\infty} \frac{2(n+1)}{n+2} z^{n+1} z^{n+1}. \quad (\text{VIII.46})$$

ii) If $\rho = -1$, we obtain $c_n \equiv 0$, $n \geq 1$, therefore

$$w_2(z) = \frac{1}{z}. \quad (\text{VIII.47})$$

(2) Note $z = 0$ is a regular singular point of the equation, and $a_0 = -5/3$, $b_0 = 7/9$. By solving the

indicial equation, we obtain $\rho_1 = 1/3$, $\rho_2 = 7/3$.

Set $w(z) = \sum_{n=0}^{\infty} c_n z^{n+\rho}$, then

$$\sum_{n=0}^{\infty} \left[c_{n+4} \left(n + \rho + \frac{11}{3} \right) \left(n + \rho + \frac{5}{3} \right) + 4c_n \right] z^{n+\rho} = 0. \quad (\text{VIII.48})$$

Thus

i) If $\rho = 1/3$,

$$c_{4n} = c_0 \cdot (-1)^n \cdot \frac{1}{(2n)!}, \quad (\text{VIII.49})$$

therefore

$$w_1(z) = z^{1/3} \sum_{n=0}^{\infty} (-1)^n \cdot \frac{1}{(2n)!} \cdot z^{4n}. \quad (\text{VIII.50})$$

ii) If $\rho = 7/3$,

$$c_{4n} = c_0 \cdot \frac{(-1)^n}{(2n+1)!}, \quad (\text{VIII.51})$$

therefore

$$w_2(z) = z^{7/3} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{4n}. \quad (\text{VIII.52})$$

(3) Note $z = 0$ is a regular singular point of the equation, and $a_0 = -1$, $b_0 = 0$. By solving the indicial equation, we obtain $\rho_1 = 0$, $\rho_2 = 2$. Set $w(z) = \sum_{n=0}^{\infty} c_n z^{n+\rho}$, then

$$c_{n+1}(n+1+\rho)(n+\rho-1) + c_n(n+1+\rho) = 0. \quad (\text{VIII.53})$$

Thus

i) If $\rho = 2$,

$$w_1(z) = z^2 \cdot \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} z^n = z^2 e^{-z}. \quad (\text{VIII.54})$$

ii) In this occasion, set

$$w_2(z) = w_1(z) \ln z + \sum_{n=0}^{\infty} d_n \cdot z^n, \quad (\text{VIII.55})$$

then

$$(n-1)d_{n+1} + d_n = \frac{(-1)^n}{(n-1)!}. \quad (\text{VIII.56})$$

Set $e_n = \frac{d_n}{(n-2)!} \cdot (-1)^n$, then

$$e_{n+1} = e_n - \frac{1}{n-1}. \quad (\text{VIII.57})$$

therefore

$$w_2(z) = z^2 e^{-z} \ln z - 1 - z - \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} [\psi(n+1) - \psi(1)] \cdot z^{n+2}. \quad (\text{VIII.58})$$

(4) Note $z = 0$ is a regular singular point of the equation, and $\rho_1 = \rho_2 = 1/2$. Set $w(z) = \sum_{n=0}^{\infty} c_n z^{n+\rho}$, then

$$n(n-1)c_n + (n+1)^2 c_{n+1} = 0. \quad (\text{VIII.59})$$

Thus $c_n = 0$, $n \geq 1$. Therefore

$$w_1(z) = z^{1/2}. \quad (\text{VIII.60})$$

Set

$$w_2(z) = w_1(z) \ln z + \sum_{n=0}^{\infty} d_n \cdot z^{n+\rho}, \quad (\text{VIII.61})$$

then

$$d_n(4n^2 - 4n) + d_{n+1} \cdot 4(n+1)^2 = 0, \quad n \geq 2, \quad (\text{VIII.62})$$

$d_1 = 1$, $d_2 = 0$, $d_n \equiv 0$, $n \geq 2$. Therefore

$$w_2(z) = \sqrt{z} \ln z + d_0 \sqrt{z} + z \sqrt{z}. \quad (\text{VIII.63})$$

(5) Note $z = 0$ is a regular singular point of the equation, and $\rho_1 = 2$, $\rho_2 = 3$. Set $w(z) = \sum_{n=0}^{\infty} c_n z^{n+\rho}$, then

$$c_{n+1}(n+\rho-1)(n+\rho-2) = c_n(n+\rho-3)^2. \quad (\text{VIII.64})$$

i) If $\rho = 3$, then $c_n \equiv 0$, $n \geq 1$. Therefore

$$w_1(z) = z^3. \quad (\text{VIII.65})$$

ii) If $\rho = 2$, set

$$w_2(z) = w_1(z) \ln z + \sum_{n=0}^{\infty} d_n \cdot z^{n+\rho}, \quad (\text{VIII.66})$$

therefore $d_0 = 1$, $d_n \equiv 0$, $n \geq 2$. Therefore

$$w_2(z) = z^3 \ln z + z^2 + d_1 z^3. \quad (\text{VIII.67})$$

4. Find two power series solutions of the equation

$$\frac{d^2 u}{dz^2} + \frac{2}{z} \frac{du}{dz} + m^2 u = 0 \quad (\text{VIII.68})$$

in a neighborhood of $z = 0$.

Note $z = 0$ is a regular singular point of the equation, and $a_0 = 2$, $b_0 = 0$. By solving the indicial equation, we obtain two exponents $\rho_1 = 0$, $\rho_2 = -1$. Set $w(z) = \sum_{n=0}^{\infty} c_n z^{n+\rho}$, then

$$\begin{aligned} \frac{d^2 u}{dz^2} + \frac{2}{z} \frac{du}{dz} + m^2 u = \\ (\rho+1)(\rho+2)c_1 z^{\rho-1} + \\ \sum_{n=0}^{\infty} [(\rho+n+2)(\rho+n+3)c_{n+2} + m^2 c_n] z^{n+\rho} = 0. \end{aligned} \quad (\text{VIII.69})$$

Then

a) If n is even,

$$c_{n+2} = -\frac{m^2}{(\rho+n+2)(\rho+n+3)} c_n. \quad (\text{VIII.70})$$

b) If n is odd, $c_n \equiv 0$.

Thus

a) If $\rho = 0$,

$$c_{2n} = (-1)^n \cdot \frac{m^{2n}}{(2n+1)!} c_0, \quad (\text{VIII.71})$$

therefore

$$w_1(z) = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{m^{2n}}{(2n+1)!} z^{2n}. \quad (\text{VIII.72})$$

b) If $\rho = -1$,

$$c_{2n} = (-1)^n \cdot \frac{m^{2n}}{(2n)!} c_0, \quad (\text{VIII.73})$$

therefore

$$w_2(z) = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{m^{2n}}{(2n)!} z^{2n-1}. \quad (\text{VIII.74})$$

5. Find two linearly independent solutions of the equation

$$\frac{d^2 w}{dz^2} + \frac{1}{z} \frac{dw}{dz} - m^2 w = 0 \quad (\text{VIII.75})$$

in a neighborhood of $z = 0$.

Note $z = 0$ is a regular singular point of the equation, and $a_0 = 1$, $b_0 = -m^2$. By solving the indicial equation, we obtain two exponents $\rho_1 = \rho_2 = 0$. Set $w_1(z) = \sum_{n=0}^{\infty} c_n z^n$, then $c_1 = 0$, $c_{n+2} = \frac{m^2}{(n+2)^2} c_n$. Thus if n is odd, $c_n \equiv 0$. Moreover,

$$c_{2n} = c_0 \cdot \left(\frac{m}{2}\right)^{2n} \cdot \frac{1}{(n!)^2}. \quad (\text{VIII.76})$$

Therefore

$$w_1(z) = \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \cdot \left(\frac{mz}{2}\right)^{2n}. \quad (\text{VIII.77})$$

Again set

$$w_2(z) = w_1(z) \ln z + \sum_{n=0}^{\infty} d_n \cdot z^n, \quad (\text{VIII.78})$$

we obtain

$$(n+2)^2 d_{n+2} - m^2 d_n + 2(n+2)c_{n+2} = 0. \quad (\text{VIII.79})$$

Thus

$$d_{2n+1} = d_1 \cdot \frac{m^{2n}}{[(2n+1)!!]^2}. \quad (\text{VIII.80})$$

Moreover,

$$(2n+2)d_{2n+2} - m^2 d_{2n} + 4 \cdot \left(\frac{m}{2}\right)^{2n+2} \cdot \frac{1}{n!(n+1)!} = 0. \quad (\text{VIII.81})$$

This equals to

$$[(n+1)!]^2 \left(\frac{2}{m}\right)^{2n+2} d_{2n+2} = \left(\frac{2}{m}\right)^{2n} [n!]^2 \cdot d_{2n} - \frac{1}{n+1}. \quad (\text{VIII.82})$$

Define e_{2n} as $e_{2n} = \left(\frac{2}{m}\right)^{2n} [n!]^2 \cdot d_{2n}$, then

$$e_{2n} = e_0 - [\psi(n+1) - \psi(1)]. \quad (\text{VIII.83})$$

Therefore

$$w_2(z) = \ln z \cdot \sum_{n=0}^{\infty} \frac{1}{(n!)^2} \cdot \left(\frac{mz}{2}\right)^{2n} + \sum_{n=0}^{\infty} \frac{\psi(n) - \psi(n+1)}{(n!)^2} \cdot \left(\frac{mz}{2}\right)^{2n} + \sum_{n=0}^{\infty} \frac{m^{2n}}{[(2n+1)!!]^2} \cdot z^{2n+1}. \quad (\text{VIII.84})$$

6. In quantum mechanics, when discussing the Stark effect, we encounter the ordinary differential equation

$$\frac{d}{dz} \left(z \frac{dw}{dz} \right) + \left(\frac{1}{2} E z + \alpha - \frac{m^2}{4z} - \frac{1}{3} F z^2 \right) w = 0, \quad (\text{VIII.85})$$

Find the bounded solution of this equation at the point $z = 0$, where α , m^2 , E and F are known constants. You may assume $m \geq 0$. (Hint: Find up to the third term of the series solution.)

Note $z = 0$ is a regular singular point of the equation, and $a_0 = 1$, $b_0 = -m^2/4$. By solving the indicial equation, we obtain two exponents $\rho_1 = m/2$, $\rho_2 = -m/2$. For a bounded solution, we take $\rho = m/2$. Now equate coefficients of $z^{n+\rho-1}$ for $n \geq 0$. The first sum gives $(n+\rho)^2 c_n$, and the $-\frac{m^2}{4z}$ term gives $-\frac{m^2}{4} c_n$, so their combination is

$$[(n+\rho)^2 - \frac{m^2}{4}] c_n = n(n+m) c_n, \quad (\text{VIII.86})$$

since $\rho = m/2$ implies $\rho^2 = m^2/4$ and $(n+\rho)^2 - m^2/4 = n^2 + mn = n(n+m)$.

The remaining terms are shifted in index:

$$\alpha \sum_{n=0}^{\infty} c_n z^{n+\rho} \rightarrow \alpha \sum_{n=1}^{\infty} c_{n-1} z^{n+\rho-1}, \quad (\text{VIII.87})$$

$$\frac{1}{2} E \sum_{n=0}^{\infty} c_n z^{n+\rho+1} \rightarrow \sum_{n=2}^{\infty} \frac{1}{2} E c_{n-2} z^{n+\rho-1}, \quad (\text{VIII.88})$$

$$-\frac{1}{3} F \sum_{n=0}^{\infty} c_n z^{n+\rho+2} \rightarrow \sum_{n=3}^{\infty} -\frac{1}{3} F c_{n-3} z^{n+\rho-1}. \quad (\text{VIII.89})$$

Thus for $n \geq 1$, the recurrence relation is

$$n(n+m) c_n + \alpha c_{n-1} + \frac{1}{2} E c_{n-2} - \frac{1}{3} F c_{n-3} = 0, \quad (\text{VIII.90})$$

with the convention $c_{-1} = c_{-2} = c_{-3} = 0$.

We compute the first few coefficients explicitly.

For $n = 1$:

$$1(1+m) c_1 + \alpha c_0 = 0 \Rightarrow c_1 = -\frac{\alpha}{1+m} c_0. \quad (\text{VIII.91})$$

For $n = 2$:

$$2(2+m) c_2 + \alpha c_1 + \frac{1}{2} E c_0 = 0. \quad (\text{VIII.92})$$

Substituting c_1 :

$$2(2+m)c_2 - \frac{\alpha^2}{1+m}c_0 + \frac{1}{2}Ec_0 = 0, \quad (\text{VIII.93})$$

$$c_2 = \frac{\frac{\alpha^2}{1+m} - \frac{1}{2}E}{2(2+m)}c_0. \quad (\text{VIII.94})$$

For $n = 3$:

$$3(3+m)c_3 + \alpha c_2 + \frac{1}{2}Ec_1 - \frac{1}{3}Fc_0 = 0. \quad (\text{VIII.95})$$

Substituting c_2 and c_1 :

$$\alpha c_2 = \alpha \cdot \frac{\frac{\alpha^2}{1+m} - \frac{1}{2}E}{2(2+m)}c_0, \quad (\text{VIII.96})$$

$$\frac{1}{2}Ec_1 = \frac{1}{2}E \left(-\frac{\alpha}{1+m} \right) c_0 = -\frac{\alpha E}{2(1+m)}c_0. \quad (\text{VIII.97})$$

Thus

$$3(3+m)c_3 = \left[-\alpha \cdot \frac{\frac{\alpha^2}{1+m} - \frac{1}{2}E}{2(2+m)} + \frac{\alpha E}{2(1+m)} + \frac{1}{3}F \right] c_0, \quad (\text{VIII.98})$$

$$c_3 = \frac{\frac{-\alpha^3/(1+m) + \alpha E/2}{2(2+m)} + \frac{\alpha E}{2(1+m)} + \frac{1}{3}F}{3(3+m)}c_0. \quad (\text{VIII.99})$$

Therefore, the bounded solution near $z = 0$ is

$$w(z) = z^{m/2} [c_0 + c_1 z + c_2 z^2 + c_3 z^3 + \mathcal{O}(z^4)], \quad (\text{VIII.100})$$

with c_0 an arbitrary normalization constant, and c_1, c_2, c_3 given above.

7. Find two linearly independent solutions of the equation

$$w'' + \frac{1}{z^2}w' - \frac{2}{z^2}w = 0 \quad (\text{VIII.101})$$

in the neighborhood of $z = 0$.

We first attempt a power series solution, and it is not hard to obtain

$$w_1(z) = 1 + 2z + 2z^2. \quad (\text{VIII.102})$$

Then, following the method of §8.6, let $w(z) = e^{Q(z)} \cdot v(z)$. Thus

$$v''(z) + \left(2Q'(z) + \frac{1}{z^2} \right) v' + \left(Q''(z) + [Q'(z)]^2 + \frac{Q'(z)}{z^2} - \frac{2}{z^2} \right) v = 0. \quad (\text{VIII.103})$$

Set $Q(z) = 1/z$, then

$$v''(z) - \frac{1}{z^2}v'(z) + \left(\frac{2}{z^3} - \frac{2}{z^2} \right) v(z) = 0. \quad (\text{VIII.104})$$

Therefore let $v(z) = z^2$, we obtain

$$w_2(z) = z^2 e^{1/z}. \quad (\text{VIII.105})$$

8. Find a series solution of the equation

$$z^4 \frac{d^2 w}{dz^2} + 2z(z^2 - 1) \frac{dw}{dz} + 2w = 0 \quad (\text{VIII.106})$$

in the neighborhood of the point at infinity.

Set $t = 1/z$, thus

$$\frac{dw}{dz} = -\frac{1}{z^2} \frac{dw}{dt}, \quad (\text{VIII.107})$$

$$\frac{d^2 w}{dz^2} = \frac{d}{dt} \left(\frac{dw}{dz} \right) \frac{dz}{dt} = \frac{2}{z^3} \frac{dw}{dt} + \frac{1}{z^4} \frac{d^2 w}{dt^2}. \quad (\text{VIII.108})$$

Thus the equation transforms into

$$\frac{d^2 w}{dt^2} + 2t \frac{dw}{dt} + 2w = 0. \quad (\text{VIII.109})$$

Let $w = \sum_{n=0}^{\infty} c_n t^n$, then

$$\frac{d^2 w}{dt^2} + 2t \frac{dw}{dt} + 2w = \sum_{n=0}^{\infty} [(n+2)(n+1)c_{n+2} + 2(n+1)c_n] t^n = 0. \quad (\text{VIII.110})$$

Thus $c_{n+2} = -\frac{2}{n+2}c_n$.

a) For even indices:

$$c_{2n} = (-1)^n \cdot \frac{1}{n!} c_0, \quad (\text{VIII.111})$$

therefore

$$w_1(z) = e^{-1/z^2}. \quad (\text{VIII.112})$$

b) For odd indices:

$$c_{2n+1} = (-1)^n \cdot \frac{2^n}{(2n+1)!!}, \quad (\text{VIII.113})$$

therefore

$$w_2(z) = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{2^n}{(2n+1)!!} \left(\frac{1}{z} \right)^{2n+1}. \quad (\text{VIII.114})$$

9. Reduce the following equation to the hypergeometric equation, and thus express the solutions in terms of the Riemann P-symbol.

The standard form of the Gauss hypergeometric equation is

$$z(1-z)y'' + [c - (a+b+1)z]y' - aby = 0, \quad (\text{VIII.115})$$

with regular singular points at $z = 0, 1, \infty$. For the standard hypergeometric equation, the Riemann P -symbol is

$$P_t \left\{ \begin{matrix} 0 & 1 & \infty \\ 0 & 0 & a \\ 1-c & c-a-b & b \end{matrix} \right\} t. \quad (\text{VIII.116})$$

- (1) Let $t = \frac{z+1}{2}$. Then

$$\frac{dw}{dz} = \frac{1}{2} \frac{dw}{dt}, \quad \frac{d^2 w}{dz^2} = \frac{1}{4} \frac{d^2 w}{dt^2}. \quad (\text{VIII.117})$$

Computing the coefficients in the new variable gives us

$$t(1-t)\ddot{w} - \left(t - \frac{1}{2}\right)\dot{w} + w = 0. \quad (\text{VIII.118})$$

Comparing with (VIII.115), we identify

$$c - (a + b + 1)t = -\left(t - \frac{1}{2}\right), \quad (\text{VIII.119})$$

thus in the variable t , the equation is the hypergeometric equation with parameters $a = 1$, $b = -1$, $c = \frac{1}{2}$. Substituting our values, we have

$$w = P \begin{Bmatrix} 0 & 1 & \infty \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & -1 \end{Bmatrix} \frac{z+1}{2}. \quad (\text{VIII.120})$$

(2) Let $t = 1/2(1 + iz)$. Then

$$\frac{dw}{dz} = \frac{i}{2} \frac{dw}{dt}, \quad \frac{d^2w}{dz^2} = -\frac{1}{4} \frac{d^2w}{dt^2}. \quad (\text{VIII.121})$$

Computing the coefficients in the new variable gives us

$$(t - t^2) \frac{d^2w}{dt^2} + (1/2 - t) \frac{dw}{dt} + 4\nu^2 w = 0, \quad (\text{VIII.122})$$

therefore $c = 1/2$, $a = 2\nu$, $b = -2\nu$. Substituting the values, we obtain

$$w = P \begin{Bmatrix} 0 & 1 & \infty \\ 0 & 0 & 2\nu \\ \frac{1}{2} & \frac{1}{2} & -2\nu \end{Bmatrix} \frac{iz+1}{2}. \quad (\text{VIII.123})$$

(3) Let $t = z - 1$. Then

$$\frac{dw}{dz} = \frac{dw}{dt}, \quad \frac{d^2w}{dz^2} = \frac{d^2w}{dt^2}. \quad (\text{VIII.124})$$

Computing the coefficients in the new variable gives us

$$t(1-t) \frac{d^2w}{dt^2} + (2t-1) \frac{dw}{dt} - 2t = 0. \quad (\text{VIII.125})$$

Therefore $a = -2$, $b = c = -1$. Substituting our values, we have

$$w = P \begin{Bmatrix} 0 & 1 & \infty \\ 0 & 0 & -2 \\ 2 & 2 & -1 \end{Bmatrix} z - 1. \quad (\text{VIII.126})$$

(4) Let $t = (z - 1)/2z$. Then

$$\begin{aligned} \frac{dw}{dz} &= \frac{(2t-1)^2}{2} \frac{dw}{dt}, \\ \frac{d^2w}{dz^2} &= \frac{(2t-1)^2}{2} \left[2(2t-1) \frac{dw}{dt} + \frac{1}{2} (2t-1)^2 \frac{d^2w}{dt^2} \right]. \end{aligned} \quad (\text{VIII.127})$$

Computing the coefficients in the new variable gives us

$$t(1-t) \frac{d^2w}{dt^2} + (2-4t) \frac{dw}{dt} - 2w = 0, \quad (\text{VIII.128})$$

thus $a = 2$, $b = 1$, $c = 2$. Substituting our values, we have

$$w = P \begin{Bmatrix} 0 & 1 & \infty \\ 0 & 0 & 2 \\ -1 & -1 & 1 \end{Bmatrix} (z-1)/2z. \quad (\text{VIII.129})$$

10. Using the invariants of the differential equation, reduce the following equation to a hypergeometric equation and find all its solutions (expressed in terms of the Riemann P-symbol).

The invariant of the hypergeometric equation is

$$\begin{aligned} C(z) &= \frac{1}{z} \left\{ -\alpha\beta - \frac{\gamma}{2} [\gamma - (\alpha + \beta + 1)] \right\} \\ &+ \frac{1}{1-z} \left\{ -\alpha\beta - \frac{\gamma}{2} [\gamma - (\alpha + \beta + 1)] \right\} \\ &+ \frac{\gamma(2-\gamma)}{4} \frac{1}{z^2} - \frac{(\gamma - \alpha - \beta)^2 - 1}{4} \frac{1}{(1-z)^2}. \end{aligned} \quad (\text{VIII.130})$$

(1) Note $p(z) = \frac{3z-1}{z(z-1)}$, $q(z) = \frac{1}{z^2(z-1)}$, therefore the invariant of the equation is

$$C_1(z) = q(z) - \frac{1}{2} p'(z) - \frac{1}{4} p^2(z) = -\frac{3}{4z^2}. \quad (\text{VIII.131})$$

thus $\alpha = 1$, $\beta = -1$, $\gamma = -1$. The the equation can be transformed into the hypergeometric equation via the following relation:

$$\begin{aligned} w(z) \cdot \exp \left\{ \frac{1}{2} \int^z \frac{3\varepsilon - 1}{\varepsilon(\varepsilon - 1)} d\varepsilon \right\} \\ = \mathcal{W}(z) \cdot \exp \left\{ \frac{1}{2} \int^z \left[\frac{\gamma}{\zeta} + \frac{\gamma - (\alpha + \beta + 1)}{1 - \zeta} \right] d\zeta \right\}. \end{aligned} \quad (\text{VIII.132})$$

Finally

$$w(z) = \frac{1}{z} \mathcal{W}(z) = \frac{1}{z} P \begin{Bmatrix} 0 & 1 & \infty \\ 0 & 0 & 1 \\ 2 & -1 & -1 \end{Bmatrix} z. \quad (\text{VIII.133})$$

(2) Note $p(z) = 1/z$, $q(z) = \frac{1}{z^2(z-1)}$, therefore the invariant of the equation is

$$C_2(z) = q(z) - \frac{1}{2} p'(z) - \frac{1}{4} p^2(z) = \frac{-1}{z} + \frac{1}{z-1} - \frac{3/4}{z^2}. \quad (\text{VIII.134})$$

Therefore $\alpha = \beta = \gamma = -1$. Then the equation can be transformed into the hypergeometric equation via the following relation:

$$\begin{aligned} w(z) \cdot \exp \left\{ \frac{1}{2} \int^z \frac{1}{\varepsilon} d\varepsilon \right\} \\ = \mathcal{W}(z) \cdot \exp \left\{ \frac{1}{2} \int^z \left[\frac{\gamma}{\zeta} + \frac{\gamma - (\alpha + \beta + 1)}{1 - \zeta} \right] d\zeta \right\}. \end{aligned} \quad (\text{VIII.135})$$

Finally

$$w(z) = \frac{1}{z} \mathcal{W}(z) = \frac{1}{z} P \begin{Bmatrix} 0 & 1 & \infty \\ 0 & 0 & -1 \\ 2 & 1 & -1 \end{Bmatrix} z. \quad (\text{VIII.136})$$

11. Using the invariant of the differential equation, reduce the following equation to the confluent hypergeometric equation, and express the solution in the neighborhood of the regular singular point in terms of confluent hypergeometric functions. The invariant of the confluent hypergeometric equation

$$z \frac{d^2 \mathcal{W}}{dz^2} + (\gamma - z) \frac{d\mathcal{W}}{dz} - \alpha \mathcal{W}(z) = 0 \quad (\text{VIII.137})$$

is

$$C(z) = -\frac{1}{4} + \frac{\gamma - 2\alpha}{2} \frac{1}{z} - \frac{\gamma(\gamma - 2)}{4} \frac{1}{z^2}, \quad (\text{VIII.138})$$

and its solution is

$$\begin{aligned} \mathcal{W}_1(z) &= F(\alpha; \gamma; z) \equiv \sum_{n=0}^{\infty} \frac{1}{n!} \frac{(\alpha)_n}{(\gamma)_n} z^n, \\ \mathcal{W}_2(z) &= z^{1-\gamma} F(\alpha - \gamma + 1; 2 - \gamma; z). \end{aligned} \quad (\text{VIII.139})$$

- (1) By directly comparing the equation with the confluent hypergeometric equation, we obtain $\alpha = -1$, $\gamma = 0$, thus the solutions are

$$w_1(z) = F(-1; 0; z), \quad w_2(z) = zF(0; 2; z). \quad (\text{VIII.140})$$

- (2) Note $p(z) = 1 - 1/z$, $q(z) = 1/z$, therefore the invariant of the equation is

$$C_2(z) = q(z) - \frac{1}{2} p'(z) - \frac{1}{4} p^2(z) = -\frac{1}{4} + \frac{3}{2z} - \frac{3}{4z^2}. \quad (\text{VIII.141})$$

thus $\alpha = -2$, $\gamma = -1$. Then the equation can be transformed into the confluent hypergeometric equation via the following relation:

$$\begin{aligned} w(z) \cdot \exp \left\{ \frac{1}{2} \int^z \left(1 - \frac{1}{\varepsilon} \right) d\varepsilon \right\} \\ = \mathcal{W}(z) \cdot \exp \left\{ \frac{1}{2} \int^z \left(-1 + \frac{\gamma}{\zeta} \right) d\zeta \right\}. \end{aligned} \quad (\text{VIII.142})$$

Finally

$$\begin{aligned} w_1(z) &= e^{-z} F(-2; -1; z), \\ w_2(z) &= e^{-z} z^2 F(0; 3; z). \end{aligned} \quad (\text{VIII.143})$$

12. Find the ordinary differential equation satisfied by the following function, and use the power series method for ordinary differential equations to obtain its power series expansion in the neighborhood of the specified point.

(1) Note

$$\begin{aligned} f'(z) &= -z \cdot (1 + z^2)^{-3/2} \cdot (\sqrt{1 + z^2} + z)^{2\nu} \\ &\quad + \frac{2\nu}{1 + z^2} (\sqrt{1 + z^2} + z)^{2\nu}. \end{aligned} \quad (\text{VIII.144})$$

Similarly,

$$\begin{aligned} g'(z) &= -z(1 + z^2)^{-3/2} (\sqrt{1 + z^2} + z)^{-2\nu} \\ &\quad - \frac{2\nu}{1 + z^2} (\sqrt{1 + z^2} + z)^{-2\nu}. \end{aligned} \quad (\text{VIII.145})$$

Therefore

$$W(f, g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = -\frac{4\nu}{(1 + z^2)^{3/2}}. \quad (\text{VIII.146})$$

Thus

$$W'[f, g] = \frac{d}{dz} W[f, g] = \frac{12\nu z}{(1 + z^2)^{5/2}}. \quad (\text{VIII.147})$$

Moreover,

$$\begin{aligned} f''(z) &= (\sqrt{1 + z^2} + z)^{2\nu} \cdot \\ &\quad \left[\frac{4\nu^2 - 1}{(1 + z^2)^{3/2}} + \frac{3z^2}{(1 + z^2)^{5/2}} - \frac{6\nu z}{(1 + z^2)^2} \right], \end{aligned} \quad (\text{VIII.148})$$

$$\begin{aligned} g''(z) &= (\sqrt{1 + z^2} + z)^{-2\nu} \cdot \\ &\quad \left[\frac{4\nu^2 - 1}{(1 + z^2)^{3/2}} + \frac{3z^2}{(1 + z^2)^{5/2}} + \frac{6\nu z}{(1 + z^2)^2} \right]. \end{aligned} \quad (\text{VIII.149})$$

Therefore

$$W[f', g'] = \frac{4\nu(4\nu^2 - 1)}{(1 + z^2)^{5/2}}. \quad (\text{VIII.150})$$

Following the lemma in 1, we obtain

$$p(z) = \frac{3z}{1 + z^2}, \quad q(z) = \frac{1 - 4\nu^2}{1 + z^2}. \quad (\text{VIII.151})$$

Thus the ordinary differential equation satisfied by $f(z)$ and $g(z)$ is

$$(1 + z^2)w'' + 3zw' + (1 - 4\nu^2)w = 0, \quad (\text{VIII.152})$$

where $z = 0$ is an ordinary point of the equation.

Set $w = \sum_{n=0}^{\infty} c_n z^n$, then

$$\begin{aligned} (1 + z^2)w'' + 3zw' + (1 - 4\nu^2)w &= \\ \sum_{n=0}^{\infty} [(n+2)(n+1)c_{n+2} + (n+1+2\nu)(n+1-2\nu)c_n] z^n &= 0. \end{aligned} \quad (\text{VIII.153})$$

Using the initial conditions of $f(z)$ and $g(z)$, we obtain

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} \frac{1}{n!} \frac{\Gamma(\nu + (1+n)/2)}{\Gamma(\nu + (1-n)/2)}, \\ g(z) &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{\Gamma(\nu + (1+n)/2)}{\Gamma(\nu + (1-n)/2)} (2z)^n. \end{aligned} \quad (\text{VIII.154})$$

- (2) Using the method from Example 8.10 in the text-book, we obtain

$$\begin{aligned} f(z) &= -2\nu \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{\Gamma(2n-2\nu)}{\Gamma(n-2\nu+1)} \cdot \left(\frac{1}{2z}\right)^{2n-2\nu}, \\ g(z) &= 2\nu \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \frac{\Gamma(2n+2\nu)}{\Gamma(n+2\nu+1)} \cdot \left(\frac{1}{2z}\right)^{2n+2\nu}. \end{aligned} \quad (\text{VIII.155})$$

13. Using the Euler transformation, find an integral solution of the equation

$$(1+z^2)w'' + zw' - 4\nu^2 w = 0, \quad (\text{VIII.156})$$

where ν is a known constant.

Note that $p_0(z) = z^2 + 1$, $p_1(z) = z$, $p_2(z) = -4\nu^2$. Then the equation satisfied by μ is

$$a_0\mu(\mu-1) + b_1\mu + c_2 = 0, \quad (\text{VIII.157})$$

where $a_0 = 1$, $b_1 = 1$, $c_2 = -4\nu^2$. Therefore let $\mu = 2\nu$, thus

$$\begin{aligned} v(t) &= A[p_0(t)]^{-\mu} \cdot \exp\left\{-\int^t \frac{p_1(\zeta)}{p_0(\zeta)} d\zeta\right\} \\ &= \frac{1}{(1+t^2)^{2\nu+1/2}}. \end{aligned} \quad (\text{VIII.158})$$

Therefore

$$w(z) = \int \frac{(z-t)^{2\nu}}{(1+t^2)^{2\nu+1/2}} dt, \quad (\text{VIII.159})$$

where the integration path is a figure-eight contour, circling $t = i$ once in the positive direction and circling $t = -i$ once in the negative direction, with the branch point $t = z$ located outside the contour.

IX. Fourier transform

1. Find the Fourier transform of the following functions.

- (1) By definition,

$$\begin{aligned} F_1(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{2} \int_{-a}^a \cos(\omega+1)x + \cos(\omega-1)x dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{\sin(\omega+1)a}{\omega+1} + \frac{\sin(\omega-1)a}{\omega-1} \right]. \end{aligned} \quad (\text{IX.1})$$

- (2) By definition,

$$\begin{aligned} F_2(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\ &= \frac{i}{\sqrt{2\pi}} \cdot \frac{1}{2} \int_{-a}^a \cos(\omega+1)x - \cos(\omega-1)x dx \\ &= \frac{i}{\sqrt{2\pi}} \left[\frac{\sin(\omega+1)a}{\omega+1} - \frac{\sin(\omega-1)a}{\omega-1} \right]. \end{aligned} \quad (\text{IX.2})$$

- (3) By definition,

$$\begin{aligned} F_3(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\int_0^{\infty} e^{-(1+i\omega)x} dx + \int_{-\infty}^0 e^{(1-i\omega)x} dx \right] \\ &= \sqrt{\frac{2}{\pi}} \cdot \frac{1}{1+\omega^2}. \end{aligned} \quad (\text{IX.3})$$

- (4) By definition,

$$\begin{aligned} F_4(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \\ &= \frac{1}{\sqrt{2\pi}} \left[\int_0^{\infty} e^{-(1+i\omega)x} dx - \int_{-\infty}^0 e^{(1-i\omega)x} dx \right] \\ &= \sqrt{\frac{2}{\pi}} \cdot \frac{-i\omega}{1+\omega^2}. \end{aligned} \quad (\text{IX.4})$$

2. Find the Fourier transform of the following functions.

- (1) By definition,

$$F_1(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{1+x^2} e^{-i\omega x} dx. \quad (\text{IX.5})$$

Set $I(\omega) = \int_{-\infty}^{\infty} \frac{1}{1+x^2} e^{-i\omega x} dx$. Let us evaluate it using the residue theorem. The integrand has two simple poles at $x = i$ and $x = -i$. For $\omega < 0$, we close the contour in the upper half-plane (thus $-\omega > 0$, and we can apply the Jordan's Lemma). The only pole enclosed is at $x = i$. The residue is

$$\text{Res}_{x=i} \frac{e^{-i\omega x}}{(x-i)(x+i)} = \frac{e^{\omega}}{-2i}. \quad (\text{IX.6})$$

Hence, by the residue theorem,

$$I(\omega) = 2\pi i \cdot \frac{e^{\omega}}{-2i} = \pi e^{\omega}. \quad (\text{IX.7})$$

For $\omega > 0$, we close the contour in the lower half-plane. Since the contour is traversed clockwise, we pick up a minus sign. The pole enclosed is at $x = i$, with residue

$$\text{Res}_{x=i} \frac{e^{-i\omega x}}{(x-i)(x+i)} = \frac{e^{-\omega}}{2i}. \quad (\text{IX.8})$$

Thus,

$$I(\omega) = -2\pi i \cdot \frac{e^{-\omega}}{-2i} = \pi e^{-\omega}. \quad (\text{IX.9})$$

Combining both cases, we obtain for all real ω

$$I(\omega) = \pi e^{-|\omega|}. \quad (\text{IX.10})$$

Therefore $F_1(\omega) = \sqrt{\frac{\pi}{2}} e^{-|\omega|}$.

- (2) Suppose $F(\omega) = \mathcal{F}f(x)$, then

$$i \frac{d}{d\omega} F(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x f(x) e^{-i\omega x} dx = \mathcal{F}(x f(x)). \quad (\text{IX.11})$$

Therefore

$$F_2(\omega) = i \frac{d}{d\omega} \left\{ \sqrt{\frac{\pi}{2}} e^{-|\omega|} \right\} = -i \sqrt{\frac{\pi}{2}} e^{-|\omega|} \cdot \text{sgn}(\omega). \quad (\text{IX.12})$$

(3) By definition,

$$\begin{aligned} F_3(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\sin x \cdot \cos \omega x}{x} dx \\ &= \frac{1}{\sqrt{2\pi}} \cdot \frac{1}{2} \left[\int_{-\infty}^{\infty} \frac{\sin(\omega+1)x}{x} - \frac{\sin(\omega-1)x}{x} dx \right] \\ &= \sqrt{\frac{\pi}{2}} \cdot [\eta(\omega+1) - \eta(\omega-1)]. \end{aligned} \quad (\text{IX.13})$$

(4) By definition,

$$\begin{aligned} F_4(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\sin x}{x} \cdot \text{sgn}(x) \cdot (-i \sin \omega x) dx \\ &= \frac{i}{\sqrt{2\pi}} \int_0^{\infty} \frac{\cos(\omega+1)x - \cos(\omega-1)x}{x} dx \\ &= \frac{i}{\sqrt{2\pi}} \cdot \ln \left(\left| \frac{1-\omega}{1+\omega} \right| \right). \end{aligned} \quad (\text{IX.14})$$

Note that we used the Frullani integral in the last equality. (See §10.3.)

3. Use either the Parseval's formula or the convolution formula for Fourier transforms to evaluate the following integrals.

(1) It is not hard to obtain

$$\int_{-\infty}^{\infty} \left(\frac{1}{1+x^2} \right)^2 dx = \int_{-\infty}^{\infty} F(\omega) G(-\omega) d\omega = \frac{\pi}{2}, \quad (\text{IX.15})$$

where $f = g = 1/(1+x^2)$, thus $F(\omega) = G(\omega) = \sqrt{\frac{\pi}{2}} e^{-|\omega|}$. Moreover,

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) g(x) e^{-i\omega x} dx &= \int_{-\infty}^{\infty} F(\sigma) G(\omega - \sigma) d\sigma \\ &= \frac{\pi}{2} (1 + |\omega|) e^{-|\omega|}. \end{aligned} \quad (\text{IX.16})$$

(2) Similarly,

$$\int_{-\infty}^{\infty} \left(\frac{x}{1+x^2} \right)^2 dx = \int_{-\infty}^{\infty} F(\omega) G(-\omega) d\omega = \frac{\pi}{2}. \quad (\text{IX.17})$$

where $f(x) = g(x) = x/(1+x^2)$, thus $F(\omega) = G(\omega) = \sqrt{\frac{\pi}{2}} e^{-|\omega|} \text{sgn}(\omega)$. Moreover,

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) g(x) e^{-i\omega x} dx &= \int_{-\infty}^{\infty} F(\sigma) G(\omega - \sigma) d\sigma \\ &= \frac{\pi}{2} (1 - |\omega|) e^{-|\omega|}. \end{aligned} \quad (\text{IX.18})$$

4. Given that the ground state spatial wave function of the hydrogen atom is

$$\psi(\mathbf{r}) = \left(\frac{1}{\pi a_0^3} \right)^{1/2} e^{-r/a_0}, \quad (\text{IX.19})$$

where a_0 is the Bohr radius and $r = |\mathbf{r}|$, compute its Fourier transform

$$g(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \iiint \psi(\mathbf{r}) e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} d^3r, \quad (\text{IX.20})$$

where $\hbar h$ is the reduced Planck constant and $d^3r = dx dy dz$.

Using spherical coordinates with the polar axis along $\mathbf{p}$, we have $\mathbf{p} \cdot \mathbf{r} = pr \cos \theta$ and $d^3r = r^2 \sin \theta dr d\theta d\phi$. Substituting the wave function:

$$\begin{aligned} g(p) &= \frac{1}{(2\pi\hbar)^{3/2}} \left(\frac{1}{\pi a_0^3} \right)^{1/2} \int_0^{\infty} e^{-r/a_0} r^2 dr \\ &\quad \int_0^{\pi} e^{-ipr \cos \theta / \hbar} \sin \theta d\theta \int_0^{2\pi} d\phi. \end{aligned} \quad (\text{IX.21})$$

The ϕ integral gives 2π . The θ integral evaluates to

$$\int_0^{\pi} e^{-ipr \cos \theta / \hbar} \sin \theta d\theta = \frac{2 \sin(pr/\hbar)}{pr/\hbar}. \quad (\text{IX.22})$$

Thus,

$$g(p) = \frac{4\pi}{(2\pi\hbar)^{3/2} \sqrt{\pi a_0^3}} \cdot \frac{\hbar}{p} \int_0^{\infty} e^{-r/a_0} r \sin \left(\frac{pr}{\hbar} \right) dr. \quad (\text{IX.23})$$

The radial integral is standard (see proof):

$$\int_0^{\infty} e^{-r/a_0} r \sin \left(\frac{pr}{\hbar} \right) dr = \frac{2a_0^3(p/\hbar)}{[1 + (a_0 p/\hbar)^2]^2}. \quad (\text{IX.24})$$

Substituting and simplifying yields

$$g(p) = \frac{8\sqrt{\pi} a_0^{3/2}}{(2\pi)^{3/2} \hbar^{3/2}} \cdot \frac{1}{[1 + (a_0 p/\hbar)^2]^2}. \quad (\text{IX.25})$$

Introducing the characteristic momentum $p_0 \equiv \hbar/a_0$, we obtain the final result

$$g(p) = \frac{2^{3/2}}{\pi} \frac{p_0^{5/2}}{(p_0^2 + p^2)^2}. \quad (\text{IX.26})$$

Proof. Set $\alpha = \frac{1}{a_0} > 0$, $\beta = \frac{p}{\hbar} > 0$. Then the integral becomes $I = \int_0^{\infty} e^{-\alpha r} r \sin(\beta r) dr$. Start from the standard integral

$$J(\alpha, \beta) = \int_0^{\infty} e^{-\alpha r} \sin(\beta r) dr = \frac{\beta}{\alpha^2 + \beta^2}. \quad (\text{IX.27})$$

Differentiate J with respect to α :

$$\frac{\partial J}{\partial \alpha} = \int_0^{\infty} (-r) e^{-\alpha r} \sin(\beta r) dr. \quad (\text{IX.28})$$

Hence

$$\int_0^{\infty} r e^{-\alpha r} \sin(\beta r) dr = -\frac{\partial J}{\partial \alpha}. \quad (\text{IX.29})$$

Compute the derivative:

$$\frac{\partial}{\partial \alpha} \left(\frac{\beta}{\alpha^2 + \beta^2} \right) = -\frac{2\alpha\beta}{(\alpha^2 + \beta^2)^2}. \quad (\text{IX.30})$$

Thus

$$I = - \left(-\frac{2\alpha\beta}{(\alpha^2 + \beta^2)^2} \right) = \frac{2\alpha\beta}{(\alpha^2 + \beta^2)^2}. \quad (\text{IX.31})$$

□

5. The charge distribution function of the atomic nucleus is given by $\rho(r)$, and its Fourier transform (referred to as the form factor) is

$$F(k) = \frac{1}{(2\pi)^{3/2}} \iiint \rho(r) e^{-i\mathbf{k} \cdot \mathbf{r}} d^3r = \frac{1}{(2\pi)^{3/2}} \left(1 + \frac{k^2}{a^2} \right)^{-1}, \quad (\text{IX.32})$$

where $r = |\mathbf{r}|$, $k = |\mathbf{k}|$, $d^3r = dx dy dz$, a is a known constant. Find $\rho(r)$.

Since $F(k)$ depends only on k , $\rho(r)$ is spherically symmetric. The inverse Fourier transform is

$$\rho(r) = \frac{1}{(2\pi)^{3/2}} \iiint F(k) e^{i\mathbf{k} \cdot \mathbf{r}} d^3k. \quad (\text{IX.33})$$

In spherical coordinates in $\mathbf{k}$ -space, $d^3k = k^2 \sin \theta dk d\theta d\phi$, and choosing the z -axis along $\mathbf{r}$, $\mathbf{k} \cdot \mathbf{r} = kr \cos \theta$. The angular integration yields

$$\int_0^{2\pi} d\phi \int_0^\pi e^{ikr \cos \theta} \sin \theta d\theta = \frac{4\pi \sin(kr)}{kr}. \quad (\text{IX.34})$$

Thus,

$$\rho(r) = \frac{4\pi}{(2\pi)^{3/2}} \frac{1}{r} \int_0^\infty k \sin(kr) F(k) dk. \quad (\text{IX.35})$$

Substitute $F(k)$:

$$\begin{aligned} \rho(r) &= \frac{4\pi}{(2\pi)^{3/2}} \frac{1}{r} \int_0^\infty k \sin(kr) \cdot \frac{1}{(2\pi)^{3/2}} \frac{1}{1 + k^2/a^2} dk \\ &= \frac{1}{2\pi^2 r} \int_0^\infty \frac{k \sin(kr)}{1 + k^2/a^2} dk. \end{aligned} \quad (\text{IX.36})$$

Rewrite the denominator using $1 + k^2/a^2 = \frac{k^2 + a^2}{a^2}$:

$$\int_0^\infty \frac{k \sin(kr)}{1 + k^2/a^2} dk = a^2 \int_0^\infty \frac{k \sin(kr)}{k^2 + a^2} dk. \quad (\text{IX.37})$$

The known integral (using the residue theory)

$$\int_0^\infty \frac{k \sin(kr)}{k^2 + m^2} dk = \frac{\pi}{2} e^{-mr}, \quad m > 0, \quad (\text{IX.38})$$

gives

$$a^2 \int_0^\infty \frac{k \sin(kr)}{k^2 + a^2} dk = a^2 \cdot \frac{\pi}{2} e^{-ar}. \quad (\text{IX.39})$$

Therefore,

$$\rho(r) = \frac{1}{2\pi^2 r} \cdot \frac{\pi a^2}{2} e^{-ar} = \frac{a^2}{4\pi r} e^{-ar}. \quad (\text{IX.40})$$

6. Prove the following properties of the delta function. To prove the properties of the delta function, we must

interpret it in the sense of an integral. We suppose f is a continuously differentiable function.

- (1) Note

$$\begin{aligned} \int_{-\infty}^\infty f(x) \delta(x) dx &= f(0), \\ \int_{-\infty}^\infty f(x) \delta(-x) dx &= \int_{-\infty}^\infty f(-u) \delta(u) du = f(0). \end{aligned} \quad (\text{IX.41})$$

Therefore $\delta(x) = \delta(-x)$.

- (2) Note

$$\begin{aligned} \int_{-\infty}^\infty f(x) \cdot (x\delta(x)) dx &= \int_{-\infty}^\infty (xf(x)) \delta(x) dx \\ &= (xf(x))|_{x=0} = 0, \end{aligned} \quad (\text{IX.42})$$

therefore $x\delta(x) = 0$.

- (3) Similarly,

$$\begin{aligned} \int_{-\infty}^\infty f(x) \cdot (g(x)\delta(x)) dx &= \int_{-\infty}^\infty (f \cdot g) \delta(x) dx \\ &= f(0)g(0), \\ \int_{-\infty}^\infty f(x) \cdot (g(0)\delta(x)) dx &= g(0) \int_{-\infty}^\infty f(x) \delta(x) dx \\ &= f(0)g(0). \end{aligned} \quad (\text{IX.43})$$

Therefore $g(x)\delta(x) = g(0)\delta(x)$.

- (4) Note

$$\int_{-\infty}^\infty f(x) \cdot (x\delta'(x)) dx = -(xf(x))'|_{x=0} = -f(0) \quad (\text{IX.44})$$

(In the first equality, we used integration by parts.)

Moreover,

$$\int_{-\infty}^\infty f(x) \cdot (-\delta(x)) dx = -f(0), \quad (\text{IX.45})$$

therefore $x\delta'(x) = -\delta(x)$.

- (5) Set $ax = t$, we obtain

$$\int_{-\infty}^\infty f(x) \delta(ax) dx = \frac{1}{a} \int_{-\infty}^\infty f(t/a) \delta(t) dt = \frac{1}{a} f(0). \quad (\text{IX.46})$$

Therefore $\delta(ax) = 1/a\delta(x)$.

- (6) Note

$$\begin{aligned} \int_{-\infty}^\infty f(x) \cdot (g(x)\delta'(x)) dx &= -(f(x)g(x))'|_{x=0} \\ &= -f(0)g'(0) - f'(0)g(0). \end{aligned} \quad (\text{IX.47})$$

Moreover,

$$\begin{aligned} \int_{-\infty}^\infty f(x) \cdot [g(0)\delta'(x) - g'(0)\delta(x)] dx \\ = g(0)(-f(x))'|_{x=0} - g'(0)f(0) = -f(0)g'(0) - f'(0)g(0). \end{aligned} \quad (\text{IX.48})$$

Therefore $g(x)\delta'(x) = g(0)\delta'(x) - g'(0)\delta(x)$.

(7) Note

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \delta(x^2 - a^2) dx &= \int_0^{\infty} \delta(x^2 - a^2) f(x) dx + \\ \int_{-\infty}^0 f(x) \delta(x^2 - a^2) dx. \end{aligned} \quad (\text{IX.49})$$

Set $t = x^2$, we obtain

$$\int_0^{\infty} \delta(x^2 - a^2) f(x) dx = \frac{1}{2a} f(a). \quad (\text{IX.50})$$

Similarly,

$$\int_{-\infty}^0 f(x) \delta(x^2 - a^2) dx = \frac{1}{2a} f(-a). \quad (\text{IX.51})$$

Therefore $\delta(x^2 - a^2) = 1/2a[\delta(x - a) + \delta(x + a)]$.
(In fact),

$$\delta(g(x)) = \sum_{\substack{x_i \\ g(x_i)=0 \\ g'(x_i) \neq 0}} \frac{\delta(x - x_i)}{|g'(x_i)|} \quad (\text{IX.52})$$

7. Use the delta function to evaluate the following integrals.

(1) Since $\cos x = \frac{e^{ix} + e^{-ix}}{2}$, we have

$$I = \frac{1}{2} \int_{-\infty}^{\infty} \frac{e^{ix}}{1 + x^2} dx + \frac{1}{2} \int_{-\infty}^{\infty} \frac{e^{-ix}}{1 + x^2} dx. \quad (\text{IX.53})$$

In the second integral, substitute $x \rightarrow -x$:

$$\int_{-\infty}^{\infty} \frac{e^{-ix}}{1 + x^2} dx = \int_{-\infty}^{\infty} \frac{e^{iu}}{1 + u^2} du, \quad (\text{IX.54})$$

which is identical to the first integral. Therefore,

$$I = \int_{-\infty}^{\infty} \frac{e^{ix}}{1 + x^2} dx. \quad (\text{IX.55})$$

We know the Fourier cosine transform:

$$\frac{1}{1 + x^2} = \frac{1}{2} \int_{-\infty}^{\infty} e^{-|t|} e^{ixt} dt. \quad (\text{IX.56})$$

Substitute this into I :

$$I = \int_{-\infty}^{\infty} \left[\frac{1}{2} \int_{-\infty}^{\infty} e^{-|t|} e^{ixt} dt \right] e^{ix} dx. \quad (\text{IX.57})$$

Interchanging the order of integration (justified in the sense of tempered distributions),

$$I = \frac{1}{2} \int_{-\infty}^{\infty} e^{-|t|} \left[\int_{-\infty}^{\infty} e^{ix(t+1)} dx \right] dt. \quad (\text{IX.58})$$

The inner integral is the Fourier representation of the Dirac delta:

$$\int_{-\infty}^{\infty} e^{ix(t+1)} dx = 2\pi \delta(t + 1). \quad (\text{IX.59})$$

Thus,

$$I = \frac{1}{2} \int_{-\infty}^{\infty} e^{-|t|} \cdot 2\pi \delta(t+1) dt = \pi \int_{-\infty}^{\infty} e^{-|t|} \delta(t+1) dt. \quad (\text{IX.60})$$

Therefore $I = \pi/e$.

(2) Recall the Fourier transform of the characteristic function of $[-1, 1]$:

$$\int_{-1}^1 e^{ixt} dt = \frac{e^{ix} - e^{-ix}}{ix} = \frac{2 \sin x}{x}. \quad (\text{IX.61})$$

Thus

$$\frac{\sin x}{x} = \frac{1}{2} \int_{-1}^1 e^{ixt} dt. \quad (\text{IX.62})$$

Square the representation, we obtain

$$\left(\frac{\sin x}{x} \right)^2 = \frac{1}{4} \int_{-1}^1 \int_{-1}^1 e^{ix(t+s)} dt ds. \quad (\text{IX.63})$$

Integrate over x , we have

$$I = \frac{1}{4} \int_{-\infty}^{\infty} \int_{-1}^1 \int_{-1}^1 e^{ix(t+s)} dt ds dx. \quad (\text{IX.64})$$

Interchange the order of integration (justified in distribution theory):

$$I = \frac{1}{4} \int_{-1}^1 \int_{-1}^1 \left[\int_{-\infty}^{\infty} e^{ix(t+s)} dx \right] dt ds. \quad (\text{IX.65})$$

The inner x -integral is the Fourier representation of the Dirac delta:

$$\int_{-\infty}^{\infty} e^{ixu} dx = 2\pi \delta(u), \quad \text{where } u = t + s. \quad (\text{IX.66})$$

Therefore

$$I = \frac{1}{4} \int_{-1}^1 \int_{-1}^1 2\pi \delta(t+s) dt ds = \frac{\pi}{2} \int_{-1}^1 \int_{-1}^1 \delta(t+s) dt ds. \quad (\text{IX.67})$$

For fixed s , integrate over t :

$$\int_{-1}^1 \delta(t+s) dt = \begin{cases} 1, & \text{if } -1 < -s < 1, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{IX.68})$$

Thus, for $s \in (-1, 1)$, the inner integral equals 1; otherwise 0. So,

$$I = \frac{\pi}{2} \int_{-1}^1 1 ds = \pi. \quad (\text{IX.69})$$

(3) Let $t = x - 1$, then

$$\begin{aligned} I &= \int_{-\infty}^{\infty} \frac{\cos(t+1)}{t^2 + 1} dt = \cos 1 \int_{-\infty}^{\infty} \frac{\cos t}{t^2 + 1} dt \\ &\quad - \sin 1 \int_{-\infty}^{\infty} \frac{\sin t}{t^2 + 1} dt. \end{aligned} \quad (\text{IX.70})$$

The second integrand is odd, so its integral over $\mathbb{R}$ is zero. Hence,

$$I = \cos 1 \int_{-\infty}^{\infty} \frac{\cos t}{t^2 + 1} dt. \quad (\text{IX.71})$$

Using the result of (1), we obtain $I_3 = \cos 1 \cdot \pi/e$.

(4) Let $t = x - 1$, then

$$\begin{aligned} I &= \int_{-\infty}^{\infty} \frac{(t+1)[\sin t \cos 1 + \cos t \sin 1]}{t^2 + 1} dt \\ &= \cos 1 \int_{-\infty}^{\infty} \frac{t \sin t}{t^2 + 1} dt + \sin 1 \int_{-\infty}^{\infty} \frac{\cos t}{t^2 + 1} dt. \end{aligned} \quad (\text{IX.72})$$

From the previous problem, we have $\int_{-\infty}^{\infty} \frac{\cos t}{t^2 + 1} dt = \pi/e$. Consider

$$L = \int_{-\infty}^{\infty} \frac{te^{it}}{t^2 + 1} dt = i \int_{-\infty}^{\infty} \frac{t \sin t}{t^2 + 1} dt = iK. \quad (\text{IX.73})$$

Define (using the same method as (1)):

$$F(a) = \int_{-\infty}^{\infty} \frac{\cos(at)}{t^2 + 1} dt = \pi e^{-|a|}. \quad (\text{IX.74})$$

Differentiate with respect to a :

$$F'(a) = - \int_{-\infty}^{\infty} \frac{t \sin(at)}{t^2 + 1} dt = -\pi \operatorname{sgn}(a) e^{-|a|}, \quad (\text{IX.75})$$

therefore $I_4 = \pi(\cos 1 + \sin 1)/e$.

8. Integral representation of the Bessel function: Find the function $T(t)$ and determine the endpoints of the integration path such that the complex integral

$$w(z) = z^{1/2} \int_a^b e^{izt} T(t) dt \quad (\text{IX.76})$$

is a solution to the Bessel equation $z^2 \frac{d^2 w}{dz^2} + z \frac{dw}{dz} + (z^2 - \nu^2)w = 0$.

Note: For details, see Exercise 39 in Chapter 7 of Wang Zhuxi et al., Introduction to Special Functions. Lemma:

$$J_\nu(z) = \frac{e^{-(\nu+1/2)\pi i/2}}{\pi \Gamma(1/2)} \left(\frac{z}{2}\right)^{1/2} \int_{i\infty}^{\infty} e^{izt} Q_{\nu-1/2}(t) dt, \quad (\text{IX.77})$$

where $Q_{\nu-1/2}(t)$ is the Legendre function of the second kind, and $-\pi/2 + \omega < \arg z < \pi/2 + \omega$ with $|\omega| < \pi/2$. For $t > 1$ on the real axis, we set $\arg t = 0$. Note

$$\begin{aligned} &\left[z^2 \frac{d^2}{dz^2} + z \frac{d}{dz} + (z^2 - \nu^2) \right] \left(z^{1/2} \int_a^b e^{izt} T(t) dt \right) \\ &= \left[z^{1/2} e^{izt} (1 - t^2) \left(\frac{dT}{dt} - izT \right) \right]_a^b \\ &\quad - z^{1/2} \int_a^b e^{izt} \left\{ \frac{d}{dt} \left[(1 - t^2) \frac{dT}{dt} \right] + \left(\nu^2 - \frac{1}{4} \right) T \right\} dt. \end{aligned} \quad (\text{IX.78})$$

Hence, as long as $T(t)$ is a solution of the Legendre equation of order $\nu - 1/2$, and a suitable integration path is chosen such that the integrated terms vanish (for example, let $a = -1$, $b = 1$), the desired representation holds.

X. Laplace transform

1. Find the Laplace transform of the following functions

(1) By definition,

$$F_1(s) = \int_0^{\infty} (1 - e^{-t})^n e^{-st} dt. \quad (\text{X.1})$$

Set $x = e^{-t}$, then

$$F_1(s) = \int_0^1 (1-x)^n x^{s-1} dx = B(n+1, s) = \frac{\Gamma(s) \cdot n!}{\Gamma(n+1+s)}. \quad (\text{X.2})$$

(2) Similarly,

$$\begin{aligned} F_2(s) &= \int_0^{\infty} (1 - e^{-t})^\alpha e^{-st} dt \\ &= \int_0^1 u^\alpha (1-u)^{s-1} du = B(\alpha+1, s) = \frac{\Gamma(\alpha+1)\Gamma(s)}{\Gamma(\alpha+1+s)}. \end{aligned} \quad (\text{X.3})$$

(3) Note $\mathcal{L}\{\sin \omega t\} = \omega/(\omega^2 + s^2)$, thus

$$F_3(s) = \frac{\omega}{(s-\lambda)^2 + \omega^2}. \quad (\text{X.4})$$

(4) Using the technique of converting a single integral into a double integral, we obtain

$$\begin{aligned} F_4(s) &= \int_0^{\infty} \frac{\sin \omega t}{t} e^{-st} dt = \int_0^{\infty} \sin \omega t \int_s^{\infty} e^{-qt} dq \\ &= \int_s^{\infty} dq \int_0^{\infty} \sin \omega t e^{-qt} dt = \int_s^{\infty} \frac{\omega}{q^2 + \omega^2} dq \\ &= \frac{\pi}{2} - \arctan\left(\frac{s}{\omega}\right). \end{aligned} \quad (\text{X.5})$$

(5) Using the inversion formula for the integral of the image function, and note

$$\mathcal{L}\left\{ \frac{t - \sin t}{t} \right\} = \frac{1}{s} - \frac{\pi}{2} + \arctan s, \quad (\text{X.6})$$

we obtain

$$\begin{aligned} F_5(s) &= \int_s^{\infty} \frac{1}{q} - \frac{\pi}{2} + \arctan q dq \\ &= -1 + s \arctan\left(\frac{1}{s}\right) + \frac{1}{2} \ln\left(1 + \frac{1}{s^2}\right). \end{aligned} \quad (\text{X.7})$$

(6) By definition,

$$\begin{aligned} F_6(s) &= \int_0^{\infty} e^{-st} dt \cdot \int_t^{\infty} \frac{\cos \tau}{\tau} d\tau \\ &= \int_0^{\infty} d\tau \int_0^{\tau} e^{-st} \cdot \frac{\cos \tau}{\tau} dt \\ &= \frac{1}{s} \int_0^{\infty} (1 - e^{-s\tau}) \cdot \frac{\cos \tau}{\tau} d\tau \\ &= \frac{1}{s} \int_0^{\infty} \cos \tau \int_0^s e^{-p\tau} dp \\ &= \frac{1}{s} \int_0^s dp \int_0^{\infty} e^{-p\tau} \cdot \cos \tau d\tau \\ &= \frac{1}{s} \int_0^s \frac{p}{p^2 + 1} dp = \frac{\ln(1 + s^2)}{2s}. \end{aligned} \quad (\text{X.8})$$

2. If $f(t)$ is a periodic function with period α , i.e., $f(t + \alpha) = f(t)$, and if the Laplace transform of $f(t)\eta(t)$ exists, prove that the image function is

$$F(s) = \frac{1}{1 - e^{-\alpha s}} \int_0^\alpha e^{-st} f(t) dt. \quad (\text{X.9})$$

Proof: By definition,

$$\begin{aligned} F(s) &= \sum_{n=0}^{\infty} \int_{n\alpha}^{(n+1)\alpha} f(t) \cdot e^{-st} dt \\ &= \sum_{n=0}^{\infty} \int_0^\alpha f(u) \cdot e^{-su} \cdot e^{-sn\alpha} du \\ &= \frac{1}{1 - e^{-\alpha s}} \int_0^\alpha e^{-st} f(t) dt. \end{aligned} \quad (\text{X.10})$$

□

3. Find the Laplace transform of the following functions

- (1) Applying the Laplace transform technique for periodic functions, we obtain

$$\begin{aligned} F_1(p) &= \frac{1}{1 - e^{-\pi p/\omega}} \int_0^{\pi/\omega} |\cos \omega t| e^{-pt} dt \\ &= \frac{p}{p^2 + \omega^2} + \frac{\omega}{\omega^2 + p^2} \cdot \frac{2e^{-p\pi/(2\omega)}}{1 - e^{-p\pi/\omega}}. \end{aligned} \quad (\text{X.11})$$

- (2) Note $[t] = n$, $n \leq t < n + 1$, thus

$$\begin{aligned} F_2(p) &= \int_0^\infty [t] e^{-pt} dt = \sum_{n=0}^{\infty} \int_n^{n+1} n \cdot e^{-pt} dt \\ &= \sum_{n=0}^{\infty} \frac{n}{p} \left(e^{-pn} - e^{-p(n+1)} \right) \\ &= \frac{1}{p} \sum_{n=1}^{\infty} e^{-pn} = \frac{1}{p} \cdot \frac{e^{-p}}{1 - e^{-p}}. \end{aligned} \quad (\text{X.12})$$

4. Determine the original function of the following expression

- (1) Decompose $F_1(p)$ as

$$F_1(p) = \frac{1}{p} - \frac{a^2}{(p+a)^3} - \frac{a}{(p+a)^2} - \frac{1}{p+a}, \quad (\text{X.13})$$

thus

$$f_1(t) = \eta(t) - \left[\frac{a^2}{2} t^2 e^{-at} + a t e^{-at} + e^{-at} \right] \eta(t). \quad (\text{X.14})$$

- (2) Decompose $F_2(p)$ as

$$F_2(p) = \frac{1}{\omega} \cdot \left[\frac{1}{p} - \frac{p}{p^2 + \omega^2} \right], \quad (\text{X.15})$$

thus

$$f_2(t) = \frac{1}{\omega} \eta(t) - \frac{1}{\omega} \cos \omega t \eta(t). \quad (\text{X.16})$$

- (3) Decompose $F_3(p)$ as

$$F_3(p) = \frac{1}{p} + \frac{5}{3} \frac{1}{p+1} - 3 \frac{1}{p+1/2} + \frac{1}{3} \frac{1}{p-1/2}, \quad (\text{X.17})$$

thus

$$f_3(t) = \eta(t) + \frac{5}{3} e^{-t} \eta(t) - 3 e^{-t/2} \eta(t) + \frac{1}{3} e^{t/2} \eta(t). \quad (\text{X.18})$$

- (4) Decompose $F_4(p)$ as

$$F_4(p) = \frac{1}{2} \frac{1}{(p-\omega)^2} + \frac{1}{2} \frac{1}{(p+\omega)^2}, \quad (\text{X.19})$$

thus

$$f_4(t) = \frac{t}{2} e^{\omega t} \eta(t) + \frac{t}{2} e^{-\omega t} \eta(t). \quad (\text{X.20})$$

- (5) Set $F(p) = 1/(p^2 + 1)^2$. Applying the general inversion formula for the Laplace transform, we obtain

$$\begin{aligned} f(t) &= \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{e^{pt}}{(p^2 + 1)^2} dp \\ &= \text{Res}_{p=i} \left[\frac{e^{pt}}{(p^2 + 1)^2} \right] + \text{Res}_{p=-i} \left[\frac{e^{pt}}{(p^2 + 1)^2} \right] \\ &= \frac{1}{2} \sin t - \frac{1}{2} t \cos t. \end{aligned} \quad (\text{X.21})$$

Using the time-shifting property of the Laplace transform,

$$f_5(t) = \left[\frac{1}{2} \sin(t - \tau) - \frac{1}{2} (t - \tau) \cos(t - \tau) \right] \eta(t - \tau). \quad (\text{X.22})$$

- (6) Note

$$\mathcal{L}\{\sin t\} = \frac{1}{1 + p^2} \quad \mathcal{L}\{1/2 t \sin t\} = \frac{p}{(p^2 + 1)^2}, \quad (\text{X.23})$$

thus

$$f_6(t) = \frac{t - \tau}{2} \sin(t - \tau) \eta(t - \tau). \quad (\text{X.24})$$

5. Using the Laplace transform to solve the following differential equations or integral equation

- (1) Take the Laplace transform of both sides of the equation, we obtain

$$p^2 Y_1(p) - 2p - 1 - 4Y_1(p) = 0. \quad (\text{X.25})$$

By solving it and decomposing $Y_1(p)$, we obtain

$$Y_1(p) = \frac{5}{4} \cdot \frac{1}{p-2} + \frac{3}{4} \cdot \frac{1}{p+2}, \quad (\text{X.26})$$

so

$$y_1(t) = \frac{5}{4} e^{-2t} \eta(t) + \frac{3}{4} e^{-2t} \eta(t). \quad (\text{X.27})$$

- (2) Take the Laplace transform of both sides of the equation, we obtain

$$p^2 Y_2(p) - p + 6Y_2(p) - 7 = 0. \quad (\text{X.28})$$

By solving it, we obtain

$$Y_2(p) = \frac{p+7}{p^2+6}, \quad (\text{X.29})$$

so

$$y_2(t) = \cos(\sqrt{6}t)\eta(t) + \frac{7}{\sqrt{6}}\sin(\sqrt{6}t)\eta(t). \quad (\text{X.30})$$

- (3) Take the Laplace transform of both sides of the equation, we obtain

$$[p^2Y_3(p)-p+1]+2[pY_3(p)-1]+2Y_3(p)=0. \quad (\text{X.31})$$

By solving it, we obtain

$$Y_3(p) = \frac{p+1}{(p+1)^2+1}, \quad (\text{X.32})$$

so

$$y_3(t) = e^{-t} \cos t \eta(t) \quad (\text{X.33})$$

- (4) Take the Laplace transform of both sides of the equation, we obtain

$$[p^2Y_4(p)-1]+2[pY_4(p)]+2Y_4(p)=\frac{2}{p}. \quad (\text{X.34})$$

By solving it and decomposing $Y_4(p)$, we obtain

$$Y_4(p) = \frac{1}{p} - \frac{p+1}{(p+1)^2+1}, \quad (\text{X.35})$$

so

$$y_4(t) = \eta(t) - e^{-t} \cos t \eta(t). \quad (\text{X.36})$$

6. Using the Laplace transform to solve the following integral equation or integro-differential equation.

- (1) Take the Laplace transform of both sides of the equation, we obtain

$$Y_1(p) = \frac{a}{p^2+1} - 2Y_1(p) \cdot \frac{p}{p^2+1}. \quad (\text{X.37})$$

By solving it, we obtain

$$Y_1(p) = \frac{a}{(p+1)^2}, \quad (\text{X.38})$$

thus

$$y_1(t) = ate^{-t}\eta(t). \quad (\text{X.39})$$

- (2) Take the Laplace transform of both sides of the equation, we obtain

$$Y_2(p) + 2Y_2(p) \cdot \frac{p}{p^2+1} = \frac{9}{p-2}. \quad (\text{X.40})$$

By solving it and decomposing $Y_2(p)$, we obtain

$$Y_2(p) = \frac{5}{p-2} - \frac{6}{(p+1)^2} + \frac{4}{p+1}, \quad (\text{X.41})$$

thus

$$y_2(t) = 5e^{2t}\eta(t) - 6te^{-t}\eta(t) + 4e^{-t}\eta(t). \quad (\text{X.42})$$

- (3) Take the Laplace transform of both sides of the equation, we obtain

$$pY_3(p)-1-Y_3(p) \cdot \frac{p}{p^2+1} = \frac{1}{p} (1-e^{-2p}). \quad (\text{X.43})$$

By solving it and decomposing $Y_3(p)$, we obtain

$$Y_3(p) = \frac{1}{p} + \frac{1}{p^3} + \left(\frac{1}{p^2} + \frac{1}{p^4} \right) (1-e^{-2p}), \quad (\text{X.44})$$

thus

$$y_3(t) = \eta(t) + \frac{1}{2}t^2\eta(t) + \left(t + \frac{1}{6}t^3\right)\eta(t) - \left[(t-2) + \frac{1}{6}(t-2)^3\right]\eta(t-2). \quad (\text{X.45})$$

7. Using the Laplace transform to solve the boundary value problem for ordinary differential equations. Take the Laplace transform of both sides of the equation, we obtain

$$p^2Y(p) - k^2Y(p) = e^{-p\tau}, \quad (\text{X.46})$$

By solving it and decomposing $Y(p)$, we obtain

$$Y(p) = \frac{1}{2k} \left(\frac{1}{p-k} - \frac{1}{p+k} \right) \cdot e^{-p\tau}, \quad (\text{X.47})$$

thus

$$y(t) = \frac{1}{2k} e^{k(t-\tau)} \eta(t-\tau) - \frac{1}{2k} e^{-k(t-\tau)} \eta(t-\tau). \quad (\text{X.48})$$

8. Using the Laplace transform to evaluate the following integral.

In this problem, we first assume that t is positive, and finally use parity (or symmetry) to obtain a general conclusion.

- (1) Set $f_1(t) = \int_0^\infty \frac{x \sin xt}{1+x^2} dx$, then

$$\begin{aligned} F_1(p) &= \int_0^\infty e^{-pt} dt \int_0^\infty \frac{x \sin xt}{1+x^2} dx \\ &= \int_0^\infty \frac{x}{1+x^2} dx \int_0^\infty e^{-pt} \sin xt dt \\ &= \int_0^\infty \frac{x}{1+x^2} \cdot \frac{x}{x^2+p^2} dx = \frac{\pi}{2} \cdot \frac{1}{p+1}. \end{aligned} \quad (\text{X.49})$$

Considering the sign of t ,

$$f_1(t) = \text{sgn}(t) \cdot e^{-|t|}. \quad (\text{X.50})$$

- (2) Set $f_2(t) = \int_0^\infty \left(\frac{\sin xt}{x} \right)^2 dx$, then

$$\begin{aligned} F_2(p) &= \int_0^\infty \frac{1}{x^2} dx \int_0^\infty \sin^2(tx) \cdot e^{-pt} dt \\ &= \int_0^\infty \frac{2}{p} \cdot \frac{1}{p^2+4x^2} dx = \frac{\pi}{2} \cdot \frac{1}{p^2}. \end{aligned} \quad (\text{X.51})$$

Considering the sign of t ,

$$f_2(t) = \frac{\pi}{2} |t|. \quad (\text{X.52})$$

(3) Set $f_3(t) = \int_0^\infty \frac{\cos xt}{(1+x^2)^2} dx$, then

$$\begin{aligned} F_3(p) &= \int_0^\infty \frac{1}{(1+x^2)^2} dx \int_0^\infty \cos xt e^{-pt} dt \\ &= \int_0^\infty \frac{1}{(1+x^2)^2} \cdot \frac{p}{p^2+x^2} dx \\ &= \frac{\pi}{4} \left(\frac{1}{p+1} + \frac{1}{(p+1)^2} \right). \end{aligned} \quad (\text{X.53})$$

Considering the sign of t ,

$$f_3(t) = \frac{\pi}{4} (1 + |t|) e^{-|t|}. \quad (\text{X.54})$$

(4) For (4), the residue method os more convenient. Take a semicircular contour in the upper half-plane and consider

$$\begin{aligned} \oint_{C_R} \frac{z \cdot e^{izt}}{(1+z^2)^2} dz &= \int_{-R}^R \frac{x e^{itx}}{(1+x^2)^2} + \int_{C_R} \frac{z \cdot e^{izt}}{(1+z^2)^2} \\ &= 2\pi i \text{Res}_{z=i} \left[\frac{z \cdot e^{izt}}{(1+z^2)^2} \right]. \end{aligned} \quad (\text{X.55})$$

Note

$$\lim_{z \rightarrow \infty} \frac{z}{(1+z^2)^2} = 0, \quad (\text{X.56})$$

thus the large arc vanishes by Jordan's Lemma. Considering the sign of t ,

$$f_4(t) = \frac{\pi}{4} |t| \cdot e^{-|t|} \cdot \text{sgn}(t). \quad (\text{X.57})$$

9. Evaluate the following Frullani integral

(1) Note

$$\mathcal{L}\{(e^{-ax} - e^{-bx}) \cdot \cos cx\} = \frac{p+a}{(p+a)^2+c^2} - \frac{p+b}{(p+b)^2+c^2}. \quad (\text{X.58})$$

Thus

$$\begin{aligned} &\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} \cdot \cos cx dx \\ &= \int_0^\infty \left(\frac{p+a}{(p+a)^2+c^2} - \frac{p+b}{(p+b)^2+c^2} \right) dp \\ &= \frac{1}{2} \ln \left(\frac{b^2+c^2}{a^2+c^2} \right). \end{aligned} \quad (\text{X.59})$$

(2) Note

$$\begin{aligned} F_2(p) &= \mathcal{L}\{(1 - \cos ax) \cdot \cos bx\} \\ &= \frac{p}{p^2+b^2} - \frac{1}{2} \frac{p}{p^2+(a+b)^2} - \frac{1}{2} \frac{p}{p^2+(a-b)^2}. \end{aligned} \quad (\text{X.60})$$

Thus

$$I_2 = \int_0^\infty F(p) dp = \frac{1}{2} \ln \left(\frac{a^2-b^2}{a^2} \right). \quad (\text{X.61})$$

(3) Note

$$F_3(p) = \mathcal{L}\{\sin^2 ax - \sin^2 bx\} = \frac{1}{2} \left(\frac{p}{p^2+4b^2} - \frac{p}{p^2+4a^2} \right). \quad (\text{X.62})$$

Thus

$$I_3 = \int_0^\infty \frac{1}{2} \left(\frac{p}{p^2+4b^2} - \frac{p}{p^2+4a^2} \right) dp = \frac{1}{2} \ln \left(\frac{a}{b} \right). \quad (\text{X.63})$$

(4) Note

$$\begin{aligned} \mathcal{L}\left\{ \frac{\sin ax}{ax} \right\} &= \int_0^\infty \frac{\sin ax}{ax} \cdot e^{-px} dx \\ &= \int_0^\infty \frac{\sin ax}{a} dx \int_p^\infty e^{-qx} dq \\ &= \int_p^\infty dq \int_0^\infty \frac{1}{a} \cdot \sin ax \cdot e^{-qx} dx \\ &= \int_p^\infty \frac{1}{q^2+a^2} dq = \frac{1}{a} \left[\frac{\pi}{2} - \arctan \left(\frac{p}{a} \right) \right]. \end{aligned} \quad (\text{X.64})$$

Similarly,

$$\mathcal{L}\left\{ \frac{\sin bx}{bx} \right\} = \frac{1}{b} \left[\frac{\pi}{2} - \arctan \left(\frac{p}{b} \right) \right]. \quad (\text{X.65})$$

Thus

$$I_4 = \int_0^\infty \left[\frac{1}{a} \arctan \left(\frac{a}{p} \right) - \frac{1}{b} \arctan \left(\frac{b}{p} \right) \right] dp = \ln \left(\frac{b}{a} \right). \quad (\text{X.66})$$

10. Find the inverse Laplace transform of the following function using the general inversion formula.

(1) Decompose $F_1(p)$ as

$$\begin{aligned} F_1(p) &= \frac{1}{(a-b)(a-c)} \frac{1}{p-a} + \frac{1}{(b-a)(b-c)} \frac{1}{p-b} \\ &\quad + \frac{1}{(c-a)(c-b)} \frac{1}{p-c}, \end{aligned} \quad (\text{X.67})$$

thus

$$\begin{aligned} f_1(t) &= \frac{1}{(a-b)(a-c)} e^{at} + \frac{1}{(b-a)(b-c)} e^{bt} \\ &\quad + \frac{1}{(c-a)(c-b)} e^{ct}. \end{aligned} \quad (\text{X.68})$$

(2) Decompose $F_2(p)$ as

$$F_2(p) = \left(\frac{1}{p^2} - \frac{1}{p^2+1} \right) e^{-p\tau}, \quad (\text{X.69})$$

thus

$$f_2(t) = (t-\tau)\eta(t-\tau) - \sin(t-\tau)\eta(t-\tau). \quad (\text{X.70})$$

(3) It is easy to obtain

$$f_3(t) = \eta(t - \alpha). \quad (\text{X.71})$$

(4) By definition,

$$\begin{aligned} f_4(t) &= \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} \frac{1}{p} \frac{e^{(l-x)\sqrt{p}} + e^{(x-l)\sqrt{p}}}{e^{l\sqrt{p}} + e^{-l\sqrt{p}}} e^{pt} dp \\ &= \text{Res}_{p=0} \left[F_4(p) e^{pt} \right] + \sum_{n=0}^{\infty} \text{Res}_{p=p_n} \left[F_4(p) e^{pt} \right], \end{aligned} \quad (\text{X.72})$$

where $p_n = -\left(\frac{2n+1}{2l}\pi\right)^2$. Moreover,

$$\text{Res}_{p=0} \left[F_4(p) e^{pt} \right] = \lim_{p \rightarrow 0} p \cdot F_4(p) e^{pt} = 1, \quad (\text{X.73})$$

$$\begin{aligned} \text{Res}_{p=p_n} \left[F_4(p) e^{pt} \right] &= \lim_{p \rightarrow p_n} (p - p_n) \cdot F_4(p) e^{pt} \\ &= -\frac{4}{(2n+1)\pi} \exp \left[-\left(\frac{2n+1}{2l}\pi\right)^2 t \right] \sin \left(\frac{2n+1}{2l}\pi x \right) \end{aligned} \quad (\text{X.74})$$

thus

$$\begin{aligned} f_4(t) &= 1 - \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{1}{2n+1} \exp \left[-\left(\frac{2n+1}{2l}\pi\right)^2 t \right] \\ &\quad \sin \left(\frac{2n+1}{2l}\pi x \right). \end{aligned} \quad (\text{X.75})$$

11. Find the sum of the following series.

(1) Lemma:

$$\int_0^1 \frac{u^{p-1}}{1+u} du = \frac{1}{2} \left[\psi \left(\frac{p+1}{2} \right) - \psi \left(\frac{p}{2} \right) \right]. \quad (\text{X.76})$$

Proof: Expand $(1+u)^{-1}$ as a geometric series for $0 < u < 1$:

$$\frac{1}{1+u} = \sum_{n=0}^{\infty} (-1)^n u^n. \quad (\text{X.77})$$

Thus

$$\int_0^1 \frac{u^{p-1}}{1+u} du = \sum_{n=0}^{\infty} (-1)^n \int_0^1 u^{p+n-1} du. \quad (\text{X.78})$$

The integral evaluates to $1/(p+n)$, so

$$\int_0^1 \frac{u^{p-1}}{1+u} du = \sum_{n=0}^{\infty} \frac{(-1)^n}{p+n}. \quad (\text{X.79})$$

Now recall the series representation of the digamma function:

$$\psi(z) = -\gamma + \sum_{k=0}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+z} \right). \quad (\text{X.80})$$

Hence

$$\psi \left(\frac{p+1}{2} \right) - \psi \left(\frac{p}{2} \right) = \sum_{k=0}^{\infty} \left(\frac{1}{k+p/2} - \frac{1}{k+(p+1)/2} \right). \quad (\text{X.81})$$

Note

$$\sum_{k=0}^{\infty} \left(\frac{1}{k+p/2} - \frac{1}{k+(p+1)/2} \right) = 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{n+p}. \quad (\text{X.82})$$

Hence the statement is proved. $\square$

Now let us find the sum of the series. It is not hard to obtain

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{(-1)^n}{3n+1} &= \sum_{n=0}^{\infty} (-1)^n \int_0^{\infty} e^{-(3n+1)x} dx \\ &= \int_0^{\infty} \left(\sum_{n=0}^{\infty} (-1)^n e^{-(3n+1)x} \right) dx = \int_0^{\infty} \frac{e^{-x}}{1+e^{-3x}} dx \end{aligned} \quad (\text{X.83})$$

Set $e^{-3x} = u$, then

$$S_1 = \frac{1}{3} \int_0^1 u^{-2/3} \frac{1}{1+u} du = \frac{1}{6} \left[\psi \left(\frac{2}{3} \right) - \psi \left(\frac{1}{6} \right) \right]. \quad (\text{X.84})$$

Note

$$\psi(2z) = \frac{1}{2} \psi(z) + \frac{1}{2} \psi \left(z + \frac{1}{2} \right) + \ln 2, \quad (\text{X.85})$$

thus

$$S_1 = \frac{1}{6} \left[2\psi \left(\frac{2}{3} \right) - 2\psi \left(\frac{1}{3} \right) + 2\ln 2 \right] = \frac{\pi}{3\sqrt{3}} + \frac{\ln 2}{3}. \quad (\text{X.86})$$

(2) Similarly,

$$\begin{aligned} S_2 &= \sum_{n=0}^{\infty} \int_0^{\infty} (-1)^n e^{-(4n+1)x} dx \\ &= \int_0^{\infty} e^{-x} \sum_{n=0}^{\infty} (-e^{-4x})^n dx = \int_0^{\infty} \frac{e^{-x}}{1+e^{-4x}} dx \\ &= \frac{1}{4} \int_0^1 \frac{u^{-3/4}}{1+u} du = \frac{1}{8} \left[\psi \left(\frac{5}{8} \right) - \psi \left(\frac{1}{8} \right) \right] \\ &= \frac{1}{\sqrt{2}} [\ln(1+\sqrt{2}) + \pi/2]. \end{aligned} \quad (\text{X.87})$$

(3) Decompose the series as

$$S_3 = \sum_{n=0}^{\infty} \frac{1}{2} \cdot \frac{(-1)^n}{3n+1} - \frac{(-1)^n}{3n+2} + \frac{1}{6} \cdot \frac{(-1)^n}{n+1} \quad (\text{X.88})$$

We already know

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{3n+1} = \frac{\pi}{3\sqrt{3}} + \frac{\ln 2}{3}, \quad \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} = \ln 2. \quad (\text{X.89})$$

Moreover, using the same trick as (1), we can easily obtain

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{3n+2} = \int_0^{\infty} \frac{e^{-2x}}{1+e^{-3x}} dx = \frac{\pi}{3\sqrt{3}} - \frac{\ln 2}{3}. \quad (\text{X.90})$$

Thus

$$S_3 = \frac{2\ln 2}{3} - \frac{\pi}{6\sqrt{3}}. \quad (\text{X.91})$$

(4) Recall 10(1) of section Gamma Function, thus

$$\begin{aligned} S_4 &= \sum_{n=0}^{\infty} \left(\frac{1}{6} \cdot \frac{1}{n+1/3} - \frac{1}{3} \frac{1}{n+2/3} + \frac{1}{6} \frac{1}{n+1} \right) \\ &= - \left[\frac{1}{6} \psi\left(\frac{1}{3}\right) - \frac{1}{3} \psi\left(\frac{2}{3}\right) + \frac{1}{6} \psi(1) \right] \\ &= \frac{\pi}{4\sqrt{3}} - \frac{\ln 3}{4}. \end{aligned} \quad (\text{X.92})$$

12. Find an integral solution for each of the following Laplace-type ordinary differential equations.

(1) The differential equation satisfied by $v_1(t)$ is:

$$-\frac{d}{dt}[(t^2-t)v_1(t)] + v_1(t) = 0, \quad (\text{X.93})$$

thus let $v_1(t) = 1/t^2$. In this occasion,

$$w_1(z) = \int_C \frac{e^{zt}}{t^2} dt \quad (\text{X.94})$$

i) If $\operatorname{Re} z > 0$, the integration path C can be taken as a contour starting from infinity below the negative real axis, going counterclockwise around $t = 0$ in the positive direction once, and returning to infinity above the negative real axis.

ii) If $\operatorname{Re} z < 0$, the integration path C can be taken as a contour starting from infinity above the positive real axis, going counterclockwise around $t = 0$ in the positive direction once, and returning to infinity below the positive real axis.

(2) The differential equation satisfied by $v_2(t)$ is:

$$-\frac{d}{dt}[(t^2+t)v_2(t)] + (-t+1)v_2(t) = 0, \quad (\text{X.95})$$

thus let $v_2(t) = 1/(t+1)^3$. In this occasion,

$$w_2(z) = \int_C \frac{e^{zt}}{(t+1)^3} dt \quad (\text{X.96})$$

i) If $\operatorname{Re} z > 0$, the integration path C can be taken as a contour starting from infinity below the negative real axis, going counterclockwise around $t = -1$ in the positive direction once, and returning to infinity above the negative real axis.

ii) If $\operatorname{Re} z < 0$, the integration path C can be taken as a contour starting from infinity above

the positive real axis, going counterclockwise around $t = -1$ in the positive direction once, and returning to infinity below the positive real axis.

XI. Equations of Mathematical Physics and Boundary Conditions

1. In the problem of transverse vibrations of a string, if the string is subject to a damping force proportional to the velocity (with proportionality constant $-\alpha$), derive the damped vibration equation of the string. Furthermore, if in addition to the damping, the string is also subject to a restoring force proportional to the displacement of the string (with proportionality constant $-k$), what is the equation governing the vibration of the string in this case?

For the first problem, we have

$$\begin{aligned} (T \cos \theta)|_{x+\Delta x} - (T \cos \theta)|_x &= 0, \\ (T \sin \theta)|_{x+\Delta x} - (T \sin \theta)|_x - \alpha \frac{\partial u}{\partial t} \Delta x &= \rho \Delta x \frac{\partial^2 u}{\partial t^2}. \end{aligned} \quad (\text{XI.1})$$

therefore

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = -\frac{\alpha}{\rho} \frac{\partial u}{\partial t}. \quad (\text{XI.2})$$

For the second problem, we have

$$\begin{aligned} (T \cos \theta)|_{x+\Delta x} - (T \cos \theta)|_x &= 0, \\ (T \sin \theta)|_{x+\Delta x} - (T \sin \theta)|_x - \alpha \frac{\partial u}{\partial t} \Delta x - ku \Delta x &= \rho \Delta x \frac{\partial^2 u}{\partial t^2}. \end{aligned} \quad (\text{XI.3})$$

therefore

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = -\frac{\alpha}{\rho} \frac{\partial u}{\partial t} - \frac{k}{\rho} u. \quad (\text{XI.4})$$

2. A uniform elastic rod of length l and cross-sectional area S has one end ($x = 0$) fixed and the other end ($x = l$) subjected to a tensile force F along the axis of the rod, thereby reaching equilibrium. At $t = 0$, the external force F is removed. Write down the governing equation, boundary conditions, and initial conditions for the longitudinal vibration of the rod.

Note that when $t = 0$,

$$ES \frac{\partial u}{\partial x} = F, \quad \forall x \in [0, l], \quad (\text{XI.5})$$

thus $u|_{t=0} = \frac{Fx}{ES}$. Therefore

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0, \\ u|_{t=0} = \frac{Fx}{ES}, \quad \frac{\partial u}{\partial t}|_{t=0} = 0, \\ u|_{x=0} = 0, \quad \frac{\partial u}{\partial x}|_{x=l} = 0. \end{cases} \quad (\text{XI.6})$$

3. A thin metal rod of length l (which can be approximately regarded as one-dimensional) carries a steady current I . If both ends of the rod ($x = 0$ and $x = l$) exchange heat with the surroundings according to Newton's law of cooling, and the ambient temperature

is u_0 , with the initial temperature distribution along the rod given by $u = u_0(1 - 2x/l)^2$. Write down the governing equation, boundary conditions, and initial conditions for the temperature field in the rod, assuming the resistance of the metal rod is R .

The heat conduction equation with a constant source term due to Joule heating is:

$$\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2} + f, \quad 0 < x < l, \quad t > 0, \quad (\text{XI.7})$$

where

$$a^2 = \frac{k}{\rho c}, \quad f = \frac{I^2 R}{\rho c l}. \quad (\text{XI.8})$$

The boundary conditions, derived from Newton's law of cooling, are:

$$\frac{\partial u}{\partial x}(0, t) = h(u(0, t) - u_0), \quad \frac{\partial u}{\partial x}(l, t) = -h(u(l, t) - u_0), \quad (\text{XI.9})$$

with $h = H/k$. The initial condition is:

$$u(x, 0) = u_0 \left(1 - \frac{2x}{l}\right)^2, \quad 0 \leq x \leq l. \quad (\text{XI.10})$$

Therefore,

$$\begin{cases} u_t = k/(\rho c)u_{xx} + I^2 R/(\rho c l), \\ u_x(0, t) = h(u(0, t) - u_0), \\ u_x(l, t) = -h(u(l, t) - u_0), \\ u(x, 0) = u_0 \left(1 - \frac{2x}{l}\right)^2 \end{cases} \quad (\text{XI.11})$$

4. In a uranium block, in addition to the diffusion motion of neutrons, there are also processes of neutron absorption and multiplication. Suppose that per unit time, the number of neutrons absorbed and multiplied per unit volume are both proportional to the neutron concentration $u(\mathbf{r}, t)$ at that time and position. Thus, the net increase in the number of neutrons can be expressed as $\alpha u(\mathbf{r}, t)$, where α is the proportionality constant. Derive the partial differential equation satisfied by $u(\mathbf{r}, t)$.

It is not hard to obtain

$$\frac{\partial u}{\partial t} = D \nabla^2 u + \alpha u. \quad (\text{XI.12})$$

5. A thin rod with non-uniform cross-section has its cross-sectional area proportional to the distance from the tip of the rod. Write down the equation for longitudinal vibrations of this rod.

Take a cross-section at x for analysis.

$$ES|_{x+\Delta x} \frac{\partial u}{\partial x} \Big|_{x+\Delta x} - ES_x \frac{\partial u}{\partial x} \Big|_x = \rho S \Delta x \frac{\partial^2 u}{\partial t^2}. \quad (\text{XI.13})$$

thus

$$\rho \cdot S \frac{\partial^2 u}{\partial t^2} = E \cdot \frac{\partial}{\partial x} \left(S \frac{\partial u}{\partial x} \right). \quad (\text{XI.14})$$

We may assume that $S(x) = kx^2$, therefore

$$\frac{\partial^2 u}{\partial t^2} = \frac{a^2}{x^2} \frac{\partial}{\partial x} \left(x^2 \frac{\partial u}{\partial x} \right). \quad (\text{XI.15})$$

6. A string of length l is immersed in a medium with damping coefficient α . One end is fixed, and the other end is subjected to a driving force $A \sin \omega t$. Initially, the string is at rest in its equilibrium position. Write down the equation for transverse vibrations of the string, along with the boundary conditions and initial conditions.

From 1 we obtain the equation for transverse vibrations of the string is

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = -\frac{\alpha}{\rho} \frac{\partial u}{\partial t}. \quad (\text{XI.16})$$

Moreover, the boundary condition at $x = l$ is derived from the transverse force balance at that end. The external driving force $A \sin \omega t$ acts transversely at $x = l$. For small vibrations, the transverse component of the string tension T is approximately $T \frac{\partial u}{\partial x}(l, t)$. Balancing forces on the endpoint (assuming negligible mass) yields:

$$A \sin \omega t - T \frac{\partial u}{\partial x}(l, t) = 0, \quad (\text{XI.17})$$

which simplifies to

$$\frac{\partial u}{\partial x}(l, t) = \frac{A}{T} \sin \omega t. \quad (\text{XI.18})$$

Therefore

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} + \alpha \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}, \\ u|_{x=0} = 0, \quad \frac{\partial u}{\partial x} \Big|_{x=l} = \frac{A}{T} \sin \omega t, \\ u|_{t=0} = 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0. \end{cases} \quad (\text{XI.19})$$